From Friends to 'Family': Schools, Neighborhoods, and Gang Recruitment*

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Abstract

This paper examines how gang-related criminal behavior spreads through extended family and peer networks, using newly linked Swedish administrative data covering over 18,000 individuals with confirmed gang affiliation. Within-family correlations reveal intergenerational transmission of gang-related criminal behavior, where parental criminal history accounts for only half of a broader latent 'crime-family' effect, and with older siblings explaining most of the remaining influence. Based on this, I study what I term 'gateway peers' – students whose older siblings have committed gang-related crimes – and estimate peer effects using quasi-random variation in cohort-to-cohort composition. Exposure in grade 7 significantly increases both general and gang-related offending in early adulthood, with effects concentrated within schools, not neighborhoods, and strongest when gateway peers share the students background. Excluding regions where gangs are not present, exposure also significantly increases the probability of being confirmed as gang affiliated after compulsory school. I provide additional evidence of a recruitment mechanism beyond just peer influence: spillovers disappear when gateway siblings are absent during the expected recruitment window, and exposed students are more likely to co-offend with gateway peers relatives and confirmed gang members. Exposure further reduces academic performance and increases the diagnosis of mood disorders among adolescents. The findings highlight the role of sibling-linked peer networks in shaping criminal trajectories and suggest that preventive interventions targeting school environments can help disrupt recruitment to organized crime.

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1 Introduction

Over the past two decades, Sweden has experienced a sharp rise in gang-related crime, raising concerns that the country may be at the forefront of a broader expansion of organized criminal activity in Western Europe (Göteborgs universitet 2023; Krüsselmann et al. 2023; Rostami and Mondani 2023; Suonpää et al. 2024; UNODC 2023).[1] The share of 18-24-year-olds suspected of narcotics offenses is now three times higher than it was twenty years ago, and four times higher among 15-17-year-olds.[2] Between 2013 and 2023, suspicions for murder or attempted murder increased eight-fold among 18-24-year-olds, and nearly forty-fold among 15-17-year-olds.[3]

Although Swedish crime rates remain relatively low compared to the Americas (UNODC 2023), the escalation in narcotics trafficking, contract killings, and credible reports of foreign state involvement (Polismyndigheten 2021, 2025; Säkerhetspolisen 2024) has caused major policy changes.[4] The electoral rise of the Swedish Democrats[5] has coincided with a legislative pivot toward more punitive reforms, including expanded surveillance powers, the introduction of security zones, and tougher sentencing laws (Polismyndigheten 2023, 2024; Regeringskansliet 2021, 2024). While such measures may reduce violence in the short term, they entail substantial fiscal, privacy, and civil liberty costs.[6]

Furthermore, anecdotal evidence suggests that many gangs are rooted in extended families and rely on recruitment processes that follow age-based progression: older members draw in younger teenagers – and, in some cases, children – with some groups reportedly returning to their former schools to identify receptive recruits (SVT 2025). These age-structured and kin-based recruitment channels are likely central to the expansion of organized crime groups, yet remain largely overlooked in public policy debates.

This paper examines how gang-related behavior diffuses through family and peer networks in Sweden, with a focus on who I term "gateway peers": students whose older siblings – or other relatives – have history of gang-related offenses. It addresses five main questions: (i) Who becomes involved in gang crime, and what types of offenses do they commit? (ii) How are gang-related activities transmitted within families, distinguishing vertical links (parents, aunts, and uncles) from horizontal ones (siblings and cousins)? (iii) How does exposure to school and neighborhood "gateway" peers affect youth without direct family ties to such networks? (iv) Do the resulting peer effects reflect systematic gang recruitment?

[1] See Figure A1 for trends in interpersonal violence in Europe.

[2] See Figure A2 and Figure A3 for gang-related crime trends since the 2000s.

[3] The murder-suspicion rate among 15-17-year-olds was near zero in 2013; the large percentage increase reflects a rise from a low baseline.

[4] For media reporting see e.g., (CNN 2025), (SVT 2024a), (SVT 2024b), (NRK 2024), (DR 2024).

[5] The Sweden Democrats are a right-wing populist party that first entered parliament in 2010. In the 2022 general election, they became the second largest party, campaigning primarily on stricter policies targeting crime and immigration.

[6] Funding for the justice system increased from approximately SEK 40 billion in 2015 (4.6 percent of the central government budget) to SEK 76 billion in 2024 (5.7 percent), an increase of nearly 90 percent (Sveriges Riksdag 2014, 2023).

(v) How are students' educational and mental health outcomes, which have been reported to have indicators for susceptible youth, affected?

The analysis uses newly compiled Swedish police data that identify more than 18,000 individuals with confirmed gang affiliation, based on encrypted communications, surveillance, and police intelligence. These records are linked to population-wide administrative registers with detailed demographic, educational, and criminal information, including convictions, suspicions, and complete kinship ties to parents, siblings, aunts, uncles, and cousins. This combination of data makes it possible to trace criminal behavior vertically across generations and horizontally within generations, and to examine how exposure to gateway peers diffuses crime and gang activity through school and neighborhood peer networks for more than 600,000 compulsory school students.

A descriptive analysis shows that approximately 94 percent of documented gang members are male, 79 percent have a foreign background, and 64 percent have lived in a single-parent household. More than three-quarters have been arrested for at least one co-offense. A substantial proportion have been suspected of narcotics offenses (69 percent), violent (57 percent), property crimes (57 percent), or weapons possession (37 percent). Importantly, 71 percent have been suspected of a *group crime*,[7] which serves as my primary proxy for gang-related activity. Among those in the gang database with respective kinship link, 23 percent have an older sibling in the gang database, 3 percent have a parent, 6 percent have an uncle or aunt, and 10 percent have an older cousin.

Turning to family correlations, students with a parent with a criminal record are associated with a higher likelihood of committing any crime, group crimes, and being confirmed as a gang member between the ages of 16 and 21. Beyond this vertical link, within-generation effects are substantial: having an older sibling involved in a group crime strongly predicts later offending, while the influence of uncles, aunts, and cousins is notably weaker – consistent with the more frequent day-to-day contact typical of sibling relationships. Following Adermon et al. (2021), I interpret each family-type indicator as a noisy signal of a common latent "crime-family" factor. Summing the estimated marginal effects provides a lower-bound estimate of this underlying influence. The results suggest that focusing on parents alone captures only about half of the total effect. Most of the remaining share is accounted for by sibling exposure, while the contribution of more distant relatives, such as uncles, aunts, and cousins, is comparatively minor.

Motivated by reports that gang recruitment often flows from older to younger adolescents (Brottsförebyggande rådet 2023), I focus on 'gateway peers': students whose older relatives – particularly siblings—have records of group crime involvement. Leveraging cohort-to-cohort variation in grade-level peer composition within school-by-neighborhood cells, I estimate

[7] Group crimes are offenses committed with two or more co-offenders – including assault, murder, robbery, narcotics offenses, weapons possession, money-laundering, and extortion – following the classification of Esbensen and Maxson (2012), the definition used by Dustmann et al. (2023) and the criminal records of gang members (see Figure A6 and Table A1).

how exposure to such peers affects later criminal outcomes. The analysis restricts attention to focal students whose own relatives have no criminal record, thereby isolating the potential for gateway peers, through their older kin, to act as bridges and draw same-grade peers into gang-connected networks.

Exposure is measured as the share of school or neighborhood grade peers of students that are gateway peers. The analysis focuses on students observed in grade 7 between 2013 and 2019. Grade 7 is chosen for two reasons. First, many students switch schools at this stage, reshaping peer groups in ways that generate natural variation in exposure. Second, recruitment into gang activity frequently begins in early adolescence, making grade 7 – a period when most students are 13 to 14 years old – a particularly salient point at which to capture initial exposure to gang-connected peers. Focusing on this entry grade also helps avoid the re-sorting and attrition that could arise in later years of compulsory schooling where institutional induced re-shuffling is uncommon.

Although schools bring together students from a wider variety of backgrounds, neighborhood peers tend to be more similar in socioeconomic terms and are linked by long-standing local connections. Analyzing how exposure impacts differ in these contexts helps to evaluate whether the influence of gateway peers is more pronounced within the centralized structure of schools or within the locally rooted networks of neighborhoods.

The key identification challenges in measuring peer effects are reflection, common shocks, and selection. Reflection (Manski 1993; Moffit 2001) arises when the behavior of the other peers and the focal student influence each other simultaneously, making causal direction difficult to establish. Because gateway peers are defined by the criminal history of their older relatives, and focal students are unlikely to influence the prior criminal activity of their older relatives, reflection should not be a concern. Thus, the design isolates channels of influence running from the older relative to the gateway peer and then to the focal student, or potentially directly from the older relative to the focal student.

Common shocks, such as policy reforms and macroeconomic changes, are absorbed through calendar-year and birth-cohort fixed effects. To address selection into schools and neighborhoods, I adopt a cohort-to-cohort research design based on the seminal work of Hoxby (2000). By comparing adjacent cohorts within the same school-by-neighborhood cells, the design differs out time-invariant local factors such as school quality, residential composition, and other neighborhood or school attributes that parents may consider when deciding where to live and send their children. The resulting variation in gateway peer exposure is plausibly quasi-random and appears orthogonal to a wide range of student background characteristics.

The results identify peers with older siblings involved in group crimes as the primary drivers of spillovers. A 5-point increase in the share of such gateway peers in a student's school-grade cohort increases the probability of committing a group crime between ages 16 and 21 by roughly 6.3 percent relative to the mean. The corresponding estimate for neighborhood-grade exposure is smaller and is not statistically significant. A decomposition confirms that most of the effect comes from peers who share the same school but not the same neighborhood, highlighting schools as the primary environment through which these exposures take hold. This likely reflects that schools are the primary institution where previously unconnected students encounter gateway peers due to residential segregation.

The effects are strongest for violent, narcotic, and weapons offenses, categories commonly linked to gang activity, and increase with the intensity of exposure, particularly in cohorts with multiple gateway peers present. The impacts are most pronounced in settings where gangs are plausibly active: urban centers, areas with documented gang presence, and neighborhoods with a high share of foreign-born residents. Within these contexts, school-based exposure also significantly increases the likelihood of appearing in the gang database after finishing compulsory school (roughly 20 percent relative to the mean given a 5 percentage point increase is sibling based exposure). Furthermore, the effects appear to operate through gender-specific channels: gateway peers with criminal older brothers influence the criminal behavior of male students but not female students, whereas criminal older sisters primarily affect female students, with some evidence of spillovers onto male students.

A series of analyses supports the interpretation that these spillovers reflect recruitment dynamics beyond general peer disruption. Focal students are more likely to co-offend not only with gateway peers but also with their gateway peers relatives and with individuals listed in the gang database. The effects are greater when the gateway peer shares the country background of the focal student, consistent with homophily in peer formation and recruitment. The effects of exposure weaken substantially when the older sibling of the gateway peer is incarcerated, resides outside the municipality, or is substantially older than the gateway peer during the exposure window. These conditions matter more for gang-related outcomes, whereas the effects on general crime appear to be more robust to the lower presence of the older sibling. This suggests both a broad criminogenic peer effect and an additional recruitment effect that depends on the active presence of the older sibling.

Back-of-the-envelope calculations suggest that school-based gateway peer exposure could account for about 14.7% of confirmed gang members born between 2000 and 2006 in county capitals. When including co-offending with confirmed gang members, the share increases to as much as 19.3% nationwide among students from the same cohorts who had no sibling links to gangs.

Exposure to gateway peers also has broader consequences. Students assigned to more gateway peers in grade 7 perform worse on standardized tests in grade 9, receive lower merit scores, and are more likely to be diagnosed with a mood-related diagnosis during compulsory school. These effects are concentrated in the school setting and absent in neighborhood exposure.

This paper contributes to several strands of literature. First, it adds to the growing body of work on gangs and organized crime and their broader societal impacts. Much of this literature focuses on the Italian Mafia and its influence on economic outcomes, firm behavior, and political institutions (e.g., Acemoglu et al. 2019; Barone and Narciso 2015; Campedelli et al. 2024; Daniele and Dipoppa 2017; De Feo and De Luca 2017; Pinotti 2015); on Latin American cartels and the effects of narcotics trafficking, violence, and forced migration (e.g., Carvalho and Soares 2016; Dell 2015; Melnikov et al. 2020; Monteiro and Rocha 2017; Sobrino 2020; Sviatschi 2022a,b); and on gangs in relation to prisons, guns, and drugs in the US (e.g., Bruhn 2021; Cook et al. 2007; Dooley et al. 2014; Grogger 2002; Levitt and Venkatesh 2000; Skarbek 2010). In contrast to this previous work, I study a setting where gang crime might not be expected to emerge as a major societal issue. Although Sweden has strong institutions and a comprehensive welfare system, it is increasingly grappling with gang-related violence and organized criminal networks. The disconnect between Swedens traditionally robust social safety net and its growing gang culture, coupled with its potential position at the forefront of a broader European trend, makes it a particularly important and understudied context for understanding the diffusion of organized crime groups.

Second, this paper contributes to the literature on the effects of criminal peer relationships within and beyond schools and neighborhoods, and builds on broader research on peer effects in educational settings (e.g., Black et al. 2013; Hoxby 2000; Hoxby and Weingarth 2005), as well as in criminal contexts such as prisons (Bayer et al. 2009; Damm and Gorinas 2020; Johnsen and Khoury 2024; Ouss 2011; Philippe 2024; Stevenson 2017), neighborhoods (Billings et al. 2019; Billings and Schnepel 2020; Damm and Dustmann 2014; Dustmann et al. 2023), and schools (Billings and Hoekstra 2023; Carrell et al. 2018; Carrell and Hoekstra 2010; Gupta et al. 2022; Kim and Fletcher 2017; Larsen and Kristensen 2017). While this literature has documented disruptive and criminogenic peer effects, the current study proposes a more systematic recruitment-based mechanism. Specifically, it argues that peers with older relatives involved in gang crime act as "gateways" or social bridges, introducing otherwise unexposed students to gang-adjacent networks.

Third, the paper contributes to the literature on intergenerational transmission of criminal capital (or "anti-human" capital), and youth indirect exposure to the justice system (Besemer and Farrington 2012; Bhuller et al. 2018; Eriksson et al. 2016; Finlay et al. 2023; Grönqvist et al. 2024; Hjalmarsson and Lindquist 2012; Junger et al. 2013; Murray and Farrington 2008; Norris et al. 2021). Although most previous studies focus on parent-to-child transmission or sibling correlations, I link individuals to a broad set of relatives including parents, older siblings, uncles, aunts, and older cousins. This rich set of family ties allows for the examination of how criminal behavior is transmitted across a broader set of relational structures. It also enables a distinction between vertical (intergenerational) transmission and lateral (within-generation) influence. Finally, the data allow me to explore how these familial connections can transfer to school and neighborhood peer effects and serve as indirect channels through which students are introduced to organized crime groups.

The results point to gateway peers as important to the diffusion of gang activity. Students with peers whose older siblings are involved in group crime are significantly more likely to engage in such offenses themselves, even absent direct familial ties. These spillovers operate primarily through school cohorts rather than neighborhoods and are strongest when gateway peers share the focal students background. Moreover, the disappearance of effects when the older sibling is not present underscores that gateway peers function as recruitment conduits rather than only markers of criminogenic peer influence. Exposure also has consequences beyond crime, lowering educational achievement and worsening adolescent mental health. Taken together, the findings suggest that disrupting the spread of gang activity requires interventions in early adolescence, including efforts to strengthen school environments, limit the role of gateway peers as recruitment intermediaries, and expand preventive and rehabilitative programs that break the intergenerational and intragenerational transmission of gang involvement, thus reducing the likelihood that gateway peers emerge in the first place.

The rest of the paper proceeds as follows. Section 2 provides institutional context on the gang networks in Sweden and the structure of the compulsory school system and the neighborhoods. Section 3 describes the data sources and describes the construction of the sample. Section 4 presents descriptive statistics on individuals in the gang database and on the main student sample. It also introduces the gateway peer exposure measure and documents patterns of criminal transmission within extended families. Section 5 details the cohort-to-cohort design and key identifying assumptions. Section 6 presents the main empirical results, along with an evaluation of the existence of a possible systematic recruitment mechanism. Section 7 concludes.

2 Institutional Setting

2.1 Gangs and Recruitment Channels in Sweden

The following review draws on reports from the Swedish National Council for Crime Prevention (Brottsförebyggande rådet 2016, 2021a,b, 2023, 2024a,b), as well as research by Swedish sociologists and criminologists (e.g., Gerell et al. 2020, 2021; Rostami and Mondani 2023; Sturup et al. 2019; Öberg 2015; Örnlind and Forkby 2025).

Since the early 2000s, the Swedish criminal landscape has shifted away from traditional hierarchical organizations, such as motorcycle gangs and certain international networks, and has moved more toward decentralized groups embedded in socially disadvantaged neighborhoods. These emerging networks were initially described as family or friendship-based, with recruitment often occurring through relatives (Brottsförebyggande rådet 2016; Gerell et al. 2020). The Swedish police initially referred to them as “clans”, highlighting their familial structure and territorial ambitions,[8] but systematic recruitment has since broadened their base. Although strong kinship bonds remain central in some groups, others are more relying on diffuse social networks (Gerell et al. 2020). Typically, these groups are deeply rooted in disadvantaged areas, with older members often mentoring younger people who struggle academically or socially.

Drug dealing is a key recruitment channel, as criminal groups seek expendable individuals willing to take risks. Historically, minors benefited from relatively lenient sentencing,[9] so recruitment often begins with small, compensated tasks and can escalate rapidly into more serious criminal activity. According to Brottsförebyggande rådet (2023), older teens (age 16-20) actively recruit children as young as 12 to 15, perpetuating a cycle in which criminal involvement is normalized and glamorized.

As these networks have expanded, Sweden has experienced a sharp increase in fatal shootings driven by young men in vulnerable neighborhoods, distinguishing it from most other European countries, where gun-related homicide rates have declined. Studies by Sturup et al. (2019) and Gerell et al. (2021) attribute this increase to violent conflicts between criminal groups, in which shootings often provoke retaliatory attacks and cycles of violence, similar patterns have also been shown in international networks (Papachristos et al. 2012). Moreover, a shooting can quickly elevate an individuals status within the organization (Jönsson and Nilsson 2019).

Although Swedens situation is particularly severe, similar trends have appeared in other western European countries. For example, the use of hand grenades increased in Sweden in 2016, followed by increased incidents in Belgium and the Netherlands between 2017 and 2020 (Krusselmann et al. 2021; Sturup et al. 2020). Likewise, England and Wales experienced an increase in youth violence between 2015 and 2018 – especially knife-related homicides – mainly involving young men in criminal groups. This increase has been associated with socioeconomic disadvantage and the expansion of the drug trade (Morgan et al. 2020).

2.2 The Swedish Compulsory School and Neighborhoods

The institutional characteristics of the compulsory school system in Sweden shape how student reference groups form and evolve over time, influencing both endogenous and quasi-exogenous variation in peer exposure. Because the empirical strategy hinges on peer composition within schools and neighborhoods, I begin with a brief overview of the structure of compulsory schooling. Children are required to attend school from approximately ages 6 to 16, progressing through three stages: lower (grades 0-3), middle (grades 4-6), and upper (grades 7-9). Although grade configurations vary somewhat across municipalities, most students transition to a new school before grades 4 or 7. As shown in Figure A4, most school

[8] The Swedish police later withdrew this terminology following public criticism.

[9] In January 2022, the Swedish government eliminated sentencing leniency for individuals aged 18 to 20 as part of broader efforts to combat organized crime.

moves occur between grades 3-4 and 6-7, making these the key points at which peer groups are reshuffled.

Since the early 1990s, Sweden has implemented a decentralized school system. Although a national curriculum is in place – including standardized testing in grades 3, 6, and 9 – independent schools are publicly funded and manage their own admissions, often based on queue time, geographic proximity, or sibling preference. Municipality-run schools assign students strictly based on residential proximity and are required to accommodate all children in their catchment area, including newly arrived refugees (Björklund et al. 2004; Holmlund et al. 2020; Sjögren et al. 2024). Although municipalities aim to limit school segregation, the combination of residential segregation and expanded school choice has contributed to increased school-level segregation since the 1990s (Böhlmark et al. 2016; Mörtlund 2020).

The analysis focuses on grade 7, the first year of upper-compulsory school, when students are typically 13 to 14 years old. This timing is in line with police intelligence reports that place the modal age of gang recruitment between 13 and 15 (Brottsförebyggande rådet 2023). In addition, grade 7 also marks an institutional transition of the school-grade group, offering a natural reshuffling of peers in this context. This provides a clean window for observing initial exposure to gateway peers, which likely sets the trajectory for peer influence throughout the remainder of upper compulsory schooling. Later grades involve fewer structural transitions, and any peer changes that do occur are more likely to reflect endogenous sorting.

3 Data and Sample

This section describes key data sources, explains how gang affiliation is identified, and presents descriptive statistics on individuals listed in the gang database. Then it outlines how the sample is constructed.

3.1 Data Sources

The analysis merges several rich individual-level datasets maintained by the Swedish Police, the Swedish National Council for Crime Prevention (BRÅ), Statistics Sweden, and the National Board of Social Affairs and Health. Unique personal identification numbers assigned to all Swedish residents enable accurate record linkage across registers.

Gang Database and Crime Registers. A core component of the analysis is a police-administered dataset known as the *cylinder model* – implemented in 2019 to track criminal networks. For clarity, I refer to this dataset as the *gang database*. Individuals are included based on a combination of local police intelligence, documented criminal history, and encrypted communications indicating active involvement in organized crime groups (Gerell et al. 2020).

Inclusion in the gang database does not necessarily correspond to the moment that an individual joined a gang; rather, it marks the point at which law enforcement confirmed active participation in networked criminal activity. Because the database is only available from 2019 to 2024, it does not capture whether older family members were already gang-involved when focal students attended upper compulsory school. Therefore, I use the gang database primarily to describe individuals currently identified as gang members and to measure direct gang affiliation after compulsory schooling. To observe the criminal histories of older family members – and to define 'gateway' peers before the gang database was established – I rely on the greater coverage provided by other crime registers.

Two additional BRÅ registers complement the gang database. The first is the *convictions register*, available from 1975 onward, which contains detailed information on offenses, sanctions, and both offense and conviction dates. The second is the *suspects register*, available from 1995 onward, which records suspected offenses along with offense and confirmation dates. Both registers include case numbers, allowing me to identify co-offenders.

In most cases, convictions stem from a prior suspicion entry, but not always: For some offenses, no separate suspicion stage is recorded, and the offense appears only in the conviction register. Using both registers therefore provides the most comprehensive view of criminal activity. Because the timing of each offense is central to the analysis, I merge the two sources and use the offense date as a consistent measure of when the crime occurred. Throughout the paper, I refer simply to "crimes" or "offenses", without always distinguishing between suspicions and convictions. In Sweden, the age of criminal responsibility is 15, which means that offenses committed by younger individuals do not appear in the suspicion or conviction registers. Since my primary outcomes are measured after compulsory school, the individuals I track are typically 16 years or older. However, this means that I cannot observe the criminal behavior that occurs during compulsory school years.

School and Neighborhood Registers. To place students and their peers in concrete institutional and residential contexts, I use annual administrative registers (available from 2008 onward) that link every compulsory school student to a specific school, grade, and residential area *Regional Statistical Area* (RegSO). Sweden is divided into 3,363 RegSOs ranging in size from approximately 650 to 23,000 residents and were designed to study segregation. These zones are independent of school catchment areas: a single school typically draws students from multiple RegSOs, and a given RegSO may send students to several different schools. This many-to-many mapping generates both within-school variation in neighborhood background and within-neighborhood variation in school composition. Because RegSO boundaries are fixed over time, they provide a stable geographic framework to track residential context, measure local socioeconomic conditions, and analyze spatial patterns of gang activity.

Demographic and Socioeconomic Registers. I also draw on the *multigenerational register* of Statistics Sweden, which links each individual to their parents, siblings, aunts, uncles and cousins. These relational links are combined with socioeconomic data on income, education, and employment for both focal individuals and their extended family members. Additional information includes household structure, country of birth, and immigration and emigration dates.

Health Registers. Finally, I link student-level health diagnoses from the National Board of Social Affairs and Health, which cover a wide range of medical conditions, including mental health diagnoses such as depression and anxiety. I use these data to track changes in student psychological well-being during and after compulsory schooling, outcomes that can be affected by increased gang activity or exposure to local violence. Previous research has shown that individuals involved in organized crime or extremist groups in Sweden exhibit elevated rates of mental health diagnoses, though without establishing causal interpretations (Rostami et al. 2018). Other students may also experience psychological stress when exposed to violence in their school or neighborhood settings.

Bringing these data sources together allows me to observe complete criminal and demographic trajectories for cohorts of compulsory school students, while simultaneously capturing the neighborhood context defined by RegSO. Through these linked datasets, I can analyze how both family factors, such as a relative's criminal history, and local conditions, such as residence in disadvantaged or high-violence areas, shape the likelihood of early exposure to peers connected with gangs and subsequent criminal outcomes.

3.2 Sample Construction

The main analysis focuses on students observed in grade 7 (ages 12 to 14) between 2013 and 2019. As noted earlier, the rationale for focusing on grade 7 is two-fold. First, many students transition to a new school at this stage, which introduces a natural reshuffling of peer groups. Second, gang recruitment is frequently reported to begin in early adolescence, making grade 7 – when most students are 13 to 14 years old – a particularly relevant window to observe initial exposure to gateway peers. Finally, if peer sorting or selective attrition poses a concern for identification, restricting the sample to students at the entry point of the upper-compulsory school helps mitigate this issue by holding initial peer group assignments fixed.

To construct the sample, I proceed as follows. (i) I begin with the comprehensive register of compulsory school students and identify the parents, siblings, and extended relatives of each student using the multigenerational register. Students are restricted to those who remain in Sweden (and are alive) throughout grades 7 to 9. (ii) I then merge the annual socioeconomic, demographic, and crime records for each family member, together with the corresponding data for the focal student. These include indicators for sex, foreign background, household structure, and parental characteristics (e.g., income, employment status, education, and age). (iii) I link criminal records with relatives and restrict offenses to those committed between the year the focal student turns six (the start of compulsory schooling) and the year before entry into grade 7. This window excludes earlier or more distant family offenses that are unlikely to shape contemporaneous behavior and avoids overlap with the academic year in which the student attends grade 7. (iv) I track the offenses of each focal student from age 16 to age 21 – starting when students typically complete compulsory schooling and ending at the age when full sentencing is applicable under Swedish law.[10] I also track mental health diagnoses (e.g. depression or anxiety) during and after compulsory school, as well as a set of educational outcomes: standardized test scores in grades 6 and 9, the merit score,[11] and eligibility for upper secondary school (defined as passing grades in Swedish, mathematics, and English). (v) After assigning each focal student their familys criminal history and their own post-school outcomes, I merge these variables back into the main student sample. (vi) I retain only observations for students enrolled in grade 7, ensuring they are between 12 and 14 years old at the start of the academic year – thus accommodating both early starters and grade repeaters. (vii) The resulting dataset consists of repeated cross sections of grade 7 students from 2013 to 2019, covering birth cohorts from 2000 to 2006. The primary analysis sample includes 609,459 students in 2,158 schools, 3,361 neighborhoods, and 22,251 unique school-neighborhood cells.

4 Gang Members, Families, and Students

Before outlining the empirical framework, I: (i) describe the individuals listed in the gang database; (ii) explain how gateway-peer measures are constructed; (iii) present descriptive statistics for the main student sample; and (iv) document intergenerational and intragenerational patterns of criminal transmission within the student population.

4.1 Gang Members

Because the gang database only begins in 2019, it cannot capture historical gang membership. To address this limitation, I use group crimes – offenses involving two or more co-offenders – as my primary proxy for gang-related activity, following Dustmann et al. (2023) and Esbensen and Maxson (2012). In addition, I motivate this definition in Figure A6, which compares the prevalence of different types of suspicion among gang-identified and non-gang individuals. Narcotics use and possession are the most common offenses in both groups, and overall rates for most categories are similar. However, when the analysis is restricted to co-offenses, meaningful differences emerge: gang-affiliated individuals are more likely to be suspected of serious offenses such as murder, robbery, illegal firearms possession,

[10] In January 2022, a reform lowered the age threshold for full sanctioning from 21 to 18, such that individuals aged 18-21 now serve their sentences in full. This change was part of a larger legislative response to the increase in gang crime.

[11] Grades in each subject at the end of grade 9 are converted into point values and summed into a total merit score, which serves as the basis for applications to upper-secondary school and functions similarly to a GPA.

Table 1: Summary Statistics for Individuals in the Gang Database

	N	Mean	Median	Min	Max
A. Demographics					
Year of Birth	18490	1998	2000	1942	2014
Male	18490	0.94	1	0	1
Foreign Born	18490	0.41	0	0	1
Foreign Background	18490	0.79	1	0	1
B. Household Status While a Minor (Ever)					
Single Parent Home	15272	0.64	1	0	1
In Vulnerable Neighborhood	15282	0.49	0	0	1
C. Criminal History (Ever)					
Convicted	18490	0.67	1	0	1
Suspected	18490	0.77	1	0	1
Ever Suspected by Crime Type:					
Group	18490	0.71	1	0	1
Narcotics	18490	0.69	1	0	1
Violent	18490	0.57	1	0	1
Property	18490	0.57	1	0	1
Weapon	18490	0.37	0	0	1
D. Ever Solo or Co-Offended					
Solo Offense	18490	0.44	0	0	1
2+ Co-Offenders	18490	0.72	1	0	1
5+ Co-Offenders	18490	0.54	1	0	1
10+ Co-Offenders	18490	0.37	0	0	1
E. Family Gang Status					
Parents	15282	0.03	0	0	1
Older Siblings	10835	0.23	0	0	1
Uncles or Aunts	5733	0.06	0	0	1
Older Cousins	4309	0.10	0	0	1

Note.—Table reports means, medians, minimum, and maximum values for demographics, criminal history, and family gang status among individuals in the gang database. Observation counts vary across variables due to missing data; for example, 3,208 individuals lack parental identifiers, often reflecting migrants who arrived without their parents. Similar patterns apply to siblings, uncles/aunts, and cousins when those relatives do not exist or are not in Sweden.

and narcotics violations. This divergence highlights why group-crime participation offers a more informative proxy for gang activity than conventional offense classifications.

Therefore, I define group crimes as offenses committed with at least two co-offenders, including murder, robbery, burglary, theft, narcotic offenses, weapons possession (guns, knives, explosives), money laundering, and extortion. This definition serves as the basis for identifying gang-related activity throughout the analysis. Among the individuals listed in the gang database, 71 percent have been suspected of at least one group crime, further supporting the relevance of this proxy for capturing organized and networked criminal behavior.

Table 1 reports descriptive statistics for the sample of known gang-affiliated individuals. Approximately 94 percent are male, and 79 percent have a foreign background—41 percent of whom are foreign-born. Their household environments also reflect multiple forms of disadvantage: 64 percent have at some point lived in a single-parent household, and 49 percent have resided in areas classified by the police as "vulnerable," defined by elevated crime levels and weakened social cohesion. A large majority—67 percent—have been convicted of at least one offense, and 77 percent have been suspected of a crime. Narcotics-related suspicions are especially common (69 percent), followed by violent (57 percent), and property crimes (57 percent), and weapons-related offenses (37 percent). In Table A1, I compare the individuals in the gang database with the broader population of offenders by crime type. Although those in the gang database constitute a relatively small share of all offenders, they represent a disproportionately large share of serious and high-profile criminal activity.

Beyond these individual-level indicators, the co-offending patterns in Table 1 reinforce the characterization of gang crime as a networked phenomenon. While 44 percent have committed at least one offense alone, 72 percent have co-offended with at least one other person, 54 percent have co-offended in a group larger than four, and 37 percent have participated in cases involving ten or more offenders. These figures suggest that group-based criminal activity is a defining feature of this population. Family bonds further underscore the intergenerational dimension of gang participation. Among individuals with linked family records, 3 percent have a parent listed in the gang database, 23 percent have an older sibling, 6 percent have an affiliated uncle or aunt, and 10 percent have an older cousin who is also identified as a gang member. These patterns point to both vertical and lateral channels through which gang affiliation can be transmitted within extended families.

Figure A5 displays a subgraph of the gang network that links confirmed gang members through school, neighborhood, and family ties. The visualization suggests that family links often act as "bridges" connecting otherwise separate groups of individuals. Put differently, family-linked nodes exhibit high betweenness centrality – that is, a large share of the shortest paths between distant parts of the network pass through individuals connected via family. On average, these nodes are about seven times more likely to lie on a shortest path between two others than nodes without family ties (mean betweenness centrality: family = 36,125; non-family = 5,184).[12] In practical terms, this suggests that family members may not only transmit criminal norms across generations, but also serve as conduits that facilitate contact and information flow between otherwise disconnected parts of the gang network, potentially offering peers without direct connections new pathways into gang-related activity.

To conclude, these descriptive patterns show that individuals in the gang database tend to originate from disadvantaged backgrounds, engage in extensive criminal activity (often in groups), and are embedded in dense family and social networks that shape both direct involvement and indirect exposure to organized crime.

[12]Non-standardized betweenness centrality is defined as the number of distinct shortest paths between all ordered pairs of nodes that pass through the focal node.

Table 2: Variation in Exposure to Peers with Criminally Involved Older Siblings

	Mean	Std. Dev	Min	Max
Raw Gateway Peer Exposure, Defined by Older Siblings' Criminal History				
School-Grade Peers (Group Crimes)	0.048	0.036	0.000	1.000
School-Grade Peers (All Crimes)	0.097	0.054	0.000	1.000
Neighborhood-Grade Peers (Group Crimes)	0.047	0.049	0.000	1.000
Neighborhood-Grade Peers (All Crimes)	0.095	0.071	0.000	1.000
Cohort Variation over Time in Gateway Peer Exposure, Defined by Older Siblings' Criminal History				
School-Grade Peers (Group Crimes)	0.000	0.023	-0.338	0.816
School-Grade Peers (All Crimes)	0.000	0.032	-0.343	0.781
Neighborhood-Grade Peers (Group Crimes)	0.000	0.033	-0.254	0.911
Neighborhood-Grade Peers (All Crimes)	0.000	0.047	-0.321	0.896

Note.—The top panel reports unconditional variation in older sibling exposure across school-grade and neighborhood-grade peer groups, based on both group crimes and all crimes. The bottom panel reports residual variation after conditioning on calendar-year, birth-year, and school-neighborhood fixed effects.

4.2 Constructing the Gateway Peer Exposure Measure

I refer to *gateway peers* as students whose older relatives – parents, older siblings, uncles, aunts, or older cousins – have a documented criminal history. Specifically, I track each type of older relative separately, creating different indicators to determine whether a student has a relative category with a history of *any* criminal offense or a *group* crime, as recorded in suspect or conviction records. By focusing exclusively on older family members, I minimize the risk that the focal student influences the relatives criminal behavior, reinforcing the directional interpretation of exposure.

Once the gateway peer indicator is defined for each peer j in the peer group p in year t, denoted D_{jpt}, the focal students exposure is calculated as a leave-one-out mean:

$$\text{Exposure}_{ipt} = \frac{\sum_{j \neq i} D_{jpt}}{N_{pt} - 1}, \tag{1}$$

where N_{pt} is the total number of students in the peer group p in year t, and $D_{jpt} = 1$ if peer j has at least one older relative (of the specified type) with a relevant criminal history. Peer groups p are defined at the school-grade, neighborhood-grade, or school-neighborhood-grade level. Exposure_{ipt} therefore represents the fraction of peers who serve as "gateways" through the participation of an older relative in criminal activity, thus capturing the extent to which a given student is surrounded by potential channels to gang-related or delinquent behaviors.

Table 2 presents summary statistics for gateway peer exposure defined by group or any crime of older siblings, focusing on school-grade and neighborhood-grade peer groups. Although I construct measures for other family members as well,[13] These descriptives focus on exposure through older siblings, as it forms the primary treatment in the main analysis. In the top panel, the unconditional mean exposure to group crimes is around 4.8 percent in the school-grade setting and 4.7 percent in the neighborhood-grade setting. When considering all crimes, these averages rise to approximately 9.7 percent (school-grade) and 9.5 percent (neighborhood-grade).

The bottom panel displays the remaining variation after controlling for calendar-year, birth-year, and school-neighborhood fixed effects, and indicates that meaningful variation in criminal involvement of older siblings remains.

4.3 Descriptive Statistics for the Student Sample

Table 3 presents summary statistics for four groups of students: (1) the full sample (All Students), (2) those without a criminally involved older sibling (Non-Gateway Students), (3) those with a criminally involved older sibling (Gateway Students), and (4) those with an older sibling listed in the gang database (Gang Gateway Students). The main analysis focuses on the students in column 2, those without a criminally involved older sibling.

Comparisons among these groups reveal several notable patterns. Overall, 13 percent of students in the full sample commit at least one offense between ages 16 and 21 (column 1), rising to 25 percent among those with a criminally involved older sibling (column 3), and reaching 43 percent among those whose sibling appears in the gang database (column 4). The share suspected or convicted of group crimes increases from 5 percent among students without a criminal sibling to 14 percent for those with one, and to 31 percent for those with a gang-affiliated sibling. Violent, property, narcotics, and weapons-related suspicions follow a similar gradient. Educational outcomes deteriorate accordingly: eligibility for upper-secondary school falls from about 90 percent among students with no criminal sibling to 78 percent and 65 percent in the two exposed groups, while standardized test scores in grade 9 decline by approximately 0.4 and 0.7 standard deviations, respectively. Merit scores show a similar pattern, dropping from an average of 228 among non-exposed students to 191 for those with a criminal sibling, and to 166 for those with a gang-affiliated sibling.

The background characteristics also vary systematically between groups. Foreign background is more prevalent among students with a criminally involved older sibling (31 percent), and especially so among those with a sibling in the gang database (77 percent), compared to 18 percent in the non-exposed group. Students in the latter two categories are also more likely to have grown up in single-parent households, to live in more disadvantaged neighborhoods, and to have parents with lower average educational attainment and employment rates.

[13] Table A3 presents the raw and conditional variation in exposure to gateway peers by parents, uncles, aunts, and cousins.

Table 3: Descriptive Statistics for the Grade 7 Student Sample

	All Students (1)	Non-Gateway Students (2)	Gateway Students (3)	Gang-Gateway Students (4)
A. Outcomes: Post or End of Compulsory School				
Crime Activity (Age 16-21)				
Any	0.13	0.12	0.25	0.43
Group	0.06	0.05	0.14	0.31
Violent	0.04	0.03	0.09	0.23
Property	0.05	0.05	0.11	0.22
Narcotics	0.05	0.04	0.12	0.26
Weapon	0.01	0.01	0.03	0.10
Other	0.05	0.05	0.09	0.15
Ever in Gang Database	0.01	0.01	0.03	0.20
Eligiblity to High School	0.89	0.90	0.78	0.65
Merit Score	223.95	227.52	190.74	165.57
Standardized Test (Grade 9)	0.02	0.06	-0.35	-0.73
Mental Health Diagnose (Age 16-21)				
Mood	0.05	0.05	0.07	0.03
Stress	0.08	0.08	0.11	0.06
B. Background				
Male	0.51	0.51	0.51	0.52
Foreign Background	0.20	0.18	0.31	0.77
Swedish as Second Language	0.12	0.11	0.20	0.58
Single Parent Home	0.25	0.23	0.39	0.41
Standardized Test (Grade 6)	0.01	0.04	-0.34	-0.74
Father Characteristics				
Age	46.33	45.98	49.54	48.65
Above Median Labor Income	0.51	0.53	0.33	0.19
Employed, yr t-1	0.87	0.88	0.74	0.53
Yrs. of Schooling	12.21	12.35	10.92	9.94
Dead	0.01	0.01	0.02	0.03
Missing	0.03	0.03	0.02	0.06
Mother Characteristics				
Age	43.32	43.11	45.26	43.25
Above Median Labor Income	0.50	0.52	0.35	0.18
Employed, yr t-1	0.84	0.85	0.72	0.49
Yrs. of Schooling	12.82	12.97	11.42	9.66
Dead	0.00	0.00	0.01	0.01
Missing	0.00	0.00	0.00	0.01
C. Peer Groups				
Peers in school-grade	95.13	95.63	90.52	89.40
Peers in neighborhood-grade	49.33	49.51	47.64	55.08
Peers in neighborhood-school-grade	22.62	22.81	20.90	20.47
Peers with Criminally Involved Older Siblings (share)				
School-grade	0.10	0.10	0.12	0.14
Neighborhood-grade	0.10	0.10	0.12	0.16
School-neighborhood-grade	0.10	0.09	0.13	0.17
Peers with Criminally Involved Older Siblings in Group Crimes (share)				
School-grade	0.05	0.05	0.06	0.08
Neighborhood-grade	0.05	0.05	0.06	0.09
School-neighborhood-grade	0.05	0.05	0.07	0.10
Observations	675,194	609,459	65,735	5,368

Note.—Table reports means for each variable across four groups: (1) all students, (2) students without a criminally involved older sibling, (3) students with a criminally involved older sibling, and (4) students with an older sibling listed in the gang database.

Mental health diagnoses follow a milder pattern: rates of mood-related diagnoses increase modestly from 5 percent to 7 percent, while stress related diagnoses peaks at 11 percent among students with a criminally involved sibling.

Table 3 also highlights the variation in the proportion of peers, whether of the same school level, neighborhood level, or both, who have older siblings involved in crime. Among students without a criminally involved older sibling, approximately 10 percent of their peers fall into this category. For students with a criminally involved sibling, between 12 and 16 percent of their peers come from similarly exposed families. This pattern reflects the geographic and institutional clustering of criminal exposure and suggests that family and peer risk overlap often in high-risk environments.

Table A2 reports summary statistics for an additional student sample that further excludes gateway peers defined through the history of group crime of their parents, uncles, aunts, or cousins. This is the sample used when jointly estimating the impact of gateway peers across the different types of relative links to group crime.

4.4 Criminal Family Transmission

Previous work on criminal peer effects in schools and the intergenerational transmission of anti-human capital has largely focused on the role of parents and siblings (Besemer and Farrington 2012; Finlay et al. 2023; Hjalmarsson and Lindquist 2012; Junger et al. 2013). As a first step, I use rich multigenerational family links and detailed criminal records to measure the exposure of each focal student not only to parents and siblings, but also to uncles, aunts, and older cousins with documented criminal involvement. This serves both as a descriptive exercise on how multiple family ties jointly contribute to the intergenerational transmission of criminal behavior, and as an informal first-stage for the main analysis that focuses on school and neighborhood peer effects.

To examine how criminal histories within the extended family relate to subsequent offending of a student, I estimate the following model:

$$y_i = b_0 + \sum_k b_k c_{ik} + e_i \tag{2}$$

where y_i denotes the student i's own criminal behavior between ages 16 and 21, and c_{ik} is an indicator equal to one if the student has at least one relative of type k – parent, sibling, uncle, aunt, or cousin – with any group or criminal history and zero otherwise. The coefficients b_k capture the association between the criminal history of each relative type and the subsequent offending of the student, while e_i is an idiosyncratic error term. Since not all students are related to all types of relative, equation (2) also includes missing dummies for each type of relative. These take the value 1 if the focal individual is not linked to a relative of that type, and 0 otherwise.

Table 4: Family Correlations in Criminal Outcomes by Relative Type (Group Crime History)

Dependent Variables:	Any Crime (Age 16-21)				Group Crime (Age 16-21)				Ever in Gang Database			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Family Group Crime History Indicators (0/1)												
Parent	0.193**	0.176**	0.161**	0.158**	0.139**	0.126**	0.115**	0.113**	0.030**	0.026**	0.024**	0.023**
	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)
Sibling		0.122**	0.114**	0.112**		0.091**	0.085**	0.084**		0.025**	0.023**	0.023**
		(0.003)	(0.003)	(0.003)		(0.002)	(0.002)	(0.002)		(0.001)	(0.001)	(0.001)
Uncle/Aunt			0.077**	0.072**			0.051**	0.048**			0.009**	0.009**
			(0.002)	(0.002)			(0.001)	(0.001)			(0.001)	(0.001)
Cousin				0.040**				0.024**				0.002**
				(0.001)				(0.001)				(0.000)
Sum of Coefficients	0.193	0.299	0.335	0.356	0.139	0.217	0.240	0.254	0.030	0.052	0.055	0.057
	(0.003)	(0.003)	(0.004)	(0.004)	(0.002)	(0.003)	(0.003)	(0.003)	(0.001)	(0.002)	(0.002)	(0.002)
Adj. R^2	0.018	0.026	0.031	0.033	0.018	0.026	0.032	0.033	0.006	0.010	0.016	0.017
Observations	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194
Dep. mean	0.133	0.133	0.133	0.133	0.062	0.062	0.062	0.062	0.009	0.009	0.009	0.009

Note.—The table reports raw correlations between indicators of the involvement of relative group crime prior to the entry of the focal student into grade 7 and the students own criminal activity between ages 16 and 21. Estimates are based on variants of equation (2). Columns 1, 5, and 9 present baseline regressions including only parental criminal history; subsequent columns sequentially add indicators for siblings, uncles, aunts, and cousins. As each relative type is introduced, the specification also includes a dummy indicating whether that relative type is missing for the focal student. The *Sum of Coefficients* is constructed following the Lubotsky-Wittenberg method for proxy variables (Lubotsky and Wittenberg 2006). Robust standard errors are reported in parentheses; standard errors for the sum of coefficients are bootstrapped with 1,000 replications.
(**) $p < .01$, (*) $p < .05$, (+) $p < .1$.

Table 4 presents the estimates from variations of equation (2), showing how exposure to older relatives with criminal records relates to the gateway peer's own offending between ages 16-21, as well as to the probability of ever appearing in the gang database. Each panel of the table reports results for progressively richer specifications: columns 4, 8, and 12 display the fully saturated models, which include indicators for all four kin types – parents, older siblings, uncles or aunts, and older cousins – simultaneously. Relatives' criminal history is measured between the years in which the focal student starts compulsory school and until the year they start grade 6.

Table 4 focuses on relatives with a history of group crime. In the fully saturated specification, having a parent with a group crime record is associated with a 15.8 percentage point increase in the probability of committing any crime (approximately 119 percent relative to the baseline mean), a 11.3 point increase in group-crime involvement (182 percent), and a 2.3 point increase in the probability of appearing in the gang database (255 percent). Older siblings show similarly strong associations, with coefficients of 11.2, 8.4, and 2.3 points on the same outcomes, respectively, underscoring the strength of within-household transmission. The coefficients for uncles and aunts are roughly one-third as large, around 5 to 7 percentage points for the two offending measures, and about one point for gang-database inclusion. Cousin effects are smaller and go from 4 points for any crime to 0.2 points for being listed in the gang database. Table A4 repeats the analysis considering all types of crime. In these specifications, having a parent with a criminal history increases the probabilities of any crime, group crime, and gang-database appearance by 10.7, 7.0, and 1.4 points, respectively – approximately one-third smaller than the corresponding estimates in Table 4, but still representing increases of 80 percent, 112 percent, and 155 percent over the relevant baseline means. The effects of the siblings remain substantial at 8.9, 6.3, and 1.5 points, respectively. The effects of uncles and aunts shrink to 4.7-0.4 points, and cousin coefficients remain small and shrink slightly. These smaller coefficients for uncles, aunts, and cousins may reflect a more limited day-to-day interaction with the focal student or weaker shared nature and nurture factors.

Following Adermon et al. (2021), each indicator in Table 4 denotes whether the focal student has at least one older parent, sibling, uncle/aunt, or cousin with a criminal record, can be viewed as a noisy proxy for a latent 'crime family' effect: an unobserved influence transmitted through the extended family that shapes the students own criminal behavior. Summing the associated coefficients provides a lower-bound estimate of this aggregate influence. However, because the indicators differ in their variance, I use Lubotsky-Wittenberg proxies (Lubotsky and Wittenberg 2006) to construct a weighted index. This procedure yields a composite measure whose coefficient captures the effect of increased exposure to the underlying crime-family environment, rather than to any specific kinship link. Although the composite measure is not straightforward to interpret, I use it for relative comparison when including additional relative information.

Leaving siblings out leads to an understatement of this effect. Expanding the specification from only parental exposure to include both parents and siblings raises the summed coefficient from 0.193 to 0.299 for any crime (a 55 percent increase), from 0.139 to 0.217 for group crime (about 56 percent), and from 0.03 to 0.05 for gang-database inclusion (nearly 73 percent). Including uncles, aunts, and cousins further increases the sums, though by a smaller amount, suggesting that parents and older siblings account for most of the extended-family "criminal capital," while more distant relatives play a more limited role. The relationship follows the same pattern at a lower magnitude. Table A5 reports correlations separately estimated for each relative type.

The adjusted R^2 values in Table 4 are modest – ranging from about 0.5 to 3 percent – which is lower than what is typically observed in studies of human capital or income. Two factors likely contribute to this. First, crime is a relatively rare outcome that inherently limits predictive power. Second, many important determinants, such as peer influence, neighborhood context, or early contact with the justice system, are not captured in this "extended-family model".

These findings support the view that older relatives – particularly those involved in serious or group-based criminal activity – play an important role in shaping the criminal trajectories of younger family members. These effects likely operate through multiple channels, including the normalization of criminal behavior, the transmission of "criminal-capital", and direct recruitment into gang networks. Extending this logic to the peer context, one might expect that students with criminally involved relatives could, in turn, influence the behavior of their schoolmates or neighbors by introducing them to similar networks or norms.

5 Method

This section outlines the cohort-to-cohort research design, discusses key identification assumptions, and addresses potential threats to identification. The section concludes with a set of tests that assess the plausibility of these assumptions.

5.1 Empirical Strategy

Identification. Estimating spillover effects in schools and neighborhoods presents three main challenges: (i) reflection, (ii) common shocks, and (iii) selection.

Reflection (Manski 1993; Moffit 2001) arises when the behavior of a focal student and their peers mutually influence each other, complicating the causal interpretation. However, I define treatment based on the historical criminal behavior of the older relatives of the peers and argue that the direction of influence is transmitted from the older relative to the gateway peer and subsequently to the focal student (or, in some cases, directly from the older relative to the focal student). Since the criminal behavior used to define treatment predates grade 7, it is unlikely that the focal student could influence a peers older relative, limiting concerns related to the reflection problem. This age-based direction of influence is consistent with previous work showing that criminal or disruptive influences flow from older to younger individuals (Altonji et al. 2017; Eriksson et al. 2016).

Selection poses a further challenge, as families can sort into schools or neighborhoods based on socioeconomic factors that are correlated with the composition of the peer group (Holmlund et al. 2020). To address this, I adopt a cohort-to-cohort design following Hoxby (2000), leveraging variation in peer composition across adjacent cohorts within the same school-neighborhood cells. By comparing students in grade 7 within these fixed cells, I effectively differentiate out time-invariant local characteristics such as school quality, neighborhood socioeconomic conditions, long-term crime exposure, and stable characteristics of the built environment. This design has previously been applied to study criminogenic peer effects in schools, for example, by Billings and Hoekstra (2023) and Carrell et al. (2018).

Common shocks may simultaneously affect the composition of the peer group and future criminal outcomes, potentially confounding the estimated effects of exposure. To address this, I include year fixed effects to control for nationwide shocks affecting entire school- and neighborhood-cohorts. In addition, because some students are "out of sync" with their birth cohort (e.g. due to repetition of grades), I also include fixed effects for the birth year. Although these two fixed effects may overlap, there is sufficient independent variation to justify their joint inclusion. However, localized shocks that vary over time within schools or neighborhoods – such as changes in teaching staff, short-term school-based programs, or targeted policing initiatives – remain a potential concern. I explicitly assess the plausibility of quasi-random cohort-to-cohort variation in Section 5.2.

Figure 1 provides a visual illustration of the intuition behind the empirical design. The goal is to compare students who attend the same school, live in the same neighborhood, and are observed in the same grade (grade 7), but who differ in their exposure to gateway peers due to cohort-to-cohort variation. In the figure, students in different cohorts pass through the same school-neighborhood cell, but only some cohorts include a gateway peer – defined here as a student with a criminally involved older sibling (shown in green), with the gateway peer themselves highlighted in pink.

Main Regression Model. Following the discussion above, the main model I estimate is specified as:

$$Y_{i,s,n,t+k} = \beta \cdot Exposure_{i,s,n,t} + \phi_t + \lambda_{s,n} + \gamma_{yob} + \varepsilon_{i,s,n,t} \tag{3}$$

where $Y_{i,s,n,t+k}$ denotes the outcome of interest for the student i – typically a post-compulsory school criminal outcome – measured between ages 16 and 21, that is, k years after the student attends grade 7 in year t, at school s and resides in neighborhood n. The variable $Exposure_{i,s,n,t}$ captures the share of gateway peers in the focal students peer group, defined

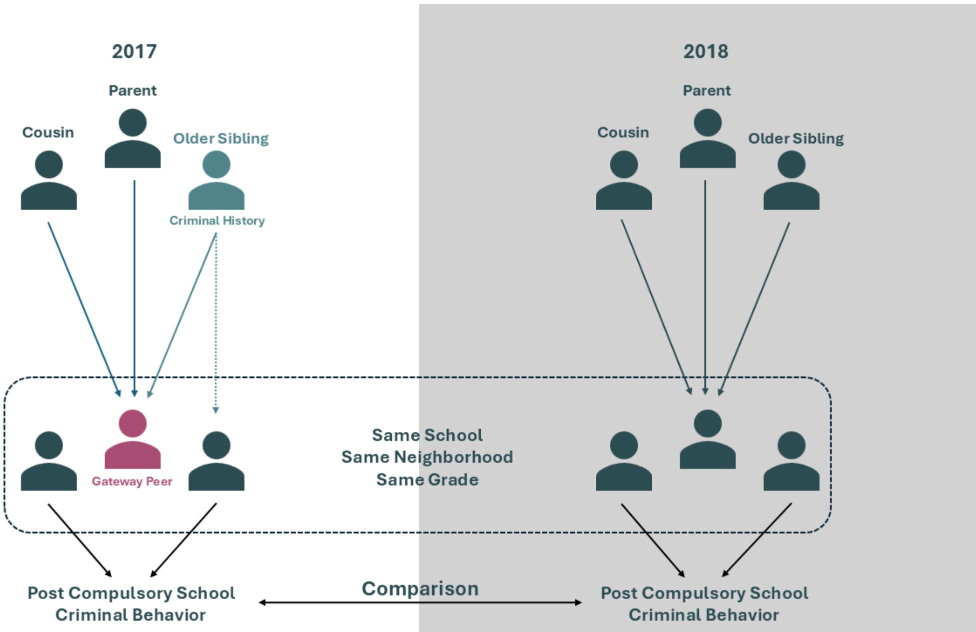


Figure 1: Visual Overview of Cohort-to-Cohort Identification Strategy

Note.—This figure illustrates the comparison estimated in (3). The gateway peer (pink) is defined as a student with a criminally involved older brother (green). The estimate captures the difference in post-compulsory outcomes across cohorts in grade 7 with varying exposure to gateway peers, within the same school-neighborhood cell.

at the school-grade level (students in the same school and grade) or at the neighborhood-grade level (students living in the same neighborhood and enrolled in grade 7, possibly at different schools). The regressions includes calendar-year fixed effects (ϕ_t) to absorb national grade-cohort shocks, school-neighborhood fixed effects ($\lambda_{s,n}$) to control for time-invariant local conditions, and birth-year fixed effects (γ_{yob}) to account for earlier starters and grade-repeaters. Standard errors are two-way clustered at the school and neighborhood level.

Following Angrist (2014) and related applied studies (Billings and Hoekstra 2023; Kim and Fletcher 2017; Larsen and Kristensen 2017), I exclude students who themselves qualify as gateway peers to avoid mechanically linking a students own characteristics to the treatment variable. That is, I estimate effects only for students without a criminally involved older sibling, ensuring that exposure captures variation in peer composition rather than individual status. To maintain consistency across different treatment definitions, students with an older sibling with any type of criminal record are also excluded. Furthermore, some specifications include exposure to gateway peers defined by their parents', uncles', aunts', or cousins' history of group crime; these types of gateway peers are also excluded from these specifications. I estimate versions of equation (3) that include controls for focal student characteristics (e.g., sex, foreign background, parental income, education, and employment) and peer group composition (e.g., the share of foreign background peers). These controls improve precision, but also serve as robustness checks. In extended versions of equation

(3), I include multiple exposure measures simultaneously – distinguishing exposure by relative type, by context (school vs. neighborhood), and by sibling characteristics such as age proximity and residence location.

5.2 Assessing Identifying Assumptions

The identification framework discussed in Section 5.1 is based on assumptions that can be partially evaluated using observable characteristics and tests of within-school and within-neighborhood variation. The assessment in the following section focuses on exposure to older siblings, as these indicators constitute the primary treatment of interest in the analysis.

Exogeneity. If exposure to gateway peers is quasi-random within school-neighborhood units, it should not systematically correlate with focal student characteristics. Although not all traits are observable, exposure can be assessed to be conditionally independent of observable student and family characteristics, providing a partial test of the exogeneity assumption.

Table 5 presents regressions of gateway exposure, based on the older siblings group and any criminal history, in schools and neighborhoods, on a set of student and family characteristics. Columns 1, 2, 5, and 6 report unconditional correlations, showing that traits such as foreign background, single parent household, and higher parental education or employment are significantly associated with exposure. However, when controlling for time-fixed effects (calendar year and year of birth) and restricting comparisons to students within the same school-neighborhood cell (columns 3, 4, 7, and 8), these associations largely disappear – becoming statistically insignificant both individually and jointly, and negligible in magnitude. These results support the assumption that the cohort-to-cohort specification mitigates concerns about selection. Table A6, Table A7, Table A8, and Table A9 present additional tests of the exogeneity of exposure to gateway peers, defined separately by siblings, parents, uncles, and cousins with a history of group crime. All tests use the same student sample, from which all four types of gateway peers are excluded.

As-random School and Neighborhood Group Formation. While school-neighborhood fixed effects account for time-invariant differences between locations, they do not rule out the possibility of endogenous changes in the composition of peer groups over time. For identification to be valid, the variation in gateway peer exposure between cohorts within a school or neighborhood must be as if random, that is, cohort formation should follow a random assignment process. To assess this assumption, I perform a cohort-shuffling exercise following Billings and Hoekstra (2023) and Carrell and West (2010), with related approaches discussed in Ammermueller and Pischke (2009). The intuition behind this test is the following. If variation in exposure across cohorts is quasi-random, then the likelihood that a student is exposed to more or fewer gateway peers should not systematically depend on the specific cohort they belong to.

Table 5: Testing Exogeneity of Gateway Peer Exposure Defined by Older Criminal Sibling

Sibling Exposure Considering:	Group Crimes				All Crimes			
In Peer Group:	School		Neighborhood		School		Neighborhood	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Male	0.000	-0.000	-0.000*	-0.000	0.000	-0.000	-0.001**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Foreign Background	0.015**	0.000*	0.024**	-0.000	0.015**	0.000+	0.029**	0.000
	(0.001)	(0.000)	(0.002)	(0.000)	(0.002)	(0.000)	(0.002)	(0.000)
Single Parent Home	0.003**	0.000	0.006**	0.000	0.003**	0.000	0.008**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Fathers Age	-0.000*	0.000	0.000	0.000	-0.000**	-0.000	-0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Fathers Income $\geq$ p50	-0.005**	0.000	-0.007**	0.000	-0.011**	-0.000	-0.014**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Father Employed	0.001**	-0.000	0.001	0.000	0.003**	-0.000	0.002**	0.000
	(0.000)	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)
Fathers Yrs. of Sch.	-0.000**	-0.000	-0.000**	-0.000	-0.001**	-0.000	-0.001**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Father Missing	-0.009**	-0.000	-0.010**	0.000	-0.014**	-0.000	-0.016**	-0.000
	(0.001)	(0.000)	(0.001)	(0.001)	(0.002)	(0.001)	(0.002)	(0.001)
Mothers Age	-0.000**	-0.000	-0.000**	-0.000	-0.001**	0.000	-0.001**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Mothers Income $\geq$ p50	-0.004**	0.000	-0.006**	0.000	-0.010**	0.000	-0.012**	0.000+
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Mother Employed	-0.001+	-0.000	-0.002**	-0.000	-0.000	-0.000	-0.003**	-0.000
	(0.000)	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)
Mothers Yrs. of Sch.	-0.001**	-0.000	-0.001**	0.000	-0.001**	-0.000	-0.001**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Mother Missing	-0.013**	0.000	-0.020**	-0.001	-0.027**	0.000	-0.036**	0.001
	(0.002)	(0.001)	(0.002)	(0.001)	(0.002)	(0.001)	(0.003)	(0.001)
Year Fixed		✓		✓		✓		✓
Year-of-Birth Fixed		✓		✓		✓		✓
School-Neighborhood Fixed		✓		✓		✓		✓
F-Stat	43.236	1.160	63.214	0.618	74.687	1.123	104.995	0.517
p(F)	0.000	0.300	0.000	0.852	0.000	0.332	0.000	0.925
Adj. R^2	0.075	0.585	0.092	0.507	0.087	0.629	0.104	0.547
Dependent Mean	0.048	0.048	0.047	0.047	0.097	0.097	0.095	0.095
Observations	609,459	609,459	609,459	609,459	609,459	609,459	609,459	609,459

Note.—Each column reports a separate regression of the listed dependent variable on a common set of control variables, estimated both with and without calendar-year, birth-year, and school-neighborhood fixed effects. The reported F-stat is the joint F-statistic for the included controls, and p(F) gives the corresponding p-value. Columns 1-4 use exposure to gateway peers with older siblings who have a history of group crime as the dependent variable. Columns 1 and 2 measure exposure in the school-grade group, while columns 3 and 4 use neighborhood-grade exposure. Columns 5-8 use exposure to gateway peers with older siblings who have any type of criminal history as the dependent variable. Columns 5 and 6 report school-grade exposure and columns 7 and 8 neighborhood-grade exposure. Standard errors are two-way clustered at the school and neighborhood levels. (**) $p < .01$, (*) $p < .05$, (+) $p < .1$.

The procedure for testing this assumption proceeds as follows: (i) For each observed cohort, I randomly draw 1,000 alternative cohorts of the same size from the pool of all students who ever attended a given school or resided in a given neighborhood in grade 7. (ii) For each simulated cohort, the share of gateway peers is calculated, defined as peers with older siblings involved in group crime. (iii) Then, a p-value is defined as the fraction of simulated cohorts whose gateway peer share is lower than that of the actual cohort.

If exposure varies as if randomly within schools and neighborhoods, the distribution of these p-values should be approximately uniform. To formally assess this, I perform both Kolmogorov-Smirnov and χ^2 uniformity tests. The results, reported in Table A10, focus on the primary treatment variable – exposure to sibling-based gateway peers. As placebo checks, I also include peer characteristics such as the share of male students and the share of students with a foreign background. Although these are not treatments of interest, systematic cohort-level variation in these characteristics would suggest underlying non-random sorting mechanisms, even if the treatment variable itself appears balanced.

At the 5% significance level, the rejection rates in both tests are generally low. Specifically, among 1,892 schools, uniformity is rejected in only 10-19 cases based on the Kolmogorov-Smirnov test and in 51-83 cases based on the χ^2 test, depending on the characteristic examined. Among 3,129 neighborhoods, the rejection counts range from 6-20 for the Kolmogorov-Smirnov test and 62-137 for the χ^2 test. Given the large number of tests conducted, some rejections are expected purely by chance. These results support the assumption that the variation in cohort peer composition is consistent with random assignment. The absence of strong or systematic deviations across all three characteristics – including the main treatment variable – reinforces the view that cohort-to-cohort changes are plausibly quasi-random, and that local shocks at the school or neighborhood level are unlikely to drive the composition of peer groups.

Threats to Identification. Sjögren et al. (2024) show that exposure to immigrant peers can induce some students to switch schools. In this setting, a related concern is that students from certain socioeconomic backgrounds – who may be at higher or lower risk of future criminal behavior – could relocate in response to gateway peer exposure, potentially biasing the estimates. For example, if students for whom exposure would be most "damaging" are more likely to leave, the effects may be understated; conversely, if students already at higher risk of criminality are more likely to remain in high-exposure cohorts, the effects could be overstated. To assess this possibility, I test whether exposure to gateway peers in grade 6 predicts switching schools or neighborhoods in grade 7. Table A11 presents results from regressions of prior exposure on subsequent moves. I find no evidence that gateway peer exposure in grade 6 predicts school switching in grade 7 – both when considering all students and when restricting to those who had the option to stay at their current school. Similarly, I do not find any indication that students move to different neighborhoods in response to exposure at the neighborhood level. These results hold whether exposure is defined by group crimes or by any type of crime.

Given these exercises, I conclude that the findings suggest that selective mobility is unlikely to bias the estimates and that the identifying assumptions underlying the empirical strategy remain credible.

6 Results

The analysis begins by estimating the reduced-form effects of exposure to "gateway" peers – students with older relatives involved in crime, including parents, uncles or aunts, cousins, and siblings – within schools and neighborhoods. Among these, students with older criminally involved siblings stand out as particularly influential in shaping peer outcomes. The remainder of the analysis therefore focuses on sibling-based exposure, investigating whether the observed effects reflect only disruptive peer influences or also indicate a systematic gang recruitment mechanism. I also examine whether exposure to gateway peers has consequences for students educational achievement and mental health.

6.1 Effect of Gateway Exposure in Schools and Neighborhoods

The first step of the analysis examines whether the strong within-family correlations extend to peer spillovers at the school and neighborhood levels. I also assess the robustness of sibling-based gateway peer effects to alternative controls and sample restrictions, and explore heterogeneity by focusing on urban areas with documented gang presence and by testing for nonlinearities across student characteristics.

6.1.1 Baseline Results

Table 6 reports estimates from equation (3) linking exposure in grade 7 to gateway peers – students whose parents, older siblings, uncles / aunts or cousins have a history of group crime – to the focal students criminal activity between ages 16 and 21. Columns 1-3 use school-grade exposure; columns 4-6 use neighborhood-grade exposure. In all columns, (3) is extended to include four exposure variables simultaneously – one for each relative type. Each treatment variable is defined as a fraction between 0 and 1, representing the share of students peers with a given type of criminally involved relative. Thus, the coefficients can be interpreted as the effect of moving from zero to full exposure. Such a shift in exposure is extreme; therefore, the estimates are interpreted in terms of a 5-percentage point (pp.) increase in exposure. In a typical peer group, this corresponds to roughly five additional peers in the school-grade setting and two to three in the neighborhood-grade setting. A 5-pp. increase also approximates one standard deviation in exposure, although the exact value varies across definitions. I apply this interpretation consistently throughout the remainder

Table 6: Effects of Exposure to Gateway Peers by Type of Older Criminal Relative

Peer Group:	School-Grade			Neighborhood-Grade		
	(1) Any Crime (Age 16-21)	(2) Group Crime (Age 16-21)	(3) Ever In Gang	(4) Any Crime (Age 16-21)	(5) Group Crime (Age 16-21)	(6) Ever In Gang
Gateway Exposure by Family Type, Defined by History of Group Crime.						
Parent	0.002	-0.003	0.003	0.019	0.010	0.005
	(0.023)	(0.017)	(0.007)	(0.015)	(0.011)	(0.004)
Sibling	0.075**	0.058**	0.006	0.019	0.017	0.003
	(0.023)	(0.018)	(0.007)	(0.015)	(0.010)	(0.004)
Uncle/Aunt	0.015	-0.005	-0.002	-0.001	0.004	-0.002
	(0.016)	(0.011)	(0.004)	(0.011)	(0.007)	(0.002)
Cousin	0.004	-0.003	0.002	0.003	-0.003	-0.003
	(0.015)	(0.010)	(0.003)	(0.010)	(0.007)	(0.002)
Adj. R^2	0.028	0.031	0.038	0.028	0.031	0.038
Observations	499,639	499,639	499,639	499,639	499,639	499,639
Dep. Mean	0.108	0.046	0.006	0.108	0.046	0.006

Note.—Each column reports estimates from a separate regression based on equation (3), where exposure to gateway peers is defined based on the group crime history of older relatives. Columns 1-3 use school-grade exposure, and columns 4-6 use neighborhood-grade exposure. Equation (3) is here extended to include one exposure variable for each relative type simultaneously—that is, exposure to gateway peers defined by: (i) parents with a history of group crime, (ii) older siblings with group crime history, (iii) uncles and aunts with group crime history, and (iv) older cousins with group crime history. All regressions include birth-year, calendar-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. Standard errors are two-way clustered at the school and neighborhood levels.
(**) $p < .01$, (*) $p < .05$, (+) $p < .1$

of the paper. Furthermore, these specifications include exposure from four types of gateway peers, who themselves are excluded from the estimation sample in Table 6.

While within-family correlations in criminal behavior suggest the relevance of the criminal histories of parents and siblings, these peer effect estimates suggest that older siblings play the primary role in driving influencing to others through gateway peers. Turning to the school-grade estimates in Table 6 (columns 1-3), a 5-pp. increase in the share of peers with older siblings involved in group crime raises the probability of committing any crime between ages 16 and 21 by roughly 3.5 percent relative to the sample mean (column 1). The likelihood of participating in group crime increases by approximately 6.3 percent (column 2), and both effects are statistically significant at the 1% level. The corresponding estimates for neighborhood-grade exposure (columns 4-6) are smaller, suggesting increases of 0.9-1.8 percent for any crime and group crime, respectively, although neither is statistically significant at conventional levels. Likewise, while the point estimates for the appearance in the gang database are positive, they remain statistically insignificant. Table A12 reports estimates that account for focal and peer group characteristics, indicating that the results remain robust to their inclusion.

Table 7: Gateway Peer Exposure by Sibling Group Crime Involvement

	Any Crime (Age 16-21)			Group Crime (Age 16-21)			Ever in Gang		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A. Gateway Exposure by Siblings, Defined by History of Group Crimes									
School Exposure	0.073**	0.069**	0.069**	0.054**	0.052**	0.052**	0.008	0.007	0.008
	(0.021)	(0.021)	(0.021)	(0.017)	(0.017)	(0.017)	(0.007)	(0.007)	(0.007)
Panel B. Gateway Exposure by Siblings, Defined by History of Group Crimes									
Neigh. Exposure	0.005	0.007	0.007	0.013	0.014	0.014	0.004	0.005	0.005
	(0.014)	(0.013)	(0.013)	(0.010)	(0.010)	(0.010)	(0.004)	(0.004)	(0.004)
Panel C. Gateway Exposure by Siblings, Defined by History of All Crimes									
School Exposure	0.047**	0.046**	0.046**	0.041**	0.039**	0.040**	0.006	0.006	0.006
	(0.015)	(0.015)	(0.015)	(0.011)	(0.011)	(0.011)	(0.004)	(0.004)	(0.004)
Panel D. Gateway Exposure by Siblings, Defined by History of All Crimes									
Neigh. Exposure	0.009	0.011	0.011	0.014*	0.015*	0.015*	0.002	0.002	0.002
	(0.010)	(0.009)	(0.009)	(0.007)	(0.007)	(0.007)	(0.003)	(0.003)	(0.003)
Focal Controls		✓	✓		✓	✓		✓	✓
Peer Group Controls			✓			✓			✓
Observations	609,459	609,459	609,459	609,459	609,459	609,459	609,459	609,459	609,459
Dep. Mean	0.120	0.120	0.120	0.054	0.054	0.054	0.007	0.007	0.007

Note.—Each cell reports the coefficient from a separate regression based on equation (3). Gateway peer exposure is defined by the criminal history of older siblings and varies by panel: *Panel A.* school-grade exposure based on the history of group crime of older siblings. *Panel B.* neighborhood-grade exposure based on the history of group crime of older siblings. *Panel C.* school-grade exposure based on any history of crime of older siblings. *Panel D.* neighborhood-grade exposure based on any history of crime of older siblings. All specifications include birth-year, calendar-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. Indicated columns additionally condition on *Focal Controls*: sex, foreign background indicator, single parent household status, parental income, employment and education, plus dummies for missing values; and *Peer-group Controls*: shares of male and foreign background students in the relevant peer group. Standard errors are two-way clustered at the school and neighborhood levels. (**) $p < .01$, (*) $p < .05$, (+) $p < .1$.

Having established that school-grade exposure effects are primarily driven by peers with criminally involved older siblings, the remainder of the analysis focuses on this specific exposure channel. Table 7 presents estimates from separate regressions of equation (3), using gateway exposure measures based solely on the criminal history of older siblings, and only excludes these types of gateway peers. The table distinguishes between group crime and any crime, and between school-grade and neighborhood-grade peer groups. Panel A reports school-grade exposure based on group crime; panel B, neighborhood-grade exposure based on group crime; panel C, school-grade exposure based on any crime; and panel D, neighborhood-grade exposure based on any crime. Columns 1, 4, and 7 of panel A replicate the baseline pattern in Table 6, excluding controls for exposure through other relative types. The remaining columns show that the estimated effects are robust to adding detailed controls for the background of the focal student, including sex, family structure, parental income, education, and employment, as well as the composition of the peer group, such as the share of male and foreign background students. School exposure based on any type of sibling criminal history (panel C) yields patterns similar to those for group crime, though with smaller effect sizes. Finally, neighborhood crime crime exposure (panel D) also predicts future group crime offending, but with magnitudes that are smaller than either school-based exposure measure.

In addition, estimates are reported that control for potential gateway exposure from older cohorts (i.e., students in grade 8 and 9). Table A13 shows that exposure to gateway peers up to two years older appears to further increase the likelihood of future criminal behavior. This pattern exists in both the school and neighborhood context. However, because older cohorts are observable for the focal student and their parents *ex ante*, these estimates should not be interpreted causally. Rather, the takeaway is that the same-grade exposure estimates remain robust when accounting for older cohort's gateway peers.

To further test whether the observed effects are specifically driven by gateway peers with criminal older siblings, and not other peer characteristics, I estimate a set of regressions based on equation (3) but defining the exposure measure on other non-criminal and peer parent traits. The results, shown in Figure A8, present the estimated effects of school- and neighborhood-grade exposure to each characteristic on future criminal outcomes. With the exception of a negative effect of neighborhood male peers on committing any crime, none of the alternative exposures yields statistically significant results. This supports the interpretation that it is gateway peers, and not any other peer characteristic, that drive these findings.

Figure A9 further show that the estimates are robust to a range of sample restrictions, including the exclusion of small and large school- and neighborhood-grade groups, specific years, and the three major metropolitan areas in Sweden (Stockholm, Göteborg, and Malmö).

Although group crime serves as the primary proxy for gang-related activity, estimates based on more conventional crime classifications are also reported. Specifically, I construct alternative gateway peer exposure measures defined by older sibling involvement in narcotic, violent, and property crimes, regardless of co-offending. Using equation (3), I estimate the effect of each sibling's crime-type-specific gateway exposure on a range of crime-type-specific outcomes. Table A14 presents the results, focusing on school-grade exposure.[14] Each panel reflects a different crime-type exposure measure and each column tracks the criminal outcomes of the focal individual between ages 16 and 21.

The results indicate that exposure to peers with siblings involved in group crime increases the likelihood of violent, narcotic, and weapon-related offenses, the categories most closely associated with gang activity at these ages. However, when exposure is defined solely by narcotics offenses, the effect is limited to future narcotics crimes. This likely reflects the heterogeneity of narcotics offenses, which range from serious trafficking to low-level possession; the latter may weaken the exposure signal, as individuals convicted for personal drug use are less likely to be embedded in organized networks. Exposure based on violent or property crimes produces statistically significant effects on a broader set of outcomes, often comparable to or larger than those observed for group crime. This is consistent with the notion that these offenses are both less common and more indicative of escalated criminal behavior, particularly in the case of violent crimes such as assault, robbery, and attempted murder.

In general, the effects are consistent between multiple types of sibling-based criminal exposure, suggesting that the peer effects identified by the design are not narrowly driven by a single category of offense.

To examine the timing of effects, I re-estimate equation (3) using cumulative indicators for whether the student has committed at least one offense by each age. As shown in Figure 2, the influence of exposure to gateway peers increases progressively from late adolescence to early adulthood. This pattern could be due to individuals who have been indoctrinated gradually increasing their criminal activities, as well as the growing likelihood of being apprehended due to prolonged engagement. In Table A16, I further examine the distribution of exposure by binarizing the exposure variable in two ways: one panel compares students with any exposure to those without, while the other divides exposure into percentile bins, 50th-74th, 75th-89th, and 90th percentile and above. The results show that the effects are concentrated among the students in the top exposure group (90th percentile and above). The pattern is intuitive since engaging in criminal acts is relatively rare and involves significant social and personal costs, with substantial exposure likely needed to lead to the adoption of serious criminal activity.

The focal students school-grade and neighborhood-grade peer groups can be thought of as two overlapping spaces in a Venn diagram. Up to this point, I have treated these

[14] Table A15 presents the corresponding analysis of neighborhood exposure.

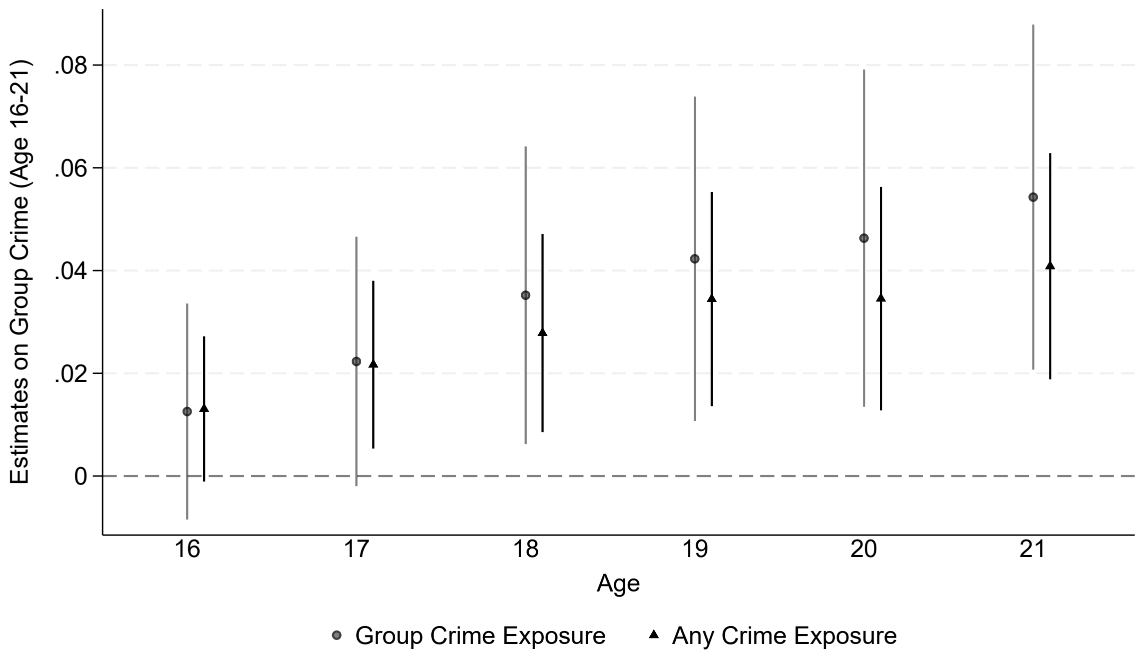


Figure 2: Cumulative Effects of Sibling-Based Peer Exposure on Group Crime by Age

Note.—Each coefficient is reported from a separate regression of based on equation (3). Gateway-peer school-exposure is constructed from the criminal records of an focal students older siblings group crime or any crime history. The dependent variables consist of cumulative group crime indicators starting from the age of 16 to the age of 21. All specifications include birth-year, calendar-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. Standard errors are two-way clustered at the school and neighborhood levels, and 95% CIs are reported.

groups separately, estimating exposure effects in schools and neighborhoods independently. In practice, however, these peer sets often intersect: some peers attend the same school but live in different neighborhoods; others live nearby but attend different schools; and a third group shares both school and neighborhood with the focal student. This overlap raises questions about the context in which peer exposure is the most influential. One possibility is that peers who share both school and neighborhood exert a stronger influence simply due to more frequent interaction between contexts. An alternative view is that schools serve as the primary contact point, bringing together students from diverse residential backgrounds and exposing them to peers they might not otherwise encounter. If so, school-based exposure may be especially important because it introduces new, non-overlapping influences relative to the home environment.

To examine these possibilities, Table A17 presents estimates in the same format as Table 7, but disaggregates exposure into three categories similar to Billings and Hoekstra (2023): peers who share only school, only neighborhood, or both. Panel A focuses on sibling-based exposure to group crime and shows that the largest and most precisely estimated effect arises from peers who attend the same school but do not live in the same neighborhood, suggesting that school-based exposure is the primary driver of the effects.

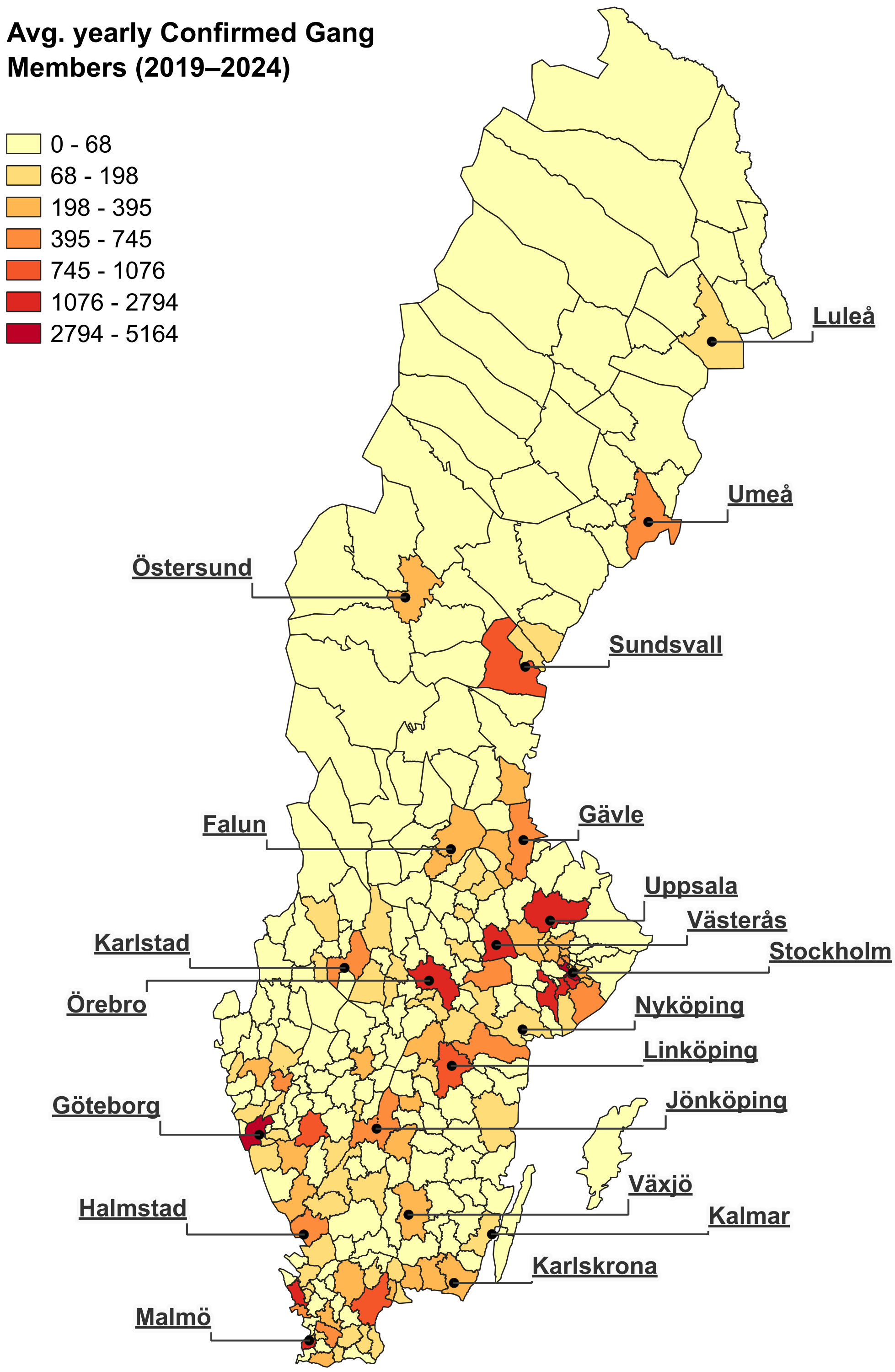


Figure 3: Municipality Distribution of Average Yearly Confirmed Gang Members

Note.—Municipality-level values are grouped using Jenks natural breaks, which minimize within-class variance and maximizes between-class variance. Average yearly confirmed gang members is defined from the count of gang members registered to be residing in each municipality, each year.

When the exposure definition is broadened to include any type of sibling crime, I find statistically significant effects across all three types of peer group contexts. Still, the largest estimates remain those for school-only peers, though the magnitudes are somewhat smaller than under the group crime definition. These results reinforce the interpretation that schools are a key setting through which exposure to criminally connected peers, specifically those with older siblings involved in group crime, translates into meaningful future criminal behavior.

6.1.2 Gang Presence and Heterogeneity

Table 6 provides one of the baseline estimates in the paper: a 5-pp. increase in school-grade exposure to peers with older siblings involved in group crime increases the probability of engaging in group crime between ages 16 and 21 by approximately 6.3 percent relative to the mean. I find that expanding the definition of exposure to include gateway peers, characterized by any form of older sibling criminal history, also results in positive and statistically significant impacts on overall criminal involvement, but smaller in magnitude. I find no evidence that neighborhood-grade exposure to gateway peers affects future criminality. Similarly, neither school- or neighborhood-based exposure significantly predicts verified gang membership after compulsory school.

However, these estimates likely mask meaningful heterogeneity. Gangs are primarily active in urban regions in Sweden, which means that the proxy for gang-related crime (group crime) in rural areas might capture more sporadic co-offending not related to organized crime. This introduces noise into the treatment variable, likely attenuating the estimated effects in Table 6, in addition to any possible non-linearities in treatment effects.

Figure 3 plots the geographic distribution of the annual average number of gang members living in each municipality between 2019 and 2024, according to the registered place of residence. Using residence as a reference point is consistent with evidence that criminal activity typically occurs near the homes of offenders (Kirchmaier et al. 2024). The figure reveals strong clustering around the major cities in each county.

To better approximate settings where gangs are likely present, I re-estimate equation (3) for individuals living in the major city of each county (labeled *County Capital*), focusing on school-based exposure. Although this restriction targets larger urban areas, it still includes neighborhoods and schools where gangs may be absent, and it may exclude gang-affected areas outside these cities. To address this, I also estimate equation (3) for two alternative subsamples: (i) neighborhoods where at least five gang members have resided between 2019 and 2024 (labeled *Gang Presence*) and (ii) areas with an above-median share of residents with a foreign background (labeled *High Foreign Share*). This approach helps to capture areas affected by gangs outside of the main cities, particularly in the south of Sweden and the areas surrounding Stockholm, as shown in Figure 3, while excluding more insulated suburban areas within larger cities. The corresponding estimates are presented in Table 8.

Table 8: Urban vs. Rural: Gateway Exposure by Siblings, Defined By History of Group Crime

Panel A. Dependent Variable: Group Crime (Age 16-21)

Geographical Restrictions:	Baseline	County Capital	Non-County Capital	Gang Presence	No Gang Presence	High Foreign Share	Low Foreign Share
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
School Exposure	0.054**	0.107**	0.029	0.108**	0.026	0.096**	-0.013
	(0.017)	(0.036)	(0.018)	(0.041)	(0.016)	(0.025)	(0.018)
Observations	609,459	189,998	419,461	118,148	491,311	290,610	318,849
Dep. Mean	0.054	0.056	0.053	0.091	0.045	0.072	0.038

Panel B. Dependent Variable: Ever In Gang Database

Geographical Restrictions:	Baseline	County Capital	Non-County Capital	Gang Presence	No Gang Presence	High Foreign Share	Low Foreign Share
	(8)	(9)	(10)	(11)	(12)	(13)	(14)
School Exposure	0.008	0.035*	-0.003	0.035+	-0.003	0.022*	-0.011**
	(0.007)	(0.017)	(0.006)	(0.021)	(0.004)	(0.011)	(0.004)
Observations	609,459	189,998	419,461	118,148	491,311	290,610	318,849
Dep. Mean	0.007	0.009	0.006	0.026	0.002	0.013	0.002

Note.—Each cell reports the coefficient from a separate regression based on equation (3), where school exposure to gateway peers is defined by whether an older sibling has a history of group crime. *Panel A.* reports effects on post-compulsory group crime offenses and *Panel B.* reports effects on ever being listed in the gang database. Sample restrictions are defined as follows: (i) *County capitals* include Stockholm, Göteborg, Malmö, Uppsala, Västerås, Nyköping, Linköping, Jönköping, Växjö, Kalmar, Karlskrona, Sundsvall, Halmstad, Karlstad, Örebro, Falun, Gävle, Östersund, Umeå, and Luleå. (ii) *Gang presence* is defined as a RegSO with at least five individuals listed in the gang database. (iii) High/low foreign share is defined relative to the median share of foreign-born residents across RegSOs. Standard errors are two-way clustered at the school and neighborhood levels. All regressions include birth-year, calendar-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. (**) $p < .01$, (*) $p < .05$, (+) $p < .1$.

Column 1 reports the baseline estimate for exposure to gateway peers defined by older siblings with a history of group crime. The effect nearly doubles in county capitals (column 2), where a 5-pp. increase in exposure leads to a 10 percent increase in group crime offending relative to the mean. Similar effect sizes are observed in areas with gang presence (column 4) and in neighborhoods with a high share of foreign-born residents (column 6). The estimates are muted and statistically insignificant in non-county capital areas (column 3), in areas without gang presence (column 5), and in neighborhoods with below-median foreign shares (column 7).

Panel B presents the corresponding results for future confirmed gang participation. Although the baseline estimate is positive but not statistically significant, the effect increases substantially in more exposed environments. In county capitals, a 5 pp. increase in school-based gateway exposure is associated with a roughly 20 percent increase relative to the mean in the probability of appearing in the gang database (column 9), significant at the 5% level.[15] Similar elevated and statistically significant effects are found in areas with presence of gangs (column 11) and in areas with a high share of foreign-born residents (column 13).

These results suggest that gateway peer exposure not only increases the risk of general and group-based criminal involvement, but also channels individuals into organized forms of criminal activity.

In addition to the geographic presence of gangs, individual characteristics likely shape how students respond to exposure to gateway peers. Factors such as gender, migration background, family structure, and academic ability can mediate the influence of peers through differences in social norms or access to resources. To examine heterogeneous effects, I split the sample by student sex, foreign background, household structure, and baseline academic performance. Table 9 reports estimates from equation (3), separately by subsample, for two outcomes: group crime (panel A) and inclusion in the gang database (panel B).

Starting with sex, columns 2 and 3 show a larger raw effect for males. However, when scaled relative to each groups baseline rate of involvement in group crime, the magnitudes are similar: A 5-pp. increase in exposure corresponds to a 4.6 percent increase for females and a 4.5 percent increase for males. In terms of background, the effect is more pronounced among students with a foreign background than among native-born peers. I also find stronger effects among students from single-parent households and those in the bottom quintile of the grade 6 test score distribution, although the latter is statistically significant only at the 10% level. Although the estimates for inclusion in the gang database (panel B) are overall not statistically significant, I find suggestive evidence of larger effects among students with a foreign background.

[15] Based on these estimates, a back-of-the-envelope calculation suggests that the effect accounts for roughly 333 individuals (0.035×0.05 (average exposure in county capitals)×189,998), with a 95% confidence interval of [171, 494]. In county capitals, this corresponds to 7.7% of confirmed gang members born between 2000 and 2006 ($n = 4,301$), or 14.7% of confirmed gang members born between 2000 and 2006 without an older sibling connected to a gang ($n = 2,270$).

Table 9: Focal Student Heterogeneity in Exposure to Gateway Peers (Group Crime Siblings)

Panel A. Dependent Variable: Group Crime (Age 16-21)

Student Restrictions:	Main	Male	Female	Foreign Background	Native Background	Single Parent Home	Two Parent Home	1st Quintile G6 Score	5th Quintile G6 Score
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
School Exposure	0.054**	0.074**	0.028+	0.125**	0.020	0.095**	0.039*	0.082+	0.026
	(0.017)	(0.028)	(0.017)	(0.045)	(0.015)	(0.037)	(0.018)	(0.045)	(0.020)
Observations	609,459	313,226	296,233	112,390	497,069	141,309	468,150	112,196	129,902
Dep. Mean	0.054	0.078	0.028	0.106	0.042	0.085	0.044	0.111	0.017

Panel B. Dependent Variable: Ever In Gang Database

Student Restrictions:	Main	Male	Female	Foreign Background	Native Background	Single Parent Home	Two Parent Home	1st Quintile G6 Score	5th Quintile G6 Score
	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
School Exposure	0.008	0.013	0.001	0.035+	-0.003	0.016	0.008	0.017	-0.010*
	(0.007)	(0.012)	(0.004)	(0.021)	(0.004)	(0.015)	(0.007)	(0.022)	(0.004)
Observations	609,459	313,226	296,233	112,390	497,069	141,309	468,150	112,196	129,902
Dep. Mean	0.007	0.012	0.001	0.026	0.002	0.011	0.006	0.020	0.001

Note.—Each cell reports the coefficient from a separate regression based on equation (3), where gateway peer school exposure is defined by whether an older sibling has a history of group crime. *Panel A.* reports effects on post-compulsory group crime offenses and *Panel B.* reports effects on ever being listed in the gang database. The sample restrictions are defined as follows: (i) Being male or female. (ii) Both parents have been born abroad or not. (iii) Living in a single or two parent home in grade 7. (iv) being at the bottom or highest quintile of grade 6 standardized test scores Standard errors are two-way clustered at the school and neighborhood levels. All regressions include birth-year, calendar-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. (**) $p < .01$, (*) $p < .05$, (+) $p < .1$.

If recruitment operates partly through gendered pathways, for example, older brothers socializing younger brothers into gang-adjacent behavior, one might expect exposure from older brothers to have a stronger effect on boys and exposure from older sisters to matter more for girls. To explore this possibility, Table A18 estimates the effect of exposure of gateway peers separately by the sex of the older sibling (brothers and sisters) and the focal student (all, male and female).

The results in Table A18 show that the strength of the gateway peer effect depends on the gender pairing between the older sibling and the focal student. When exposure is driven by brothers with a group crime history (panel A), a 5-pp. increase in school-grade exposure raises group crime involvement among male students by 4.3 percent relative to their mean. When exposure is transmitted through sisters (panel B), the largest effect is observed among girls: a 5-pp. increase in exposure raises their group crime involvement by 13 percent relative to the mean. Boys are also affected by sister-based exposure, an estimated increase of 6 percent, though this effect is less precisely estimated. In all cases, neighborhood-based exposure yields statistically insignificant estimates. These patterns are consistent with school-based exposure as the primary mechanism and suggest that peer effects are facilitated when the gateway peers' criminally involved sibling shares the focal students gender, consistent with gender-specific role models.

The surprisingly large coefficients associated with sisters should be interpreted in light of their relative and absolute rarity. Having an older sister with a documented history of group crime is far less common than having a comparable brother, so her presence likely indicates a stronger signal about the underlying criminal environment in the household. In this sense, sisters may act as a "high-salience" indicator of a particularly entrenched criminal environment, one that amplifies effects even when the focal student is male.

6.1.3 Emerging Patterns

The findings so far point to four main takeaways. First, while criminal involvement among various types of older relatives – such as parents, uncles, and cousins – is associated with later offending within families, it is specifically exposure to gateway peers with criminally involved older siblings that drives peer effects. This is consistent with anecdotal reports that indicate that gang recruitment flows from older to younger adolescents, and makes older siblings an expected origin of this influence, as they are likely close in age and interact regularly. Second, these peer effects operate through schools rather than neighborhoods, highlighting the central role of school-based peers in the diffusion of criminal behavior. Third, in settings where gangs are plausibly present, exposure effects are larger and even increase the probability of being confirmed as gang-affiliated after compulsory school. Fourth, these effects are not driven by other characteristics of peers. However, students with a foreign background, living in a single parent household, or performing poorly in school appear particularly vulnerable to such exposure.

6.2 Mechanism: Gang Recruitment or Just Bad Apples?

The preceding analysis shows that exposure to gateway peers, students with older siblings involved in group crime, increases the likelihood that students themselves offend, including in group offenses. In urban areas, this exposure also increases the probability of appearing in the gang database after finishing compulsory school. To assess whether these patterns reflect only delinquent peer effects, or also indicate a more systematic recruitment mechanism, I now examine: (i) co-offending patterns, (ii) homophily, and (iii) sibling presence.

6.2.1 Co-Offending

If exposure operates purely through peer influence, we should expect to see increased co-offending only with the gateway peers themselves. A recruitment mechanism would imply broader co-offending patterns: exposed students should also be more likely to co-offend with the criminal relatives of the gateway peer and with the individuals listed in the gang database. To test this, I re-estimate equation (3) using post-compulsory school co-offending outcomes. The results based on school- and neighborhood-grade gateway exposure defined by older siblings' history of group crimes appear in Table 10.

Panel A of Table 10 shows that exposure to gateway peers whose older siblings have a history of group crime increases co-offending with gateway peers. It also raises co-offending with gateway peers' family, primarily driven by co-offending with older siblings. These effects correspond to increases of 9-19.5 percent relative to the respective means given a 5-pp. increase in exposure. At the 10% significance level, I also find a smaller effect on co-offending with cousins of gateway peers. In addition, exposure raises the likelihood of co-offending with a documented gang member by 10 percent of the mean, consistent with a recruitment pathway linking students to broader gang networks.[16].

Panels B estimate the models using neighborhood-based exposure. In this setting, the results show that effects persist for co-offending with gateway peers and with the older siblings of gateway peers, while the effect on future co-offending with confirmed gang members disappears.

The pattern of results is consistent with what one would expect under an active recruitment mechanism: the focal student is more likely to co-offend directly with the gateway peer, but also with the source of this influence, the older sibling of the gateway peer, as well as, in school-based estimates, with confirmed gang members. The neighborhood estimates therefore reinforce the interpretation that the critical channel of recruitment and influence operates primarily within schools rather than neighborhoods.

[16] If co-offending with a confirmed gang member is taken as an indicator of gang involvement, the estimate of 0.026 implies, by a back-of-the-envelope calculation, that older-sibling group crime exposure accounts for about 792 individuals ($0.026 \times 0.05 \times 609{,}459$), with a 95% confidence interval of [488, 1097]. This corresponds to 10.1% of confirmed gang members born between 2000 and 2006 ($n = 7{,}823$), and 19.3% of those born between 2000 and 2006 and without a gang-connected older sibling ($n = 4{,}094$)

Table 10: Post-Compulsory School Co-Offending (School Exposure)

	Post-Compulsory School Co-Offense with:						
	(1) Gateway Peer	(2) Gateway Peer's Family	(3) Gateway Peer's Parent	(4) Gateway Peer's Sibling	(5) Gateway Peer's Uncle	(6) Gateway Peer's Cousin	(7) Individual in Gang Database
Panel A. Gateway Exposure by Siblings, Defined by History of Group Crimes							
School Exp.	0.039**	0.030**	0.002	0.033**	0.000	0.012+	0.026**
	(0.009)	(0.011)	(0.002)	(0.010)	(0.002)	(0.007)	(0.010)
Panel B. Gateway Exposure by Siblings, Defined by History of Group Crimes							
Neigh. Exp.	0.015**	0.007	0.001	0.011*	-0.002	0.007	0.003
	(0.005)	(0.006)	(0.001)	(0.005)	(0.001)	(0.004)	(0.006)
Observations	609,459	609,459	609,459	609,459	609,459	609,459	609,459
Dep. Mean	0.010	0.017	0.001	0.013	0.001	0.008	0.013

Note.—Each cell reports estimates from a separate regression based on equation (3), where school-exposure to gateway peers is defined based on the group crime history of older siblings. *Panel A.* reports estimates from school exposure based on older siblings with group crime history. *Panel B.* reports estimates from neighborhood exposure based on older siblings with group crime history. Dependent variables are defined as the focal student post-compulsory school co-offending with a gateway peer, a gateway peer's family, parent, sibling, uncle, aunt, or cousin, as well as, with anyone in the gang database. All regressions include birth-year, calendar year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. Standard errors are two-way clustered at the school and neighborhood levels. (**) $p < .01$, (*) $p < .05$, (+) $p < .1$.

6.2.2 The Role of Homophily and Immigrant Background

As shown in Table 1 and Figure A7, most of the individuals listed in the gang database are Swedish-born but have foreign-born parents. This overrepresentation of second-generation immigrants has been associated with residential segregation, slow integration, and a growing wealth inequality (Rostami and Mondani 2023; Örnlind and Forkby 2025). Because gangs typically emerge and operate in neighborhoods that also host ethnic enclaves, their membership often reflects the composition of local national diasporas. Homophily in social network formation – the tendency of individuals to associate with others who share salient traits (McPherson et al. 2001) – implies that students with similar backgrounds as gang members may be particularly vulnerable to exposure to gateway peers and potential recruitment.

To test whether peer effects vary by national proximity, I use parents countries of birth to classify students into mutually exclusive regional origin groups: Sweden; other Nordic countries; Western and Central Europe; Eastern Europe and the Caucasus; Central and South Asia; East and Southeast Asia; the Middle East and North Africa; Sub-Saharan Africa; Latin America; North America; and Oceania. A focal student is considered matched with a gateway peer if at least one parent shares a region of origin with the peer. Based on this classification, I construct four exposure measures: (i) exposure to gateway peers with no matching parent region, (ii) exposure to gateway peers with at least one matching region, (iii) exposure with exactly one parent region matched, and (iv) exposure with both parent regions matched.

I estimate versions of equation (3) that incorporate either a two-category split (no match vs. at least one match) or a three-category split (no match, exactly one match, both matched), to gauge the similarity of the match. Regressions are estimated separately for four samples: (i) the full student sample, (ii) students with at least one Swedish-born parent, (iii) students with no Swedish-born parents, and (iv) students with no Swedish-born parents where at least one parent is from a country frequently represented among gang-affiliated individuals (see Figure A7). Panel A in Table 11 reports baseline estimates of exposure to gateway peers, defined by having a sibling with a history of group crime. Columns 1 and 2 replicate the estimates from Table 7. The effects are small and statistically insignificant for students with at least one Swedish-born parent. For students with two foreign-born parents, the estimated effect sizes are similar in magnitude relative to the mean as those in the full sample. When restricted to students whose parents come from countries frequently represented in the gang database, the effects increase further. I find large and statistically significant effects on subsequent inclusion in the gang database.

Panel B in Table 11 examines whether exposure effects differ by national background match. In the full sample, the effects are driven by exposure to gateway peers with no shared parent-region—while exposure to peers with a shared background yields negative coefficients. These results mask considerable heterogeneity. Among students with at least one Swedish-born parent (columns 3–4), there is no evidence of effects of either type of exposure. Among students with two foreign parents (columns 5–8), the entire effect is driven by exposure to peers with shared regional backgrounds. Panel C further splits exposure into three groups (no match, one parent match, and both parents match). This helps explain the negative coefficient of the matched exposure in panel B. In the full sample, exposure to gateway peers with no matched background remains large and significant for future group crime offenses. Exposure to peers with exactly one parent match yields a positive, but statistically insignificant effect on gang listing, while exposure to peers with both parents matched generates negative effects. These patterns are largely driven by the sample with native-born parents: in this group, gateway exposure from native-to-native matches appears to reduce both group crime and gang outcomes. The positive effect is concentrated among students where only one parent is matched, suggesting heterogeneity in the way gateway peers are perceived. For both non-native subgroups, I find positive effects on future group crime from exposure to peers with either one or both parents matched, and significant effects on gang database listing when both parents share the same birth region as the gateway peer.

Table 11: Shared Background and Heterogeneous Effects of Gateway Peer Exposure

	Full Sample		Native Background		Foreign Background		Foreign Background (Countries Frequent in Gang Database)	
	(1) Group Crime (Age 16-21)	(2) Ever In Gang	(3) Group Crime (Age 16-21)	(4) Ever In Gang	(5) Group Crime (Age 16-21)	(6) Ever In Gang	(7) Group Crime (Age 16-21)	(8) Ever In Gang
Panel A. Gateway Exposure by Siblings, Defined by History of Group Crimes								
School Exp. (Baseline)	0.054**	0.008	0.020	-0.003	0.115*	0.033	0.162**	0.054*
	(0.017)	(0.007)	(0.015)	(0.004)	(0.046)	(0.022)	(0.052)	(0.027)
Panel B. Gateway Exposure by Siblings, Defined by History of Group Crimes								
School Exp. (No Shared Bkg.)	0.270**	0.055**	-0.012	-0.016	0.048	-0.008	0.090	0.020
	(0.026)	(0.010)	(0.033)	(0.012)	(0.046)	(0.021)	(0.055)	(0.027)
School Exp. (Any Parent Shared Bkg.)	-0.088**	-0.023**	0.031+	0.001	0.366**	0.177**	0.349**	0.152**
	(0.020)	(0.008)	(0.017)	(0.004)	(0.079)	(0.053)	(0.081)	(0.056)
Panel C. Gateway Exposure by Siblings, Defined by History of Group Crimes								
School Exp. (No Shared Bkg.)	0.272**	0.055**	-0.003	-0.014	0.048	-0.007	0.090	0.021
	(0.026)	(0.010)	(0.033)	(0.012)	(0.046)	(0.021)	(0.055)	(0.027)
School Exp. (One Parent Shared Bkg.)	0.117**	-0.011	0.281**	0.034**	0.280*	0.057	0.340*	0.045
	(0.028)	(0.011)	(0.028)	(0.008)	(0.140)	(0.092)	(0.151)	(0.103)
School Exp. (Both Parent Shared Bkg.)	-0.168**	-0.027**	-0.073**	-0.013**	0.384**	0.201**	0.351**	0.172**
	(0.023)	(0.009)	(0.018)	(0.004)	(0.086)	(0.059)	(0.088)	(0.061)
Observations	608,780	608,780	497,069	497,069	94,360	94,360	70,049	70,049
Dep. Mean	0.054	0.007	0.042	0.002	0.106	0.026	0.115	0.031

Note.—Each column-panel combination reports estimates from a separate regression based on equation (3), where gateway peer exposure is defined by the group crime history of older siblings. Estimates are shown for four samples: *Full Sample* (all students), *Native Background* (students with at least one Swedish-born parent), *Foreign Background* (students with no Swedish-born parent), and *Foreign Background - Countries Frequent in Gang Database* (students with foreign-born parents, at least one of whom is from a country listed in Figure A7). Equation (3) is extended to include exposure measures that vary by the degree of shared parental country of birth region between the focal student and the gateway peer. *Panel A* includes a single exposure measure based on all school-grade peers. *Panel B* distinguishes between exposure to gateway peers who (i) do not share any parental country of birth region with the focal student, and (ii) share at least one parental country of birth region. *Panel C* further disaggregates exposure into three categories: (i) no shared parental country of birth region, (ii) exactly one shared parental country of birth region, and (iii) both parents from the same country of birth region. Parental country-of-birth regions are categorized as: Sweden; Nordic countries; Western and Central Europe; Eastern Europe and the Caucasus; Central and South Asia; East and Southeast Asia; the Middle East and North Africa; Sub-Saharan Africa; Latin America; North America; and Oceania. All regressions include birth-year, calendar-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. Standard errors are two-way clustered at the school and neighborhood levels. (**) $p < .01$, (*) $p < .05$, (+) $p < .1$

These results align with the hypothesis that homophily influences school transmissions: Students who share a national background with gang members seem more vulnerable to the influence of gateway peers. The negative effects observed for native-to-native exposure likely reflect different mechanisms. For one, gateway peers in more socially integrated settings may be perceived to deviate more sharply from the prevailing norms, making their behavior more salient and potentially triggering a deterrence response. In some cases, such peers may even be ostracized, reinforcing their role as negative examples. In addition, native-born students likely have greater access to resources and support networks, increasing the opportunity cost of participating in criminal activity or joining a gang. Even when the gateway peer is native, the focal native student may face a higher implicit cost of entry as they may stand out relative to the broader composition of gangs.

6.2.3 Disruption by Non-Present Gateway Siblings

If a systematic recruitment mechanism is at work, then the absence of an older sibling should disrupt that channel – even if the gateway peers themselves act as 'bad apples' who exert criminogenic peer effects. To test this, I use information on the age, residence, and incarceration status of older siblings of gateway peers to assess whether exposure effects weaken when these siblings are not geographically present, are incarcerated, or fall outside of the typical recruiter's age.

Sibling Incarceration. If a gateway peers older sibling was incarcerated during the focal student's upper-compulsory school years (grades 6–7), the most likely link to a gang recruiter would have been interrupted. In this case, we would expect the reduced-form effect of exposure to decrease. Importantly, incarceration must be of sufficient duration to plausibly disrupt the recruitment channel; shorter sentences may instead signal more serious or entrenched criminal involvement without necessarily limiting contact.

I categorize the older siblings of the gateway peers into three mutually exclusive incarceration statuses: (i) not incarcerated while the gateway peer was in grades 6–7; (ii) incarcerated for less than one year during that period; and (iii) incarcerated for more than one year during that period. Using these classifications, I construct three separate exposure measures, each capturing the number of gateway peers in a students cohort whose sibling falls into each incarceration category. Then I estimate variants of equation (3) that include all three exposure measures simultaneously, using future offenses, listing in the gang database, and co-offending as outcomes. The corresponding estimates are shown in Table 12.

Exposure to gateway peers whose siblings were not incarcerated yields effects similar in magnitude to the baseline estimates in Table 7 and Table 10, in future offending, gang database listing, and co-offending. For gateway peers whose siblings were incarcerated for less than one year, I find larger effects on future offending and co-offending. There is also a rise in sibling and gang member co-offending, although the estimate is imprecise and not statistically distinguishable from zero.

Table 12: Sibling Incarceration

	(1) Any Crime (Age 16-21)	(2) Group Crime (Age 16-21)	(3) Ever In Gang	(4) Co-Offend with Gateway Peer	(5) Co-Offend with Gateway Sibling	(6) Co-Offend with Gang Member
Gateway Exposure by Siblings, Defined by History of Group Crimes						
School Exp. (Sibling No Inc.)	0.065**	0.051**	0.009	0.035**	0.032**	0.024*
	(0.022)	(0.017)	(0.007)	(0.008)	(0.010)	(0.010)
School Exp. (Sibling Inc. $<$ 1 Yr)	0.173*	0.112+	0.006	0.117**	0.064	0.069
	(0.087)	(0.068)	(0.027)	(0.035)	(0.041)	(0.042)
School Exp. (Sibling Inc. $\geq$ 1 Yr)	0.116	0.038	-0.013	-0.003	-0.012	-0.020
	(0.144)	(0.112)	(0.046)	(0.054)	(0.063)	(0.064)
Observations	609,459	609,459	609,459	609,459	609,459	609,459
Dep. Mean	0.120	0.054	0.007	0.010	0.013	0.013

Note.—Each column reports estimates from a separate regression based on an extended version of equation (3), where gateway peer exposure at the school-grade level is defined by the group crime history of older siblings. The specification includes three mutually exclusive exposure measures: *School Exp. (Sibling No Inc.)*: exposure to gateway peers whose older siblings were not incarcerated; *School Exp. (Sibling Inc. $<$ 1 Yr)*: exposure to gateway peers whose older siblings were incarcerated for less than one year; *School Exp. (Sibling Inc. $\geq$ 1 Yr)*: exposure to gateway peers whose older siblings were incarcerated for one year or more. Incarceration status is based on conviction dates and refers to sentences imposed while the focal student was in grades 6 to 7. All regressions include birth-year, calendar-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. Standard errors are two-way clustered at the school and neighborhood levels.
(**) $p < .01$, (*) $p < .05$, ($^{+}$) $p < .1$

Table 13: Sibling Residency

	(1) Any Crime	(2) Group Crime	(3) Ever In Gang	(4) Co-Offend with Gateway Peer	(5) Co-Offend with Gateway Sibling	(6) Co-Offend with Gang Member
Panel A. Gateway Exposure by Siblings, Defined by History of Group Crimes						
School Exp. (Baseline)	0.073**	0.054**	0.008	0.039**	0.033**	0.026**
	(0.021)	(0.017)	(0.007)	(0.009)	(0.010)	(0.010)
Panel B. Gateway Exposure by Siblings, Defined by History of Group Crimes						
School Exp. (Gateway-Local Sibling)	0.075*	0.076**	0.013	0.048**	0.050**	0.033*
	(0.029)	(0.024)	(0.010)	(0.011)	(0.014)	(0.014)
School Exp. (Gateway-Nonlocal Sibling)	0.068	0.013	-0.002	0.022	-0.000	0.012
	(0.042)	(0.030)	(0.010)	(0.015)	(0.016)	(0.015)
$p(H_0 : \beta_{Local} = \beta_{Nonlocal})$	0.901	0.142	0.341	0.175	0.026	0.336
Dep. Mean	0.120	0.054	0.007	0.010	0.013	0.013
Observations	609,459	609,459	609,459	609,459	609,459	609,459

Note.—Each column-panel combination reports estimates from a separate regression based on equation (3), where school-grade gateway peer exposure is defined by the group crime history of older siblings. *Panel A* reports estimates from the baseline version of equation (3). *Panel B* extends the model to distinguish between two exposure measures: *School Exp. (Gateway-Local Sibling)*: exposure to gateway peers whose older sibling resides in the same municipality as the focal student; *School Exp. (Gateway-Nonlocal Sibling)*: exposure to gateway peers whose older sibling resides in a different municipality. Sibling residence is measured at the time the focal student is in grade 7. All regressions include birth-year, calendar-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. Standard errors are two-way clustered at the school and neighborhood levels.
(**) $p < .01$, (*) $p < .05$, (+) $p < .1$

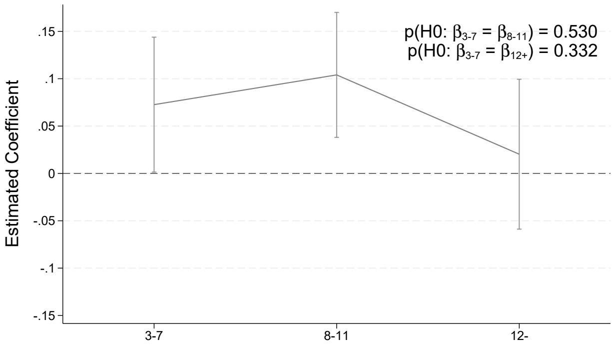
The effect on the likelihood of the focal student appearing in the gang database decreases slightly and remains statistically insignificant, but remains positive. Exposure to gateway peers whose siblings were incarcerated for more than a year leads to smaller effects on future offending (columns 1 and 2). The effects on gang listing and co-offending go towards zero. Although estimates are imprecise due to the relatively small number of siblings with long prison sentences, the pattern is consistent with a possible disruption of a recruitment mechanism, possibly driven by the absence of the sibling, a deterrence effect, or both.

Sibling Geographical Proximity. While incarceration captures both the absence of the older sibling and a potential deterrence effect through the increased perceived costs of gang involvement, it does not isolate the importance of the presence of the sibling per se. To further assess whether the physical proximity of older siblings matters, I use residential records to identify whether the older sibling lived in the same municipality as the gateway peer during the upper-compulsory school years of the focal student. Then I construct two separate exposure measures: (i) school-level exposure to gateway peers whose older siblings resided in the same municipality, and (ii) exposure to gateway peers whose siblings lived in a different municipality. I estimate variants of equation (3) that include both measures simultaneously. The results are presented in Table 13.

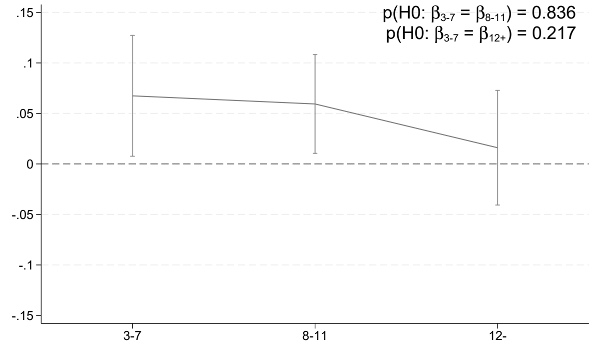
Panel A of Table 13 replicates the baseline effects of exposure to gateway peers with siblings who have a history of group crime. Panel B shows that these effects are driven by gateway peers whose older siblings reside in the same municipality as the peer and the focal student (labeled *Gateway-Local Sibling*). The estimated effects for this group increase in magnitude in all outcomes. Exposure to gateway peers whose siblings live in a different municipality yields a large, but statistically insignificant, point estimate for future criminal offenses, while the estimated effects for group crime, gang database inclusion, and co-offending drop substantially toward zero. The precision of these estimates is comparable to those in the local-sibling group, suggesting that the decline reflects a genuine reduction in effect size rather than a loss of power.

These findings reinforce the view that the physical presence of the older sibling is relevant in driving gang influence and potential recruitment. Although some behavioral spillovers can still occur through peer interactions alone, sibling proximity appears necessary to maintain effects on gang-related outcomes, including group crime and co-offending.

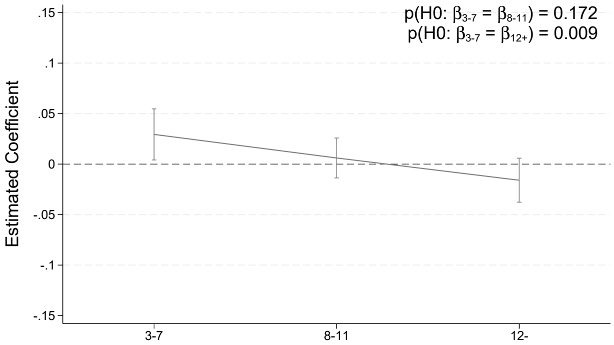
Gateway and Sibling Age Gap. A complementary way to assess the role of sibling presence is to examine the age gap between the gateway peers and their older siblings. Siblings closer in age to both the gateway peer and the focal student are more likely to resemble the modal recruiter and to interact more frequently. Siblings with larger age gaps are less likely to have regular contact with the gateway peer and their friends, and are often older than the typical recruiting age. Although these older siblings may still be involved in gangs, they are more likely to occupy senior roles and less likely to participate directly in adolescent recruitment.



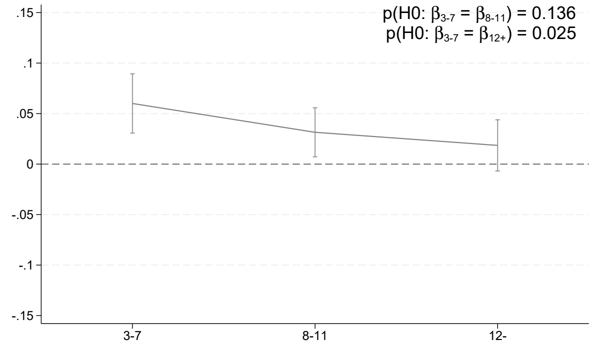
(a) Any Crime (Age 16-21)



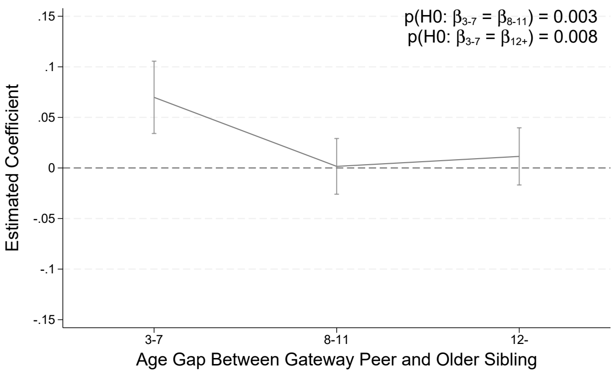
(b) Group Crime (Age 16-21)



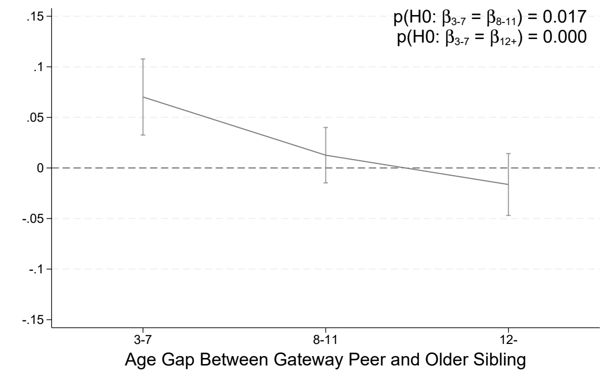
(c) Ever in Gang Database



(d) Co-Offended w/ Gateway Peer



(e) Co-Offended w/ Gateway Sibling



(f) Co-Offended w/ Gang Member

Figure 4: Age Gap between School Gateway Peers and their Siblings

Note.—Coefficients are estimated from a modified version of equation (3), where three exposure measures are included jointly. Each measure corresponds to gateway peers whose mean age gap with older siblings falls within a specific tercile of the age-gap distribution (in years). Age gaps are defined as the difference between the gateway peer and their older sibling. As the older siblings criminal offense must occur before the peer enters grade 7—typically around age 12—and the sibling must be at least 15, the age of criminal responsibility in Sweden, the minimum observable age gap of is three years. Dependent variables are indicated by the sub-figure captions. All regressions include school-neighborhood, calendar-year, and birth-year fixed effects, and exclude the gateway peers themselves from the estimation sample. Standard errors are two-way clustered at the school and neighborhood levels. 95% confidence intervals are shown.

To investigate this, I construct three exposure measures based on terciles of the age-gap distribution between gateway peers and their older siblings. Specifically, I define gateway peers by whether their sibling is (i) 3–7 years older, (ii) 8–11 years older, or (iii) 12+ years older. The minimum age gap is three years, since the sibling offense must occur when the sibling was at least 15 and the gateway peer was at most in grade 6 (typically around age 12). The youngest group corresponds to the reported modal age range for gang recruiters (Brottsförebyggande rådet 2023). I estimate equation (3) with all three school-level exposure measures included simultaneously. The resulting estimates are reported in Figure 4.

The first row of Figure 4 presents estimates for future criminal behavior. For any crime (subfigure a), the largest effects are driven by exposure to gateway peers with moderately older siblings (8–11 years older), whereas for group crime (subfigure b), both the youngest (3–7 years) and middle (8–11 years) age-gap groups generate similarly sized effects. However, when examining more gang-specific outcomes, focal students being listed in the gang database (subfigure c) or co-offending with a gateway peer (subfigure d), positive and statistically significant effects appear only for the youngest sibling group. For gateway peers with older siblings in the upper two age-gap terciles, the estimated effects are close to zero and statistically insignificant. The same pattern holds when looking at co-offending with the gateway peers sibling or with a confirmed gang member (subfigures e and f).

These results reinforce the interpretation that recruitment effects operate primarily when gateway peers have siblings within the modal age range for gang recruiters. Effects on general criminal behavior appear to be driven more broadly by exposure to gateway peers with older siblings, consistent with non-recruitment related peer spillovers.

6.2.4 Summary of Evidence on Gang-Recruitment Process

The combined patterns across co-offending, homophily, and sibling presence point toward the presence of a recruitment process, rather than a purely disruptive peer effect. Focal students are not only more likely to offend when exposed to gateway peers, but are also more likely to co-offend with the gateway peers sibling (with some indications of gateway-cousin co-offending) and with documented gang members. These effects are stronger among students who share a regional background with gateway peers, particularly when that background corresponds to regions frequently represented among gang-affiliated individuals.

Importantly, the effects on gang-related offenses and co-offending with gang members, gateway peers, and their relatives disappear when the older sibling is incarcerated for an extended period. A similar decline in effects is observed when the older sibling is not geographically present, where deterrence is less likely to explain the drop. In addition, the effects on future participation in gangs and group crime are concentrated among gateway peers whose older siblings fall within the modal recruiting age. Together, these findings suggest that older siblings play an active role in the transmission of gang involvement, consistent with a recruitment process.

6.3 Educational and Mental Health Consequences

Although the main focus of the paper is on future criminal behavior, exposure to peers with older siblings involved in crime and gang-related crime can also affect other relevant outcomes, particularly school achievement and mental health. This is relevant not only in its own right but also because anecdotal reports suggest that students who are struggling academically or socially are more likely to be targeted for gang recruitment. If peer exposure contributes to declines in educational performance or mental well-being, it may reinforce an already existing recruitment channel.

Mental illness can be triggered by violence, such as school shootings (Rossin-Slater et al. 2020), or by adverse childhood experiences in general (Blanchflower et al. 2024). Furthermore, former gang members describe depression and anxiety related to threats to themselves and their families (Brottsförebyggande rådet 2016; Örnlind and Forkby 2025), often managed by self-medication of narcotics. Such coping strategies are likely to contribute to under-reporting in medical records, especially among those more deeply embedded in organized crime. In line with this, a recent Swedish Social Services report shows relatively low rates of diagnosed mental illness in vulnerable areas, but elevated comorbidities, including substance abuse and suicide (Socialstyrelsen 2024).

Table 14 presents estimates of the impact of sibling-based exposure – defined through group crime or any crime – on three sets of outcomes: mental health diagnoses during upper-compulsory school, grade 9 academic performance, and post-compulsory school mental health diagnoses (up to age 21). Regressions for mental health outcomes additionally control for pre-upper compulsory school diagnoses.

Starting with mental health during compulsory school (columns 1-2), panel A shows that a 5-pp. increase in school-grade exposure to peers with older siblings involved in group crime raises the likelihood of mood-related diagnoses by approximately 4.7 percent, relative to a base rate of 3.7 percent. This estimate is statistically significant at the 1% level. The corresponding estimate for stress-related diagnoses (0.012) is not statistically significant. The analogous neighborhood-based effects in Panel B are smaller: a 5-pp. increase in exposure is associated with a 1.9 percent increase in mood-related diagnoses relative to the mean, significant at the 10% level, and an insignificant effect on stress-related diagnoses. When exposure is defined more broadly to include any sibling crime (panels C and D), the estimated effects are further attenuated. Overall, these results provide some evidence that peer exposure during compulsory school affects mental health, particularly mood-related diagnoses, with patterns more consistent with school-based than neighborhood-based effects.

Table 14: Educational and Mental Health Effects of Exposure to Peers with Group Crime-Involved Siblings

	Mental Health Diagnoses (Age 13-15)		Grade 9 Education Outcomes			Mental Health Diagnoses (Age 16-21)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Mood	Stress	Standardized Test Score	Merit Score	High School Eligibility	Mood	Stress
Panel A. Gateway Exposure by Siblings, Defined by History of Group Crimes							
School Exposure	0.035**	0.012	-0.147**	-8.614*	-0.022	0.004	0.023
	(0.011)	(0.013)	(0.053)	(3.686)	(0.019)	(0.013)	(0.016)
Panel B. Gateway Exposure by Siblings, Defined by History of Group Crimes							
Neigh. Exposure	0.014+	0.006	-0.034	-0.287	-0.018	0.013	0.008
	(0.007)	(0.009)	(0.027)	(2.095)	(0.011)	(0.009)	(0.011)
Panel C. Gateway Exposure by Siblings, Defined by History of All Crimes							
School Exposure	0.016*	0.013	-0.093*	-4.044	-0.019	-0.007	0.016
	(0.008)	(0.010)	(0.037)	(2.620)	(0.013)	(0.009)	(0.012)
Panel D. Gateway Exposure by Siblings, Defined by History of All Crimes							
Neigh. Exposure	0.004	0.001	-0.023	0.279	-0.008	0.003	0.001
	(0.005)	(0.006)	(0.019)	(1.494)	(0.008)	(0.006)	(0.008)
Observations	603,952	603,952	603,952	603,242	603,242	603,952	603,952
Dep. Mean	0.037	0.056	0.064	228.785	0.903	0.052	0.080

Note.—Each cell reports the coefficient from a separate regression based on equation (3). Gateway peer exposure is defined by the criminal history of older siblings and varies by panel: *Panel A.* school-grade exposure based on siblings' history of group crime. *Panel B.* neighborhood-grade exposure based on group crime history. *Panel C.* school-grade exposure based on any history of crime. *Panel D* neighborhood-grade exposure based on any history of crime. Mood and stress are defined as receiving a relevant diagnosis between ages 13-15 (columns 1-2) or ages 16-21 (columns 6-7). These regressions also control for any mental health diagnoses received before upper-compulsory school. Educational outcomes are measured at the end of grade 9: *Standardized Test Score.* standardized test scores within years in Swedish, English, and mathematics. *Merit Score.* Total sum of points based on subject grades (comparable to GPA) *High School Eligibility.* indicator for passing grades in Swedish, English, and Mathematics. All regressions include birth-year, calendar-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. Standard errors are two-way clustered at the school and neighborhood levels. (**) $p < .01$, (*) $p < .05$, (+) $p < .1$

I also find negative impacts on academic results in grade 9. School exposure to peers with older siblings involved in group crime is associated with significant declines in both standardized test scores (column 3) and merit scores (column 4, equivalent to GPA), significant at the 1% and 5% levels, respectively. The effect on test scores is sizable, 11.4 percent relative to the mean, while the impact on merit scores is negligible. There is no detectable impact on high school eligibility (column 5), likely because the eligibility threshold is low and most students qualify even with moderate academic performance. When exposure is based on any type of sibling crime (panel C), the effects are attenuated: Standardized test scores decline (significant at the 5% level), while the reduction in merit scores is smaller and not statistically significant.

Turning to post-compulsory school mental health outcomes, I find no clear evidence of worsening conditions. One interpretation is that such effects may exist but are obscured by under-reporting, particularly among students who become involved in criminal activity and are less likely to seek or receive formal care. During the compulsory school years, students have more regular contact with school health services, making symptoms more likely to be documented.

The results suggest that exposure to gateway peers generates meaningful effects on both mental health and educational outcomes during compulsory school. Although immediate mental health impacts are reflected primarily in increased mood-related diagnoses, academic consequences are pronounced in test scores; however, they do not seem to translate into larger academic consequences on merit scores and high school eligibility. Estimates on neighborhood exposure remain statistically insignificant, except for some suggestion of impact on mood-related diagnosis during compulsory school.

6.4 Policy Discussion

Based on the findings in this paper, exposure to gateway peers, students whose older siblings have committed group crime, induces negative spillovers, with implied long-run costs to society. These effects seem to be present in schools rather than in neighborhoods. This should be interpreted together with the fact that the sample consists of students with no prior family ties to crime. Despite some sorting into schools, most compulsory schools still bring together students from various neighborhoods and backgrounds. In combination with residential segregation, schools are therefore likely the first institutional setting where previously non-connected students encounter gateway peers. Hence, while school exposure should be seen as a part of a broader picture, it is likely to play a significant role in enabling expansion of organized crime groups.

Although there are different underlying mechanisms that can plausibly generate these effects, such as changes in perceived costs or deviations from social norms,[17] A cost-effective

[17] The empirical strategy in this paper assumes a linear-in-means model, where peer influence is assumed to be linear and operate through the group average. However, recent work by Boucher et al. (2024) introduces

approach is likely to target the gateway peers, i.e. addressing the origin of these negative effects rather than trying to mitigate their consequences due to the exponential nature of network spillovers. Possible strategies could include avoiding the clustering of gateway peers in the same schools or classrooms, and providing additional support and monitoring in affected schools (in line with hotspot policing). These approaches concentrate resources on the relatively small group of students who are more likely to influence others. However, such interventions should avoid stigmatization, as it may lead students to see themselves as part of an out-group, with gangs perceived as the only remaining accepting community.

In extreme cases, intervention by social services to separate children from criminally active relatives could prevent a student from becoming a gateway peer in the first place. However, such measures must be approached with care. Removing a gateway peer or their sibling from their home environment might seem like a way to disrupt negative influences, but it could backfire, both by triggering adverse reactions to the removal itself and by leading to placements that increase exposure to high-risk peers, such as youth detention or group homes. As shown, for example, by Bayer et al. (2009) and Stevenson (2017), the effects of peers in these environments can intensify future offending. Policymakers must therefore weigh the potential benefits of breaking up harmful networks against the risks of concentrating high-risk youth in settings where criminal behavior is reinforced rather than reduced.

7 Conclusion

This paper examines how gang-related behavior diffuses through family and peer networks in Sweden, focusing on gateway peers, students whose older siblings or extended relatives have committed group crimes. Using newly linked administrative data that combine verified gang membership with full kinship ties, criminal records, school and neighborhood identifiers, and educational and health outcomes, I study how exposure to gateway peers affects subsequent criminal behavior. The empirical strategy compares students across adjacent cohorts within school-by-neighborhood cells, leveraging quasi-random variation in peer composition to identify causal effects.

The analysis begins by documenting within-family transmission of criminal and gang-related behavior. Although parental criminal history predicts later offending, it explains only about half of a latent 'crime-family' effect. Older siblings account for most of the remaining share, whereas uncles, aunts, and cousins contribute relatively little.

The main analysis then turns to peer spillovers. Exposure in grade 7 to peers whose older siblings have a history of group crime increases the likelihood of general and group-

a more flexible framework and allows peer effects to be driven by spillovers or conformism. That is, gateway peers may change the perceived return to crime or gang participation, or shift the social norm about participation, and increase the cost of deviating from this norm. Boucher et al. (2024) show that depending on context, the mechanisms can differ, with implications for how to optimally design policy.

based offending by early adulthood. These effects are concentrated within schools rather than neighborhoods and are driven by peers who share school but not neighborhood with the focal student, highlighting schools as institutional spaces where networks form across otherwise segregated environments. The effects are stronger among students with a foreign background and those from more disadvantaged family or academic contexts. I also find suggestive evidence that exposure operates through gendered channels. In urban areas, exposure significantly increases the likelihood of being identified as a gang member after completing compulsory school and up to age 21. Back-of-the-envelope calculations suggest that up to 14.7% of confirmed gang members born between 2000 and 2006 in county capitals could be attributed to school-based gateway peer exposure.

Several patterns point to a recruitment mechanism rather than pure peer disruption. Focal students are more likely to co-offend not only with gateway peers, but also with the criminally involved siblings of those peers and confirmed gang members. These effects weaken when the sibling is removed from the environment (due to incarceration, residing outside the municipality, or being outside the typical recruiting age range) and strengthen when students share a country background with gateway peers. These findings suggest that physical proximity and shared identity jointly shape recruitment dynamics. If co-offending with a confirmed gang member is taken as an indicator of gang involvement, older siblings' group crime exposure could account for 10.1% of confirmed gang members born between 2000 and 2006, and up to 19.3% for those without a gang-connected older sibling.

Gateway peer exposure also has consequences beyond crime. Lower academic achievement and increased diagnoses related to mood during compulsory school, particularly in school rather than in neighborhood settings. These disruptions may both reflect and reinforce a recruitment process, as struggling students become more susceptible to social influence and less protected by institutional support. I find no sustained effects on post-compulsory mental health.

Taken together, the findings show that gang-related behavior spreads not only through one's own family, but also through peer networks in schools, where younger siblings can act as gateways to gang-affiliated networks. These dynamics are unlikely to be addressed through punitive or surveillance-based policies alone. Interventions that target school-based peer structures, particularly those that weaken the role of gateway peers as recruitment intermediaries, may offer more promising avenues to prevent the continued growth of organized crime groups and their longer-term consequences.

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A Appendix Figures

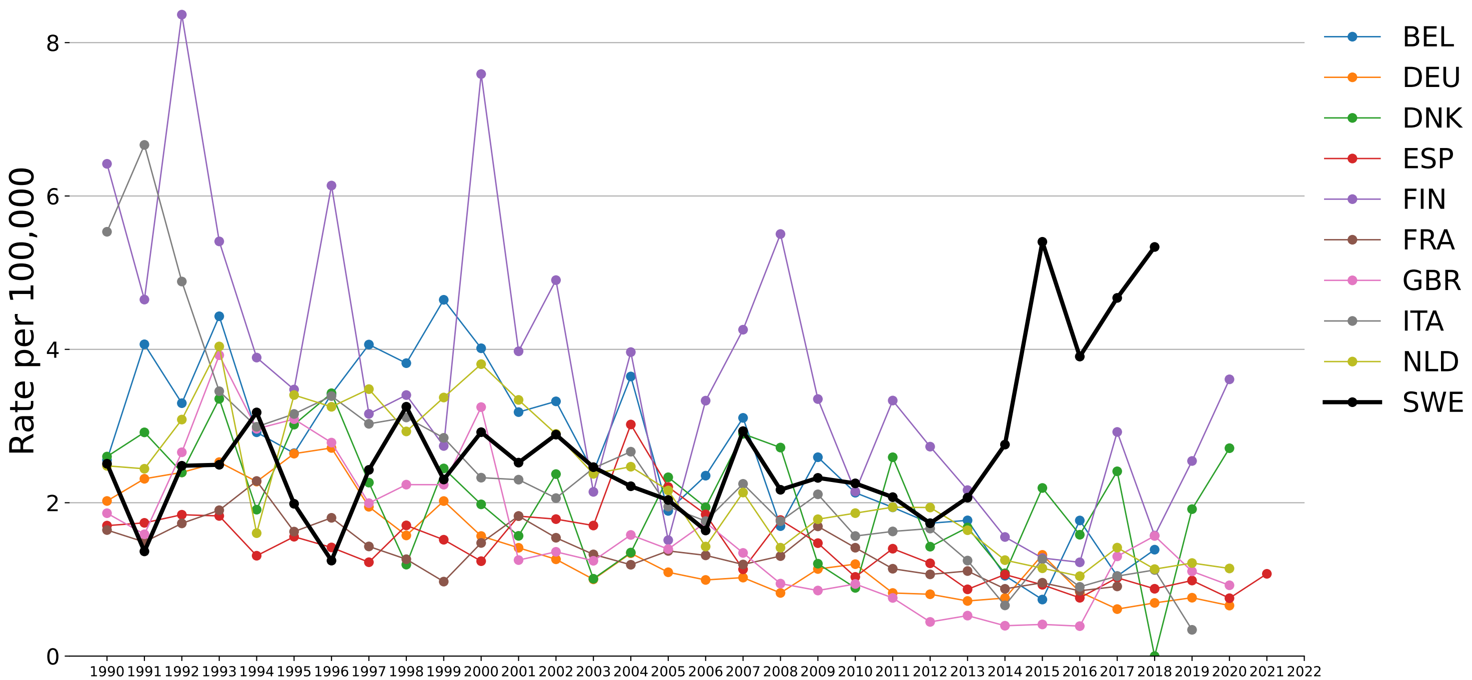


Figure A1: Mortality from Interpersonal Violence in European Countries

Note.—Figure reports deaths from interpersonal violence per 100,000 individuals across European countries. Source: WHO, 2023, Mortality Database.

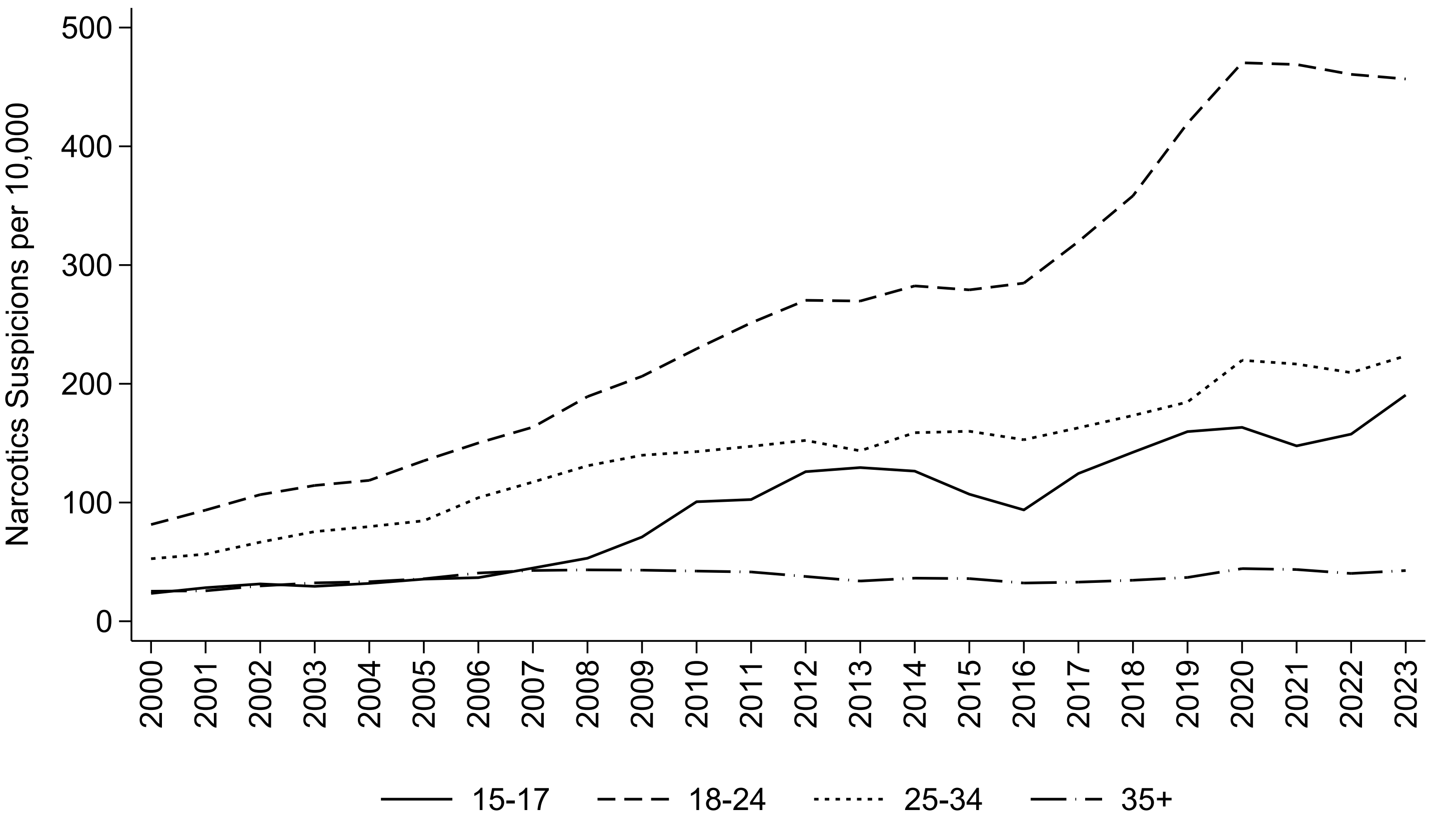


Figure A2: Narcotics Suspicions per 10,000 Individuals, by Age Group (2000-2023)

Note.—The figure reports number of suspicions per 100,000 individuals within each age bracket from 2000 to 2023, indicated by the x-axis. Suspicions are reported by date of registration.

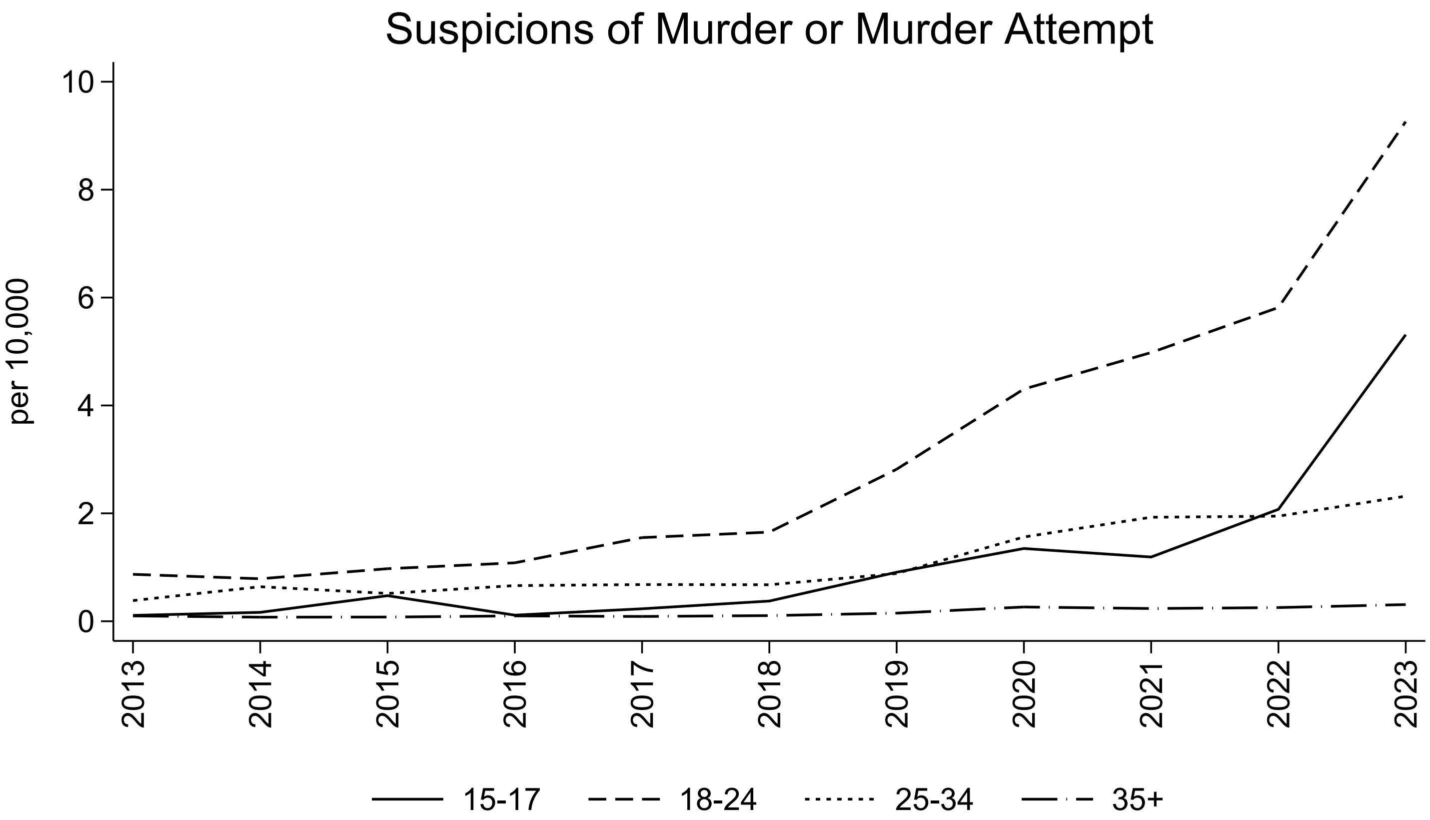


Figure A3: Murder or Attempted Murder Suspicions per 100,000 Individuals, by Age Group (2013-2023)

Note.—The figure reports number of murder or murder attempt suspicions per 100,000 individuals within each age bracket from 2000 to 2023, indicated by the x-axis. Suspicions are reported by date of registration.

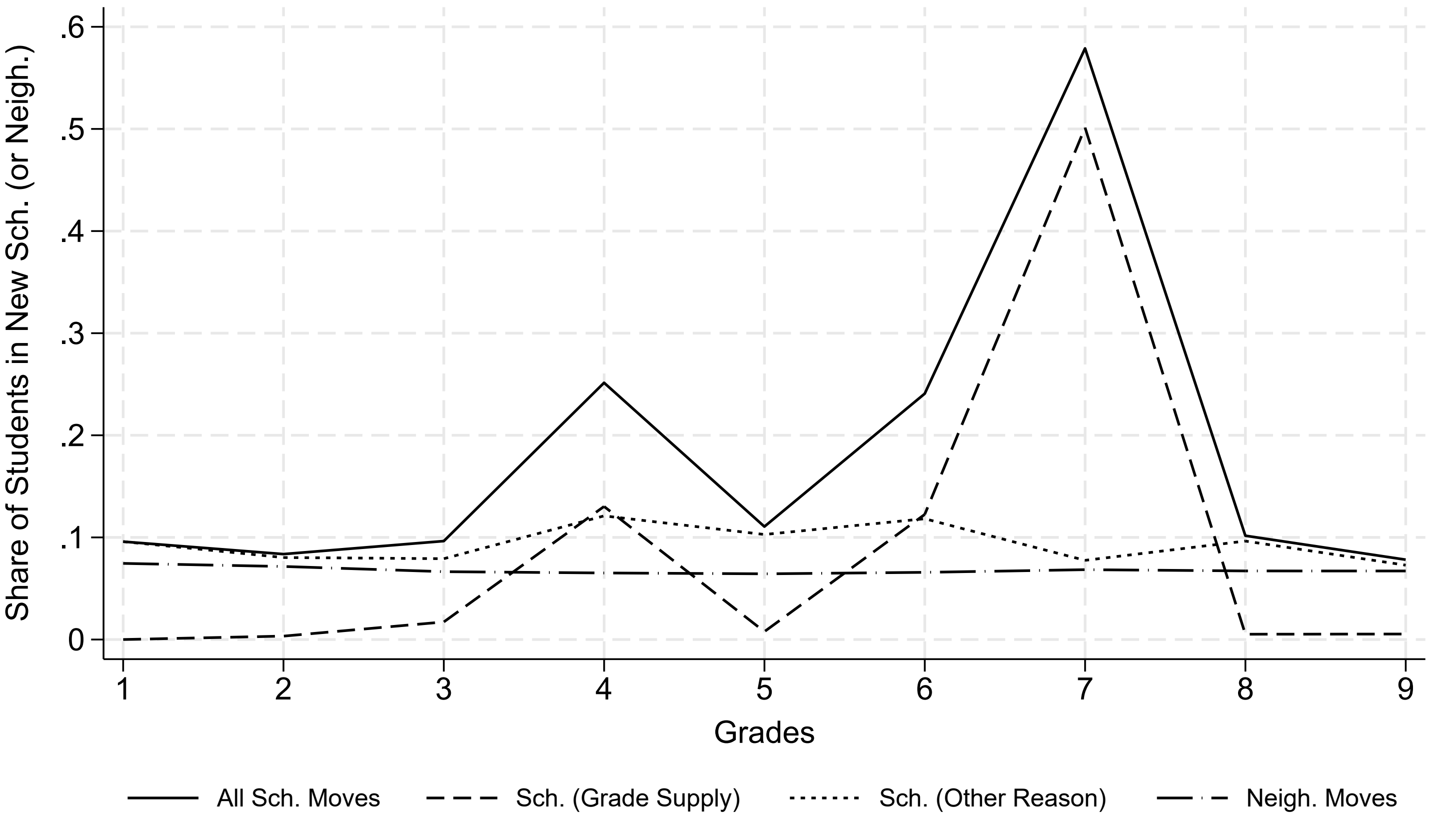


Figure A4: School and Neighborhood Mobility Across Grades (2013-2019)

Note.–Figure shows the share of students who change schools or neighborhoods across grades 1-9, pooled over the years 2013 to 2019. The y-axis indicates the share of students in a new school or neighborhood; the x-axis indicates the grade level. *All Sch. Moves* capture students who change schools for any reason. *Sch. (Grade supply)* captures moves due to the previous school not offering the next grade. *Sch. (Other reason)* captures school changes despite the availability of grades. *Neigh. Moves* show the share of students living in a new neighborhood compared to the previous grade.

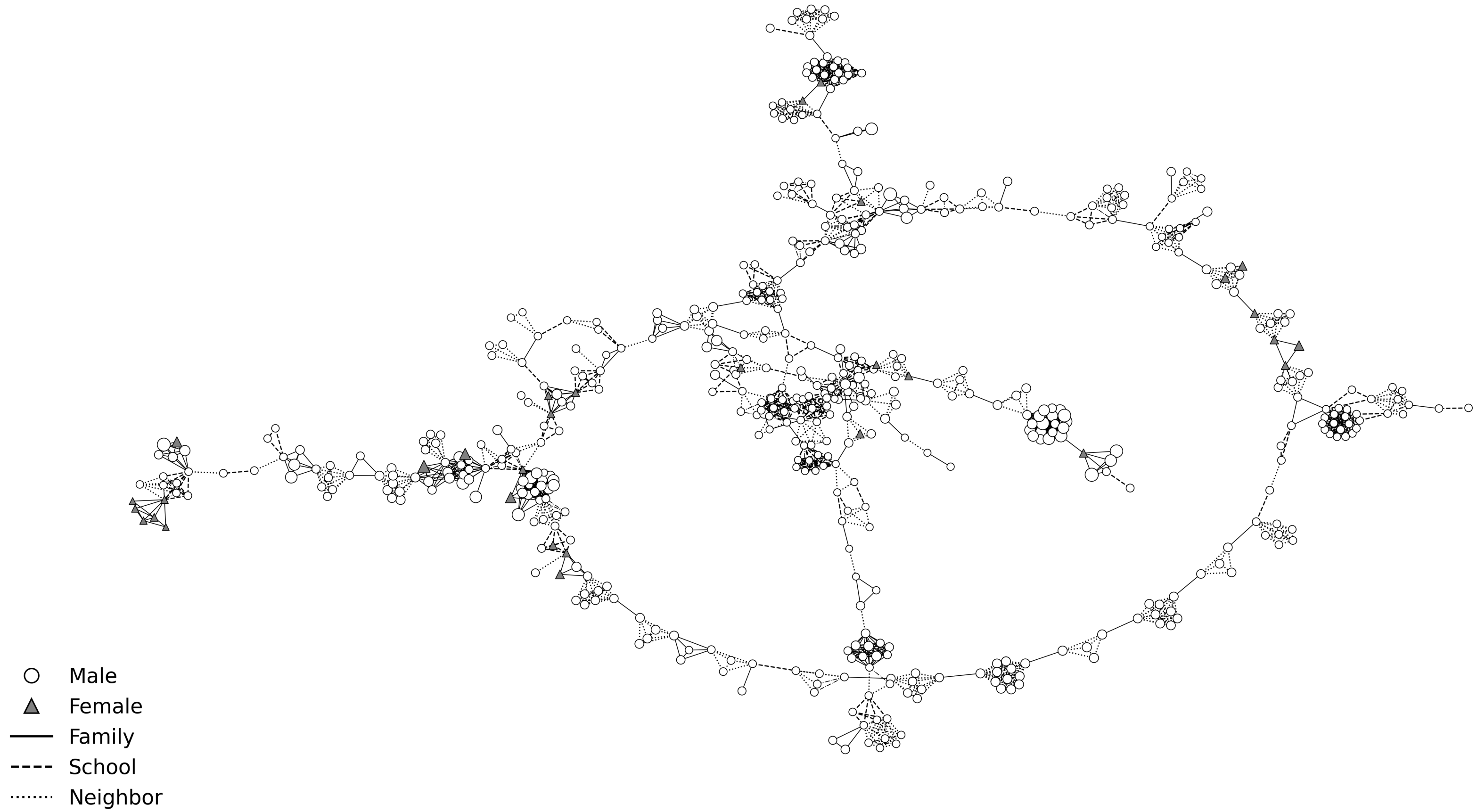
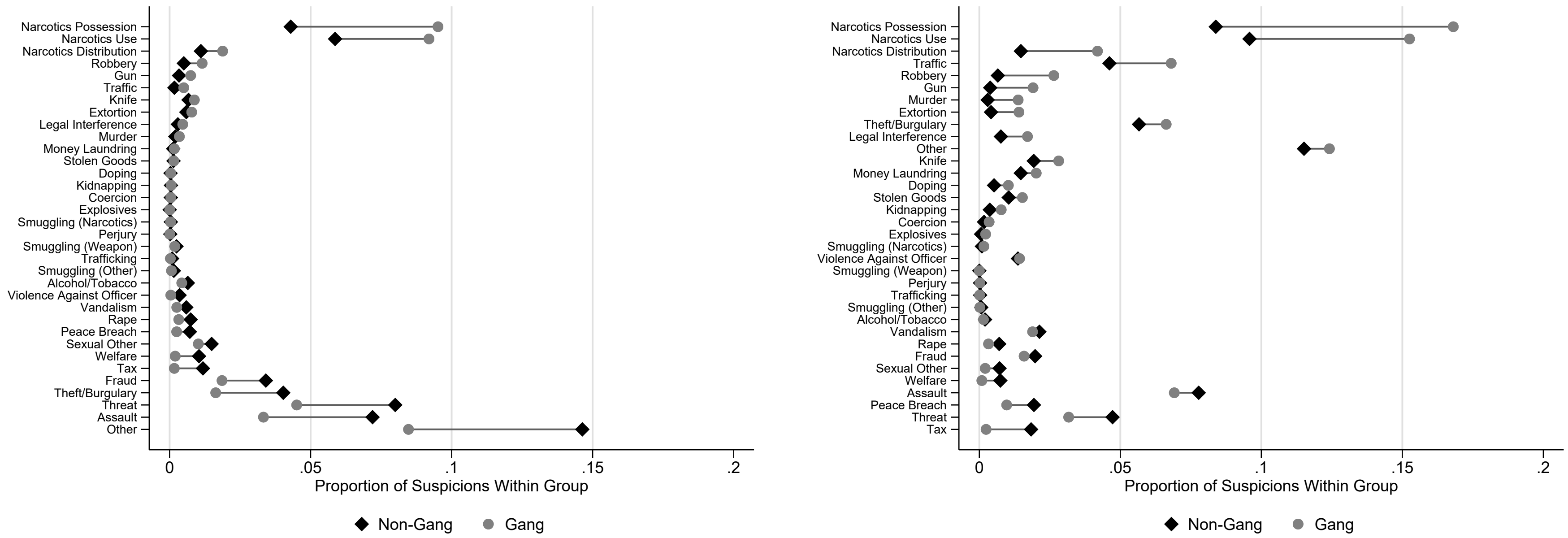


Figure A5: Network Visualization of School, Neighborhood, and Family Connections Among Gang Members

Note.—The figure presents a subgraph of individuals in the gang database. Family links include parents, siblings, uncles, aunts, or cousins. School links are defined by students in the same school-grade-year cell in grade 7; neighborhood links are based on individuals in the same neighborhood-grade-year cell in grade 7. The size of the nodes reflects the year of birth, with the larger nodes representing older individuals.

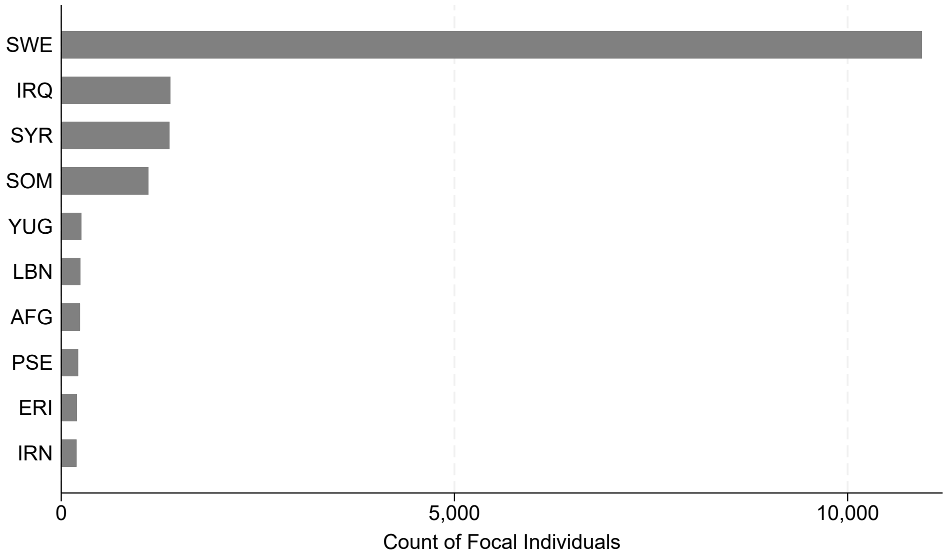


(a) Solo offenses

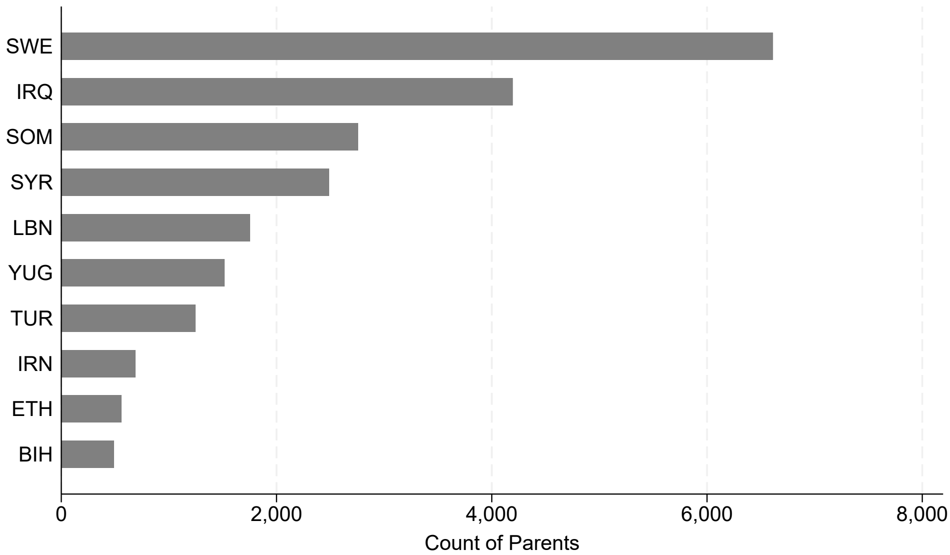
(b) Cooffenses

Figure A6: Prevalence of Crime Types Among Gang and Non-Gang Individuals: Solo vs. Co-Offenses

Note.—The figure plots, by crime category, the share of total suspicions for gang-affiliated and non-gang individuals. Categories are ordered by the difference in prevalence between the two groups (largest first). Subfigure (a) shows solo offenses; Subfigure (b) shows co-offenses (involving two or more suspects).



(a) Gang Members



(b) Parents

Figure A7: Country of Birth for Gang Members and Their Parents

Note.—Panel (a) shows the ten most common countries of birth among individuals listed in the gang database. Panel (b) shows the ten most common countries of birth for the parents (fathers and mothers) of these individuals.

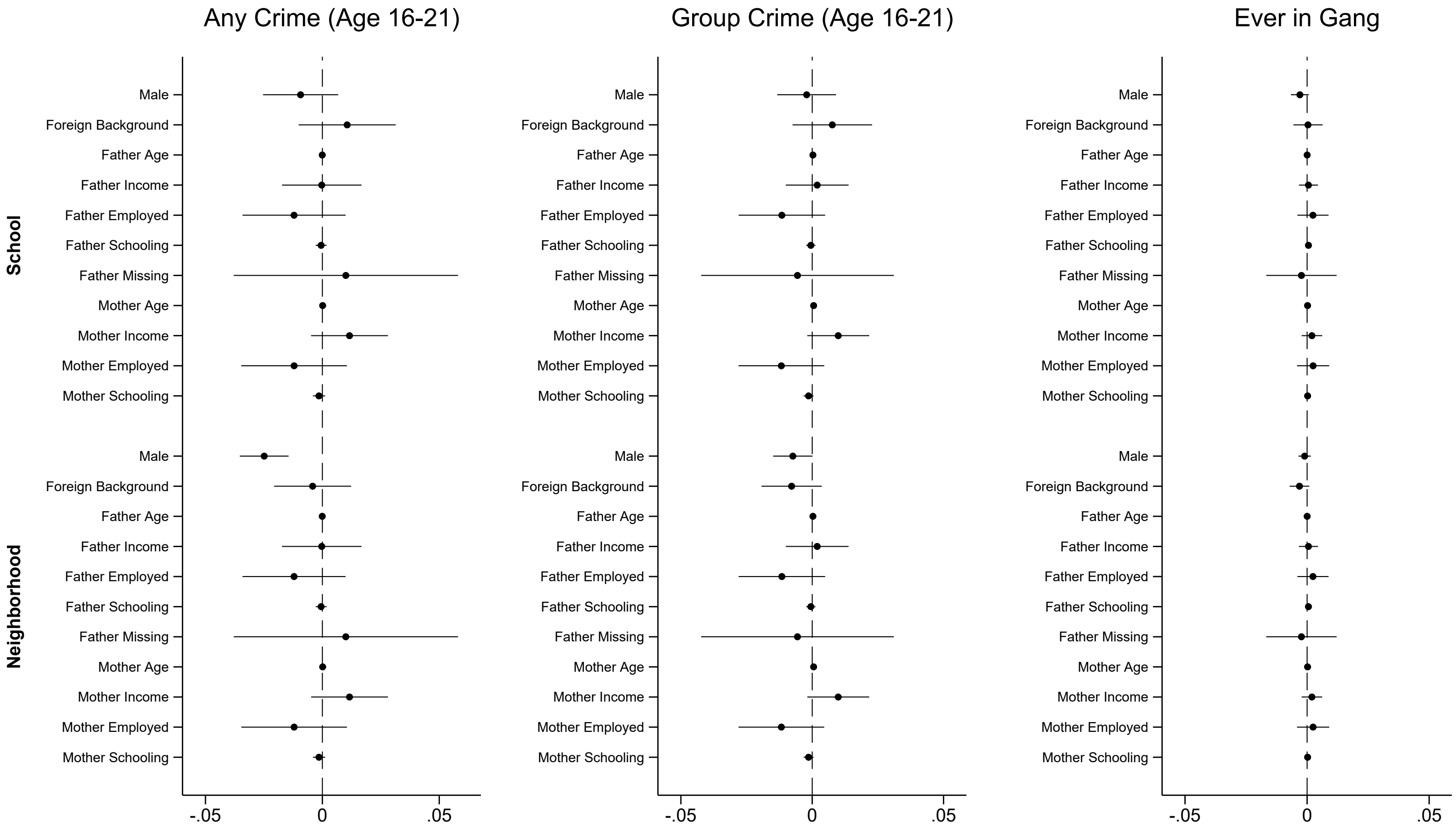


Figure A8: Effects of Alternative Peer Group Characteristics on Criminal Outcomes

Note.—Each coefficient is estimated from a separate regression based on equation (3), including school-neighborhood, year, and birth-year fixed effects. The rows indicate the characteristics of the peers that are being used as an alternative treatment exposure. The top half of the figure report estimates from school-grade exposure, and the bottom half from neighborhood-grade exposure. Standard errors are two-way clustered at the school and neighborhood levels. 95% confidence intervals are reported.

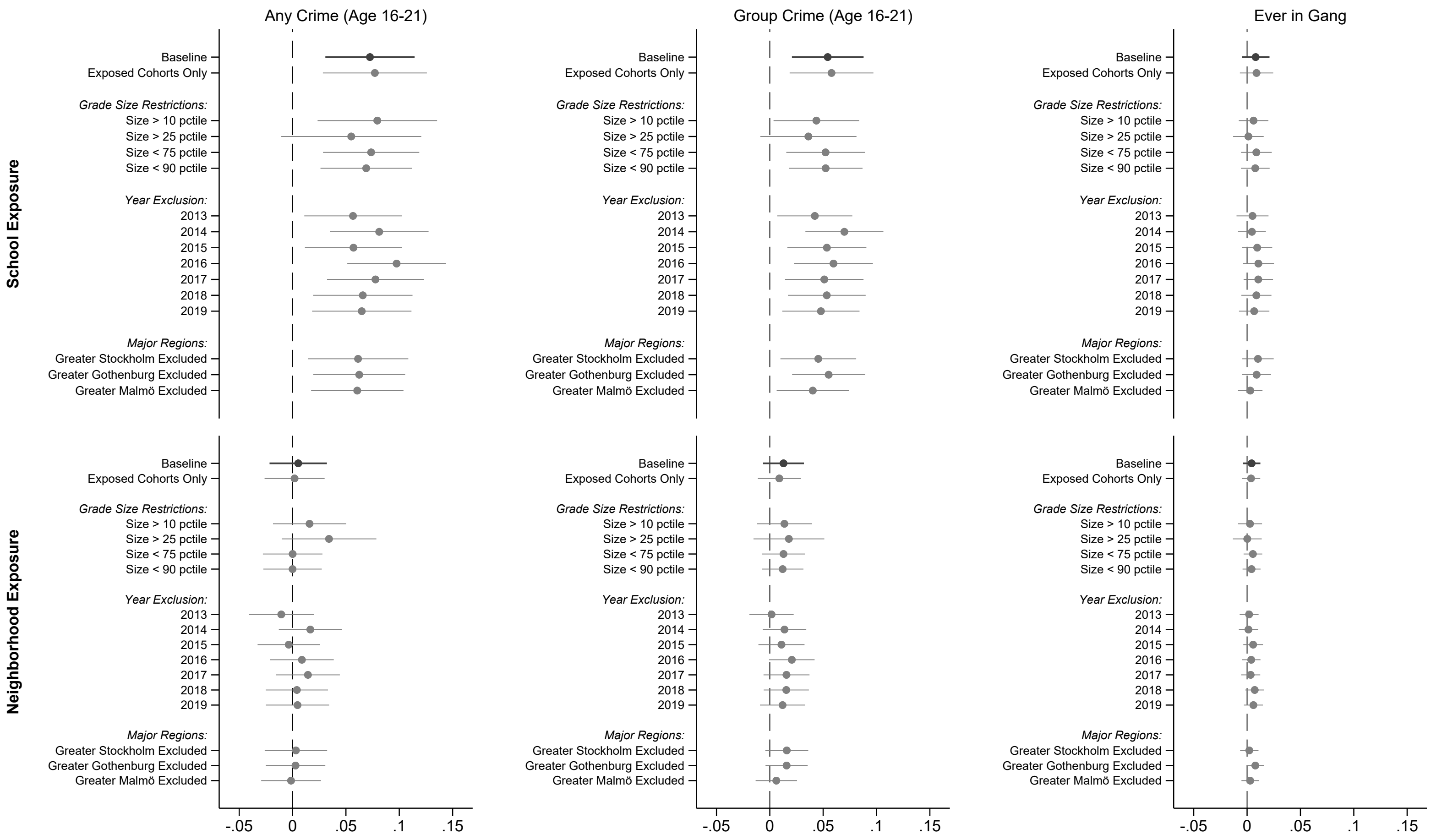


Figure A9: Sample Robustness

Note.—Each coefficient is estimated from a separate regression based on equation (3), including school-neighborhood, year, and birth-year fixed effects, and exclude the gateway peers themselves from the estimation sample.. The baseline rows report the full sample estimates reported in Table 7, each subsequent row imposes a separate sample restriction. The top half of the figure report estimates from school-grade exposure, and the bottom half from neighborhood-grade exposure. Standard errors are two-way clustered at the school and neighborhood levels. 95% confidence intervals are reported.

B Appendix Tables

Table A1: Conviction and Suspicion Type Frequencies For Individuals Listed and Not-Listed in the Gang Database

Panel A: Convictions

	Not in Gang Database						In Gang Database				
	Freq.	Age	Male	Foreign	Uniq. IDs		Freq.	Age	Male	Foreign	Uniq. IDs
Traffic	345,493	39.64	0.84	0.35	237,938	Narcotics	41,177	23.36	0.96	0.70	10,980
Narcotics	307,311	29.61	0.83	0.33	150,124	Traffic	17,830	24.98	0.97	0.71	6,986
Property	204,177	34.63	0.64	0.42	129,033	Other	11,158	23.80	0.97	0.72	6,595
Other	172,506	37.55	0.80	0.35	140,967	Property	10,296	23.84	0.95	0.69	5,366
Violent	72,311	33.68	0.84	0.40	61,192	Weapon	8,707	24.42	0.98	0.67	4,818
Weapon	56,949	33.23	0.90	0.30	37,686	Violent	7,090	22.47	0.97	0.73	4,847
Sex	15,005	35.38	0.99	0.44	14,038	Sex	375	22.81	1.00	0.82	361

Panel B: Suspicions

	Not in Gang Database						In Gang Database				
	Freq.	Age	Male	Foreign	Uniq. IDs		Freq.	Age	Male	Foreign	Uniq. IDs
Other	705,909	38.18	0.78	0.40	370,756	NarcPoss	64,123	23.00	0.96	0.72	11,993
NarcUse	420,632	28.21	0.82	0.32	187,666	NarcUse	60,852	22.72	0.96	0.72	13,197
Assault	408,974	35.76	0.75	0.47	252,395	Other	52,483	23.72	0.96	0.73	10,941
TheftBurg	375,338	33.09	0.62	0.43	151,237	TheftBurg	28,465	23.15	0.95	0.72	7,515
NarcPoss	335,881	29.38	0.82	0.35	138,740	Assault	25,814	22.85	0.96	0.74	9,286
Threat	244,532	38.43	0.78	0.44	148,603	Traffic	17,756	24.86	0.96	0.71	6,723
Traffic	143,925	34.13	0.84	0.35	88,145	Threat	12,303	24.15	0.96	0.72	5,689
Fraud	141,913	34.90	0.68	0.47	72,985	NarcDist	12,150	23.76	0.97	0.72	5,924
Vandalism	98,822	32.84	0.82	0.33	67,071	Knife	11,596	24.35	0.98	0.67	4,923
Knife	80,535	32.49	0.90	0.33	42,073	Fraud	8,518	24.77	0.95	0.69	3,418
PeaceBreach	68,274	37.74	0.78	0.44	54,120	Robbery	7,893	21.54	0.98	0.77	4,493
NarcDist	49,625	31.14	0.82	0.35	34,509	Vandalism	7,307	23.40	0.97	0.68	4,114
Tax	59,906	45.92	0.80	0.41	40,202	Gun	7,262	22.94	0.98	0.72	4,384
MoneyLaun	52,952	32.19	0.69	0.57	33,817	MoneyLaun	6,876	23.49	0.95	0.76	4,088
VioOfficer	55,059	32.33	0.81	0.38	39,702	StolGoods	5,800	25.09	0.97	0.68	3,100

Note.—The figure reports the frequency of convictions and suspicions by crime type, along with the average age, share of males, share with foreign background, and the number of unique individuals for not listed and those listed in the gang database.

Table A2: Summary Statistics for Student Sample Excluding All Four Types of Gateway Peers

	Mean	SD
A. Outcomes: Post or End of Compulsory School		
Crime Activity (Age 16-21)		
Any	0.11	0.31
Group	0.05	0.21
Violent	0.03	0.16
Property	0.04	0.20
Narcotics	0.04	0.19
Weapon	0.01	0.09
Other	0.04	0.20
Ever in Gang Database	0.01	0.08
Eligiblity to High School	0.91	0.29
Merit Score	231.59	62.08
Standardized Test (Grade 9)	0.10	0.81
Mental Health Diagnose (Age 16-21)		
Mood	0.05	0.22
Stress	0.08	0.27
B. Background		
Male	0.51	0.50
Foreign Background	0.20	0.40
Swedish as Second Language	0.12	0.33
Single Parent Home	0.21	0.41
Standardized Test (Grade 6)	0.08	0.82
Father Characteristics		
Age	46.22	5.77
Above Median Labor Income	0.56	0.50
Employed, yr t-1	0.89	0.32
Yrs. of Schooling	12.50	3.28
Dead	0.01	0.09
Missing	0.03	0.18
Mother Characteristics		
Age	43.33	5.05
Above Median Labor Income	0.53	0.50
Employed, yr t-1	0.85	0.35
Yrs. of Schooling	13.08	3.22
Dead	0.00	0.06
Missing	0.01	0.07
C. Peer Groups		
Peers in school-grade	96.08	48.42
Peers in neighborhood-grade	49.86	35.65
Peers in neighborhood-school-grade	22.96	19.40
Peers with Criminally Involved Older Siblings (share)		
School-grade	0.10	0.05
Neighborhood-grade	0.09	0.07
School-neighborhood-grade	0.09	0.12
Peers with Criminally Involved Older Siblings in Group Crimes (share)		
School-grade	0.05	0.04
Neighborhood-grade	0.05	0.05
School-neighborhood-grade	0.04	0.09
Observations	499,639	

Note.—Table reports means and standard deviation for the sample of students excluding gateway peers defined either by parents, uncles, aunts, or cousins with a history of group crime, and gateway peers defined by criminal older siblings.

Table A3: Treatment Variation in Gateway Exposure Defined by Parents, Uncles/Aunts, and Cousins

	Mean	Std. Dev	Min	Max
Raw Parent Exposure Variation (Shares)				
School-Grade Peers (Group Crimes)	0.047	0.036	0.000	1.000
School-Grade Peers (All Crimes)	0.171	0.073	0.000	1.000
Neighborhood-Grade Peers (Group Crimes)	0.047	0.049	0.000	1.000
Neighborhood-Grade Peers (All Crimes)	0.170	0.099	0.000	1.000
Parent Exposure Cohort Variation over Time (Shares)				
School-Grade Peers (Group Crimes)	0.000	0.023	-0.334	0.667
School-Grade Peers (All Crimes)	0.000	0.041	-0.526	0.773
Neighborhood-Grade Peers (Group Crimes)	0.000	0.034	-0.258	0.513
Neighborhood-Grade Peers (All Crimes)	0.000	0.060	-0.500	0.852
	Mean	Std. Dev	Min	Max
Raw Uncle and Aunt Exposure Variation (Shares)				
School-Grade Peers (Group Crimes)	0.081	0.042	0.000	1.000
School-Grade Peers (All Crimes)	0.224	0.070	0.000	1.000
Neighborhood-Grade Peers (Group Crimes)	0.080	0.057	0.000	1.000
Neighborhood-Grade Peers (All Crimes)	0.223	0.092	0.000	1.000
Uncle and Aunt Exposure Cohort Variation over Time (Shares)				
School-Grade Peers (Group Crimes)	0.000	0.029	-0.333	0.916
School-Grade Peers (All Crimes)	0.000	0.045	-0.500	0.709
Neighborhood-Grade Peers (Group Crimes)	0.000	0.044	-0.294	0.910
Neighborhood-Grade Peers (All Crimes)	0.000	0.067	-0.500	0.844
	Mean	Std. Dev	Min	Max
Raw Older Cousin Exposure Variation (Shares)				
School-Grade Peers (Group Crimes)	0.104	0.049	0.000	1.000
School-Grade Peers (All Crimes)	0.212	0.077	0.000	1.000
Neighborhood-Grade Peers (Group Crimes)	0.104	0.066	0.000	1.000
Neighborhood-Grade Peers (All Crimes)	0.211	0.098	0.000	1.000
Older Cousin Exposure Cohort Variation over Time (Shares)				
School-Grade Peers (Group Crimes)	0.000	0.033	-0.295	0.931
School-Grade Peers (All Crimes)	0.000	0.043	-0.502	0.830
Neighborhood-Grade Peers (Group Crimes)	0.000	0.049	-0.344	0.891
Neighborhood-Grade Peers (All Crimes)	0.000	0.065	-0.506	0.845

Note.—The table reports unconditional variation in exposure to parents, uncles/aunts, and cousins with criminal histories across school-grade and neighborhood-grade peer groups, using both group crimes and all crimes. It also reports residual variation after conditioning on year, year of birth, and school-neighborhood fixed effects.

Table A4: Family Correlations in Criminal Outcomes by Relative Type (Any Crime History)

Dependent Variables:	Any Crime (Age 16-21)				Group Crime (Age 16-21)				Ever in Gang Database			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Family Any Crime History Indicators (0/1)												
Parent	0.133**	0.119**	0.109**	0.107**	0.088**	0.078**	0.071**	0.070**	0.018**	0.016**	0.014**	0.014**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)
Sibling		0.096**	0.091**	0.089**		0.067**	0.064**	0.063**		0.017**	0.015**	0.015**
		(0.002)	(0.002)	(0.002)		(0.001)	(0.001)	(0.001)		(0.001)	(0.001)	(0.001)
Uncle/Aunt			0.050**	0.047**			0.030**	0.028**			0.004**	0.004**
			(0.001)	(0.001)			(0.001)	(0.001)			(0.000)	(0.000)
Cousin				0.029**				0.017**				0.002**
				(0.001)				(0.001)				(0.000)
Sum of Coefficients	0.133	0.195	0.231	0.257	0.088	0.131	0.152	0.167	0.018	0.029	0.031	0.033
	(0.001)	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)	(0.002)	(0.002)	(0.000)	(0.001)	(0.001)	(0.001)
Adj. R^2	0.025	0.033	0.037	0.039	0.022	0.029	0.034	0.035	0.007	0.010	0.015	0.016
Observations	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194
Dep. mean	0.133	0.133	0.133	0.133	0.062	0.062	0.062	0.062	0.009	0.009	0.009	0.009

Note.—The table reports raw correlations between the indicators of family involvement in any crime before the entry of the focal student into grade 7 and the students own criminal activity between ages 16 and 21. Estimates are based on variants of equation (2). Columns 1, 5, and 9 present baseline regressions including only parental criminal history; subsequent columns sequentially add indicators for siblings, uncles, aunts, and cousins. As each relative type is introduced, the specification also includes a dummy indicating whether that relative type is missing for the focal student. The *Sum of Coefficients* is constructed following the Lubotsky-Wittenberg method for proxy variables (Lubotsky and Wittenberg 2006). Robust standard errors are reported in parentheses; standard errors for the sum of coefficients are bootstrapped with 1,000 replications.
(**) $p < .01$, (*) $p < .05$, (+) $p < .1$.

Table A5: Family Correlations in Criminal Outcomes by Relative Type (Separate)

Dependent Variables:	Any Crime (Age 16-21)				Group Crime (Age 16-21)				Ever in Gang Database			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A. Family Group Crime History Indicators (0/1)												
Parent	0.193**				0.139**				0.030**			
	(0.003)				(0.002)				(0.001)			
Sibling		0.143**				0.106**				0.029**		
		(0.003)				(0.002)				(0.001)		
Uncle/Aunt			0.100**				0.068**				0.013**	
			(0.002)				(0.001)				(0.001)	
Cousin				0.063**				0.040**				0.006**
				(0.002)				(0.001)				(0.000)
Adj. R^2	0.018	0.011	0.011	0.006	0.018	0.011	0.012	0.006	0.006	0.005	0.009	0.004
Panel B. Family Any Crime History Indicators (0/1)												
Parent	0.133**				0.088**				0.018**			
	(0.001)				(0.001)				(0.000)			
Sibling		0.121**				0.084**				0.020**		
		(0.002)				(0.001)				(0.001)		
Uncle/Aunt			0.070**				0.043**				0.007**	
			(0.001)				(0.001)				(0.000)	
Cousin				0.052**				0.032**				0.004**
				(0.001)				(0.001)				(0.000)
Adj. R^2	0.025	0.012	0.011	0.006	0.022	0.012	0.011	0.006	0.007	0.005	0.009	0.004
Observations	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194	675,194
Dep. mean	0.133	0.133	0.133	0.133	0.062	0.062	0.062	0.062	0.009	0.009	0.009	0.009

Note.—The table reports raw correlations between indicators of relatives crime involvement prior to the entry of the focal student into grade 7 and the students own criminal activity between ages 16 and 21. *Panel A.* considers relatives with group crime history, and *Panel B.* relative with any history of criminal activity. Estimates are based on equation (2), where indicators for the type of criminal relative are included separately in each subsequent column. Robust standard errors are shown in parentheses. (**) $p < .01$, (*) $p < .05$, ($^+$) $p < .1$.

Table A6: Exogeneity of Gateway Peer Exposure Defined by Older Siblings

Sibling Exposure Considering:	Group Crimes			
In Peer Group:	School		Neighborhood	
	(1)	(2)	(3)	(4)
Male	0.000	0.000	-0.000^{+}	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Foreign Background	0.015**	0.000^{+}	0.024**	-0.000
	(0.001)	(0.000)	(0.002)	(0.000)
Single Parent Home	0.002**	-0.000	0.005**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Fathers Age	-0.000	0.000	0.000*	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Fathers Income $\geq$ p50	-0.005**	0.000	-0.007**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Father Employed	0.002**	-0.000	0.001^{+}	0.000
	(0.000)	(0.000)	(0.001)	(0.000)
Fathers Yrs. of Sch.	-0.000**	-0.000	-0.000**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Father Missing	-0.006**	0.000	-0.006**	0.001
	(0.001)	(0.000)	(0.001)	(0.001)
Mothers Age	-0.000**	-0.000	-0.000**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Mothers Income $\geq$ p50	-0.004**	0.000	-0.005**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Mother Employed	-0.001	-0.000	-0.002**	-0.000
	(0.000)	(0.000)	(0.001)	(0.000)
Mothers Yrs. of Sch.	-0.001**	-0.000	-0.001**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Mother Missing	-0.011**	0.000	-0.019**	-0.002^{+}
	(0.002)	(0.001)	(0.002)	(0.001)
Year Fixed		✓		✓
Year-of-Birth Fixed		✓		✓
School-Neighborhood Fixed		✓		✓
F-Stat	40.108	0.971	60.824	0.876
p(F)	0.000	0.481	0.000	0.585
Adj. R^2	0.077	0.587	0.095	0.509
Dependent Mean	0.047	0.047	0.046	0.046
Observations	499,639	499,639	499,639	499,639

Note.—Each column reports a separate regression of the listed dependent variable on a common set of control variables, estimated both with and without calendar-year, birth-year, and school-neighborhood fixed effects. The reported F-stat is the joint F-statistic for the included controls, and p(F) gives the corresponding p-value. Columns 1-4 use exposure to gateway peers with older siblings who have a history of group crime as the dependent variable. Columns 1 and 2 measure exposure in the school-grade group, while columns 3 and 4 use neighborhood-grade exposure. Gateway peers defined by parents, uncles, aunts, or cousins with a history of group crime, or gateway peers defined by older criminal siblings, are excluded from the sample. Standard errors are two-way clustered at the school and neighborhood levels.
(**) $p < .01$, (*) $p < .05$, ($^{+}$) $p < .1$.

Table A7: Exogeneity of Gateway Peer Exposure Defined by Parents

Parent Exposure Considering:	Group Crimes			
In Peer Group:	School		Neighborhood	
	(1)	(2)	(3)	(4)
Male	-0.000	-0.000	-0.000^{+}	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Foreign Background	0.017**	0.000**	0.027**	0.000
	(0.001)	(0.000)	(0.001)	(0.000)
Single Parent Home	0.003**	0.000	0.007**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Fathers Age	0.000	-0.000	0.000*	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Fathers Income $\geq$ p50	-0.003**	0.000	-0.005**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Father Employed	0.001*	-0.000	0.000	0.000
	(0.000)	(0.000)	(0.001)	(0.000)
Fathers Yrs. of Sch.	-0.000**	-0.000	-0.000**	-0.000^{+}
	(0.000)	(0.000)	(0.000)	(0.000)
Father Missing	-0.005**	-0.000	-0.007**	0.000
	(0.001)	(0.000)	(0.001)	(0.001)
Mothers Age	-0.000**	0.000	-0.000**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Mothers Income $\geq$ p50	-0.002**	-0.000*	-0.004**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Mother Employed	-0.001*	0.000	-0.002**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Mothers Yrs. of Sch.	-0.000**	-0.000	-0.000**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Mother Missing	-0.012**	-0.000	-0.014**	0.003**
	(0.001)	(0.001)	(0.002)	(0.001)
Year Fixed		✓		✓
Year-of-Birth Fixed		✓		✓
School-Neighborhood Fixed		✓		✓
F-Stat	36.667	1.993	72.694	1.660
p(F)	0.000	0.015	0.000	0.057
Adj. R^2	0.075	0.577	0.100	0.514
Dependent Mean	0.046	0.046	0.045	0.045
Observations	499,639	499,639	499,639	499,639

Note.—Each column reports a separate regression of the listed dependent variable on a common set of control variables, estimated both with and without calendar-year, birth-year, and school-neighborhood fixed effects. The reported F-stat is the joint F-statistic for the included controls, and p(F) gives the corresponding p-value. Columns 1-4 use exposure to gateway peers with older siblings who have a history of group crime as the dependent variable. Columns 1 and 2 measure exposure in the school-grade group, while columns 3 and 4 use neighborhood-grade exposure. Gateway peers defined by parents, uncles, aunts, or cousins with a history of group crime, or gateway peers defined by older criminal siblings, are excluded from the sample. Standard errors are two-way clustered at the school and neighborhood levels.
(**) $p < .01$, (*) $p < .05$, ($^{+}$) $p < .1$.

Table A8: Exogeneity of Gateway Peer Exposure Defined by Uncles and Aunts

Uncle and Aunt Exposure Considering:	Group Crimes			
In Peer Group:	School		Neighborhood	
	(1)	(2)	(3)	(4)
Male	0.000**	0.000	-0.000+	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Foreign Background	0.003**	0.000	0.005**	-0.000
	(0.001)	(0.000)	(0.001)	(0.000)
Single Parent Home	0.002**	-0.000	0.004**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Fathers Age	-0.000**	0.000	-0.000**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Fathers Income $\geq$ p50	-0.004**	0.000	-0.006**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Father Employed	0.005**	0.000	0.005**	0.000
	(0.000)	(0.000)	(0.001)	(0.000)
Fathers Yrs. of Sch.	-0.001**	-0.000	-0.001**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Father Missing	-0.008**	0.001+	-0.010**	-0.000
	(0.001)	(0.001)	(0.001)	(0.001)
Mothers Age	-0.000**	0.000	-0.000**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Mothers Income $\geq$ p50	-0.003**	-0.000	-0.004**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Mother Employed	0.002**	0.000	0.003**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Mothers Yrs. of Sch.	-0.000**	0.000	-0.000**	0.000*
	(0.000)	(0.000)	(0.000)	(0.000)
Mother Missing	-0.017**	0.001	-0.019**	0.002*
	(0.002)	(0.001)	(0.002)	(0.001)
Year Fixed		✓		✓
Year-of-Birth Fixed		✓		✓
School-Neighborhood Fixed		✓		✓
F-Stat	25.134	1.012	30.482	0.896
p(F)	0.000	0.438	0.000	0.563
Adj. R^2	0.021	0.483	0.016	0.384
Dependent Mean	0.079	0.079	0.079	0.079
Observations	499,639	499,639	499,639	499,639

Note.—Each column reports a separate regression of the listed dependent variable on a common set of control variables, estimated both with and without calendar-year, birth-year, and school-neighborhood fixed effects. The reported F-stat is the joint F-statistic for the included controls, and p(F) gives the corresponding p-value. Columns 1-4 use exposure to gateway peers with older siblings who have a history of group crime as the dependent variable. Columns 1 and 2 measure exposure in the school-grade group, while columns 3 and 4 use neighborhood-grade exposure. Gateway peers defined by parents, uncles, aunts, or cousins with a history of group crime, or gateway peers defined by older criminal siblings, are excluded from the sample. Standard errors are two-way clustered at the school and neighborhood levels.
(**) $p < .01$, (*) $p < .05$, ($^{+}$) $p < .1$.

Table A9: Exogeneity of Gateway Peer Exposure Defined by Older Cousins

Cousin Exposure Considering:	Group Crimes			
In Peer Group:	School		Neighborhood	
	(1)	(2)	(3)	(4)
Male	0.000+	-0.000	-0.000	-0.000*
	(0.000)	(0.000)	(0.000)	(0.000)
Foreign Background	-0.018**	-0.000+	-0.021**	0.000
	(0.001)	(0.000)	(0.001)	(0.000)
Single Parent Home	-0.000	0.000	-0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Fathers Age	-0.000**	0.000	-0.000**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Fathers Income $\geq$ p50	-0.007**	-0.000	-0.007**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Father Employed	0.006**	0.001**	0.007**	0.000
	(0.001)	(0.000)	(0.001)	(0.000)
Fathers Yrs. of Sch.	-0.001**	-0.000+	-0.001**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Father Missing	-0.006**	-0.001	-0.006**	-0.001
	(0.001)	(0.001)	(0.002)	(0.001)
Mothers Age	-0.000**	0.000	-0.000**	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Mothers Income $\geq$ p50	-0.007**	-0.000+	-0.008**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Mother Employed	0.005**	-0.000	0.006**	0.001+
	(0.000)	(0.000)	(0.001)	(0.000)
Mothers Yrs. of Sch.	-0.000**	0.000	-0.000**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Mother Missing	-0.016**	0.001	-0.016**	-0.000
	(0.002)	(0.001)	(0.002)	(0.001)
Year Fixed		✓		✓
Year-of-Birth Fixed		✓		✓
School-Neighborhood Fixed		✓		✓
F-Stat	57.330	1.762	62.173	1.227
p(F)	0.000	0.039	0.000	0.248
Adj. R^2	0.029	0.527	0.021	0.414
Dependent Mean	0.104	0.104	0.103	0.103
Observations	499,639	499,639	499,639	499,639

Note.—Each column reports a separate regression of the listed dependent variable on a common set of control variables, estimated both with and without calendar-year, birth-year, and school-neighborhood fixed effects. The reported F-stat is the joint F-statistic for the included controls, and p(F) gives the corresponding p-value. Columns 1-4 use exposure to gateway peers with older siblings who have a history of group crime as the dependent variable. Columns 1 and 2 measure exposure in the school-grade group, while columns 3 and 4 use neighborhood-grade exposure. Gateway peers defined by parents, uncles, aunts, or cousins with a history of group crime, or gateway peers defined by older criminal siblings, are excluded from the sample. Standard errors are two-way clustered at the school and neighborhood levels.
(**) $p < .01$, (*) $p < .05$, ($^{+}$) $p < .1$.

Table A10: Testing Within-School and Neighborhood Cohort Variation in Peer Composition

	School			Neighborhood		
	Group Crime (1)	Males (2)	Foreign Background (3)	Group Crime (4)	Males (5)	Foreign Background (6)
Rejection Frequency at the 5% level.						
χ^2	51	54	83	117	62	137
	[0.026]	[0.029]	[0.044]	[0.037]	[0.020]	[0.044]
$Kolmogorov - Smirnov$	10	5	19	20	6	18
	[0.005]	[0.003]	[0.010]	[0.006]	[0.002]	[0.006]
Count of Schools or Neighborhoods	1892	1892	1892	3129	3129	3129

Note.—The table reports the number of schools or neighborhoods rejected by the χ^2 or Kolmogorov-Smirnov tests, which assess whether variation in peer composition across cohorts follows a distribution consistent with random assignment. The brackets show the proportion of units rejected at the 5% significance level. The tests are conducted on cohort-level characteristics including the share of gateway peers (defined by older siblings involved in group crime), the share of male students, and the share of students with a foreign background.

Table A11: Grade 6 Exposure to Gateway Peers and Grade 7 School and Neighborhood Mobility

	School Moves All Students		School Moves Grade 7 Available		Neighborhood Moves All Students	
	(1)	(2)	(3)	(4)	(5)	(6)
School (Group Crime)	0.013		0.008			
	(0.027)		(0.067)			
School (Any Crime)		-0.004		0.009		
		(0.019)		(0.052)		
Neighborhood (Group Crime)					0.008	
					(0.010)	
Neighborhood (Any Crime)						-0.000
						(0.007)
Observations	520,735	520,735	253,663	253,663	520,735	520,735
Dep. Mean	0.572	0.572	0.138	0.138	0.049	0.049

Note.—Each column reports estimates based on equation (3) where the exposure variable is defined by the gateway peer exposure defined by older siblings with a history of group crime or any criminal history in grade 6. All regressions include birth-year, calendar-year, and school-neighborhood fixed effects. Columns 1-4 report the effect of school exposure on the focal student changing school in grade 7, on either all student (columns 1 and 2) or only students where their grade 6 school did also provide grade 7 (columns 3 and 4). Columns 5 and 6 estimate the effect of neighborhood exposure on the focal student moving to a new neighborhood in grade 7. Standard errors are two-way clustered at the school and neighborhood levels.
(**) $p < .01$, (*) $p < .05$, ($^{+}$) $p < .1$.

Table A12: Exposure to Gateway Peers by Type of Older Criminal Relative With Controls

Peer Group:	School-Grade			Neighborhood-Grade		
	(1) Any Crime (Age 16-21)	(2) Group Crime (Age 16-21)	(3) Ever In Gang	(4) Any Crime (Age 16-21)	(5) Group Crime (Age 16-21)	(6) Ever In Gang
Gateway Exposure by Family Type, Defined by History of Group Crime.						
Parent	-0.005	-0.007	0.002	0.016	0.009	0.005
	(0.023)	(0.017)	(0.007)	(0.015)	(0.011)	(0.004)
Sibling	0.072**	0.057**	0.006	0.021	0.017^{+}	0.003
	(0.023)	(0.018)	(0.007)	(0.015)	(0.010)	(0.004)
Uncle/Aunt	0.017	-0.004	-0.001	0.002	0.006	-0.002
	(0.016)	(0.011)	(0.004)	(0.011)	(0.007)	(0.002)
Cousin	0.006	-0.002	0.002	0.004	-0.002	-0.003
	(0.014)	(0.010)	(0.003)	(0.010)	(0.007)	(0.002)
Focal Controls	✓	✓	✓	✓	✓	✓
Peer Controls	✓	✓	✓	✓	✓	✓
Observations	499,639	499,639	499,639	499,639	499,639	499,639
Dep. Mean	0.108	0.046	0.006	0.108	0.046	0.006

Note.—Each column reports estimates from a separate regression based on equation (3), where exposure to gateway peers is defined based on the group crime history of older relatives. Columns 1-3 use school-grade exposure, and columns 4-6 use neighborhood-grade exposure. Equation (3) is here extended to include one exposure variable for each relative type simultaneously − that is, exposure to gateway peers defined by: (i) parents with a history of group crime, (ii) older siblings with group crime history, (iii) uncles and aunts with group crime history, and (iv) older cousins with group crime history. All regressions include birth-year, calendar-year, and school-neighborhood fixed effects. Focal controls include sex, foreign background, single parent household status, and parental income, employment, education, and missing value indicators. Peer controls reflect the share of male and foreign background students in the relevant school- or neighborhood-grade group. Standard errors are two-way clustered at the school and neighborhood levels. (**) $p < .01$, (*) $p < .05$, ($^{+}$) $p < .1$

Table A13: Older Cohorts

	Any Crime (Age 16-21)		Group Crime (Age 16-21)		Ever In Gang	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Gateway Exposure by Siblings, Defined by History of Group Crimes						
School Exposure:						
Grade 7 (Focal's Cohort)	0.075**	0.072**	0.051**	0.049**	0.007	0.006
	(0.022)	(0.021)	(0.017)	(0.016)	(0.006)	(0.006)
Grade 8 (1 Year Older Cohort)	0.005	0.003	0.016	0.015	0.007	0.007
	(0.018)	(0.018)	(0.014)	(0.013)	(0.005)	(0.005)
Grade 9 (2 Year Older Cohort)	0.042*	0.043**	0.011	0.011	0.002	0.002
	(0.017)	(0.016)	(0.012)	(0.012)	(0.005)	(0.005)
Panel B. Gateway Exposure by Siblings, Defined by History of Group Crimes						
Neighborhood Exposure:						
Grade 7 (Focal's Cohort)	0.005	0.006	0.012	0.013	0.003	0.003
	(0.014)	(0.014)	(0.010)	(0.010)	(0.004)	(0.004)
Grade 8 (1 Year Older Cohort)	0.006	0.007	0.011+	0.011+	0.002	0.002
	(0.008)	(0.008)	(0.006)	(0.006)	(0.002)	(0.002)
Grade 9 (2 Year Older Cohort)	0.026**	0.025**	0.021**	0.020**	0.002	0.002
	(0.008)	(0.008)	(0.005)	(0.005)	(0.002)	(0.002)
Focal Controls		✓		✓		✓
Peer Group Controls		✓		✓		✓
Observations	577,863	577,863	577,863	577,863	577,863	577,863
Dep. Mean	0.117	0.117	0.052	0.052	0.006	0.006

Note.—Each column-panel combination reports estimates from a separate regression based on equation (3), now extended to include the share of gateway peers in the two older cohorts, students in grades 8 and 9 when the focal student is in grade 7. Exposure is defined as the share of gateway peers with older siblings who have a history of group crime, measured in the school context (panel A) or in the neighborhood context (panel B). All regressions include birth-year, calender-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. Standard errors are two-way clustered at the school and neighborhood levels. Columns 2, 4, and 6 additionally control for focal and peer characteristics. Focal controls include sex, foreign background, single parent household status, and parental income, employment, education, and missing value indicators. Peer controls reflect the share of male and foreign background students in the relevant school- or neighborhood-grade group.
(**) $p < .01$, (*) $p < .05$, (+) $p < .1$.

Table A14: Effects of Sibling-Based Gateway Peer School Exposure by Type of Crime Exposure and Outcome

	Violent (Age 16-21)		Property (Age 16-21)		Narcotics (Age 16-21)		Weapon (Age 16-21)		Other (Age 16-21)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A. School Gateway Exposure by Sibling										
Group Crimes	0.027*	0.026*	0.033*	0.033*	0.046**	0.046**	0.015*	0.015*	0.013	0.013
	(0.012)	(0.012)	(0.015)	(0.015)	(0.014)	(0.013)	(0.007)	(0.007)	(0.013)	(0.013)
Panel B. School Gateway Exposure by Sibling										
Narcotics Crimes	0.012	0.011	0.006	0.006	0.039*	0.040**	0.013	0.013	0.015	0.016
	(0.013)	(0.013)	(0.017)	(0.017)	(0.015)	(0.015)	(0.008)	(0.008)	(0.015)	(0.015)
Panel C. School Gateway Exposure by Sibling										
Violent Crimes	0.034**	0.034**	0.039*	0.039*	0.043**	0.045**	0.017*	0.018*	0.024	0.026+
	(0.013)	(0.013)	(0.017)	(0.017)	(0.015)	(0.015)	(0.007)	(0.007)	(0.015)	(0.015)
Panel D. School Gateway Exposure by Sibling										
Property Crimes	0.047**	0.046**	0.034*	0.033*	0.049**	0.048**	0.014*	0.014*	0.018	0.017
	(0.011)	(0.011)	(0.015)	(0.014)	(0.014)	(0.013)	(0.007)	(0.007)	(0.013)	(0.012)
Focal Controls		✓		✓		✓		✓		✓
Peer Group Controls		✓		✓		✓		✓		✓
Obs.	609,459	609,459	609,459	609,459	609,459	609,459	609,459	609,459	609,459	609,459
Dep. Mean	0.031	0.031	0.047	0.047	0.044	0.044	0.010	0.010	0.045	0.045

Note.—Each cell reports estimates from a separate regression based on equation (3), where school-grade gateway peer exposure is defined by the criminal history of older siblings. *Panel A* uses the main proxy for gang-related activity, group crimes. *Panels B-D* define exposure based on older sibling participation in narcotic offenses, violent crimes, and property crimes, respectively. The dependent variable is an indicator for post-compulsory school criminal charges, disaggregated into five categories: violent, property, narcotics, weapon, and other crimes. All specifications include birth-year, calendar-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. Columns 2, 4, 6, 8, and 10 additionally control for focal and peer group characteristics. Focal controls include sex, foreign background, single parent household status, and parental income, employment, and education, with dummies for missing values. Peer controls capture the share of male and foreign background students in the relevant peer group. Standard errors are two-way clustered at the school and neighborhood levels. (**) $p < .01$, (*) $p < .05$, (+) $p < .1$.

Table A15: Effects of Sibling-Based Gateway Peer Neighborhood Exposure by Type of Crime Exposure and Outcome

	Violent (Age 16-21)		Property (Age 16-21)		Narcotics (Age 16-21)		Weapon (Age 16-21)		Other (Age 16-21)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A. Neighborhood Gateway Exposure by Sibling										
Group Crimes	0.007	0.008	0.010	0.011	0.010	0.011	0.006	0.006	0.002	0.002
	(0.008)	(0.008)	(0.010)	(0.010)	(0.008)	(0.008)	(0.005)	(0.005)	(0.009)	(0.009)
Panel B. Neighborhood Gateway Exposure by Sibling										
Narcotics Crimes	-0.009	-0.008	-0.007	-0.006	-0.007	-0.006	-0.000	0.000	-0.003	-0.003
	(0.009)	(0.009)	(0.010)	(0.010)	(0.010)	(0.010)	(0.005)	(0.005)	(0.010)	(0.010)
Panel C. Neighborhood Gateway Exposure by Sibling										
Violent Crimes	0.009	0.009	0.006	0.007	0.010	0.010	0.008^{+}	0.008^{+}	0.000	0.001
	(0.009)	(0.009)	(0.010)	(0.010)	(0.010)	(0.010)	(0.005)	(0.005)	(0.009)	(0.009)
Panel D. Neighborhood Gateway Exposure by Sibling										
Property Crimes	0.022^{**}	0.023^{**}	0.027^{**}	0.028^{**}	0.013	0.015^{+}	0.009^{*}	0.010^{*}	0.013	0.015^{+}
	(0.008)	(0.008)	(0.009)	(0.009)	(0.008)	(0.008)	(0.004)	(0.004)	(0.009)	(0.008)
Focal Controls		✓		✓		✓		✓		✓
Peer Group Controls		✓		✓		✓		✓		✓
Obs.	609,459	609,459	609,459	609,459	609,459	609,459	609,459	609,459	609,459	609,459
Dep. Mean	0.031	0.031	0.047	0.047	0.044	0.044	0.010	0.010	0.045	0.045

Note.—Each cell reports estimates from a separate regression based on equation (3), where neighborhood-grade gateway peer exposure is defined by the criminal history of older siblings. *Panel A* uses the main proxy for gang-related activity, group crimes. *Panels B-D* define exposure based on older sibling participation in narcotic offenses, violent crimes, and property crimes, respectively. The dependent variable is an indicator for post-compulsory school criminal charges, disaggregated into five categories: violent, property, narcotics, weapon, and other crimes. All specifications include birth-year, calendar-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. Columns 2, 4, 6, 8, and 10 additionally control for focal and peer group characteristics. Focal controls include sex, foreign background, single parent household status, and parental income, employment, and education, with dummies for missing values. Peer controls capture the share of male and foreign background students in the relevant peer group. Standard errors are two-way clustered at the school and neighborhood levels. (**) $p < .01$, (*) $p < .05$, ($^{+}$) $p < .1$.

Table A16: Gateway Peer Exposure by Sibling Group Crime: Binary Treatment Indicators

Peer Group:	School			Neighborhood		
	(1) Any Crime (Age 16-21)	(2) Group Crime (Age 16-21)	(3) Ever In Gang	(4) Any Crime (Age 16-21)	(5) Group Crime (Age 16-21)	(6) Ever In Gang
Panel A. Gateway Exposure by Siblings, Defined by History of Group Crimes						
Any Exposure (0/1)	0.002	-0.000	0.000	0.000	0.000	0.000
	(0.002)	(0.001)	(0.000)	(0.001)	(0.001)	(0.000)
Panel B. Gateway Exposure by Siblings, Defined by History of Group Crimes (Percentile Groups)						
50th - 74th Pcts (0/1)	-0.001	0.000	-0.000	-0.001	-0.000	0.000^{+}
	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.000)
75th - 89th Pcts (0/1)	0.002	0.001	-0.000	-0.000	-0.000	-0.000
	(0.002)	(0.001)	(0.000)	(0.002)	(0.001)	(0.000)
90th- Pcts (0/1)	0.008^{**}	0.006^{**}	0.000	0.001	0.002	-0.001
	(0.003)	(0.002)	(0.001)	(0.002)	(0.002)	(0.001)
Observations	609,459	609,459	609,459	609,459	609,459	609,459
Dep. Mean	0.120	0.054	0.007	0.120	0.054	0.007

Note.—Each column-panel combination reports estimates from a separate regression based on equation (3) where gateway exposure is defined by older siblings with group crime history. Exposure has here been dichotomized to no exposure (0) or any exposure (1) *Panel A.* or binarized into three exposure variables included simultaneously, 50th to 74th percentile in exposure intensity, 75th to 89th percentile in exposure intensity, and above 90th percentile in exposure intensity *Panel B.* All regressions include birth-year, calendar-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. Columns 1-3 report the effect in the school-grade context, and columns 4-6 in the neighborhood-grade context. Standard errors are two-way clustered at the school and neighborhood levels. (**) $p < .01$, (*) $p < .05$, ($^{+}$) $p < 0.1$.

Table A17: Sibling Crime Exposure: School vs. Neighborhood Peer Groups (Mutually Exclusive Definitions)

	Any Crime (Age 16-21)			Group Crime (Age 16-21)			Ever in Gang		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A. Gateway Exposure by Siblings, Defined by History of Group Crimes									
School w/o Ngh.	0.027*	0.026*	0.026*	0.026**	0.025*	0.025*	0.004	0.004	0.004
	(0.012)	(0.012)	(0.012)	(0.010)	(0.010)	(0.010)	(0.004)	(0.004)	(0.004)
Ngh. w/o School	-0.001	-0.001	-0.000	0.003	0.003	0.003	0.001	0.002	0.002
	(0.006)	(0.006)	(0.006)	(0.005)	(0.005)	(0.005)	(0.002)	(0.002)	(0.002)
School & Ngh.	0.008	0.005	0.005	0.005	0.004	0.004	-0.001	-0.001	-0.001
	(0.007)	(0.007)	(0.007)	(0.005)	(0.005)	(0.005)	(0.003)	(0.003)	(0.003)
Panel B. Gateway Exposure by Siblings, Defined by History of All Crimes									
School w/o Ngh.	0.010	0.009	0.009	0.016*	0.015*	0.015*	0.003	0.003	0.003
	(0.009)	(0.009)	(0.009)	(0.007)	(0.007)	(0.007)	(0.002)	(0.002)	(0.002)
Ngh. w/o School	0.002	0.003	0.003	0.005	0.005^{+}	0.005^{+}	0.000	0.000	0.000
	(0.005)	(0.005)	(0.005)	(0.003)	(0.003)	(0.003)	(0.001)	(0.001)	(0.001)
School & Ngh.	0.011*	0.011*	0.011*	0.008*	0.008*	0.008*	0.002	0.002	0.002
	(0.005)	(0.005)	(0.005)	(0.004)	(0.004)	(0.004)	(0.002)	(0.002)	(0.002)
Focal Controls		✓	✓		✓	✓		✓	✓
Peer Group Controls			✓			✓			✓
Observations	567,029	567,029	567,029	567,029	567,029	567,029	567,029	567,029	567,029
Dep. Mean	0.120	0.120	0.120	0.054	0.054	0.054	0.007	0.007	0.007

Note.—Each column-panel combination reports estimates from a separate regression based on equation (3) considering gateway exposure defined by older siblings with group crime history. However, here exposure groups are defined as: (i) peers who attend the same school but live in a different neighborhood, (ii) peers who live in the same neighborhood but attend a different school, and (iii) peers who share both school and neighborhood with the focal student. All three exposure measures are included simultaneously in each regression. All regressions include birth-year, calendar-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. Focal controls consist of sex, foreign background, single parent household status, and parental income, employment, and education, along with indicators of missing values. Peer controls are the share of male and foreign background students in each respective peer group. A Standard errors are two-way clustered at the school and neighborhood levels. (**) $p < .01$, (*) $p < .05$, ($^{+}$) $p < .1$

Table A18: Gateway and Sibling Gender

Student Sample:	All		Males		Females	
	(1) Group Crime	(2) Ever In Gang	(3) Group Crime	(4) Ever In Gang	(5) Group Crime	(6) Ever In Gang
Panel A. Gateway Exposure by Brothers, Defined by History of Group Crimes						
School Exp.	0.051**	0.007	0.068*	0.014	0.026	0.001
	(0.020)	(0.008)	(0.031)	(0.015)	(0.019)	(0.004)
Panel B. Gateway Exposure by Sisters, Defined by History of Group Crimes						
School Exp.	0.081**	0.004	0.095+	0.003	0.073*	0.000
	(0.029)	(0.010)	(0.049)	(0.020)	(0.030)	(0.005)
Panel C. Gateway Exposure by Brothers, Defined by History of Group Crimes						
Neighborhood Exp.	0.011	0.002	0.010	0.005	0.016	0.000
	(0.011)	(0.005)	(0.019)	(0.009)	(0.011)	(0.002)
Panel D. Gateway Exposure by Sisters, Defined by History of Group Crimes						
Neighborhood Exp.	0.030+	0.008	0.040	0.004	0.014	0.007+
	(0.018)	(0.008)	(0.030)	(0.014)	(0.019)	(0.004)
Observations	609,459	609,459	313,226	313,226	296,233	296,233
Dep. Mean	0.054	0.007	0.078	0.012	0.028	0.001

Note.—Each cell reports estimates from a separate regression based on equation (3), where gateway exposure is defined by whether an older sibling or an older sister has a history of group crime. Exposure is measured at the school-grade level in *panels A* and *B*, and at the neighborhood-grade level in *panels C* and *D*. Estimates are presented for three subsamples: (i) all students (columns 1 and 2), (ii) male students (columns 3 and 4), and (iii) female students (columns 5 and 6). All regressions include birth-year, calendar-year, and school-neighborhood fixed effects, and exclude the gateway peers themselves from the estimation sample. Standard errors are two-way clustered at the school and neighborhood levels.
(**) $p < .01$, (*) $p < .05$, (+) $p < .1$